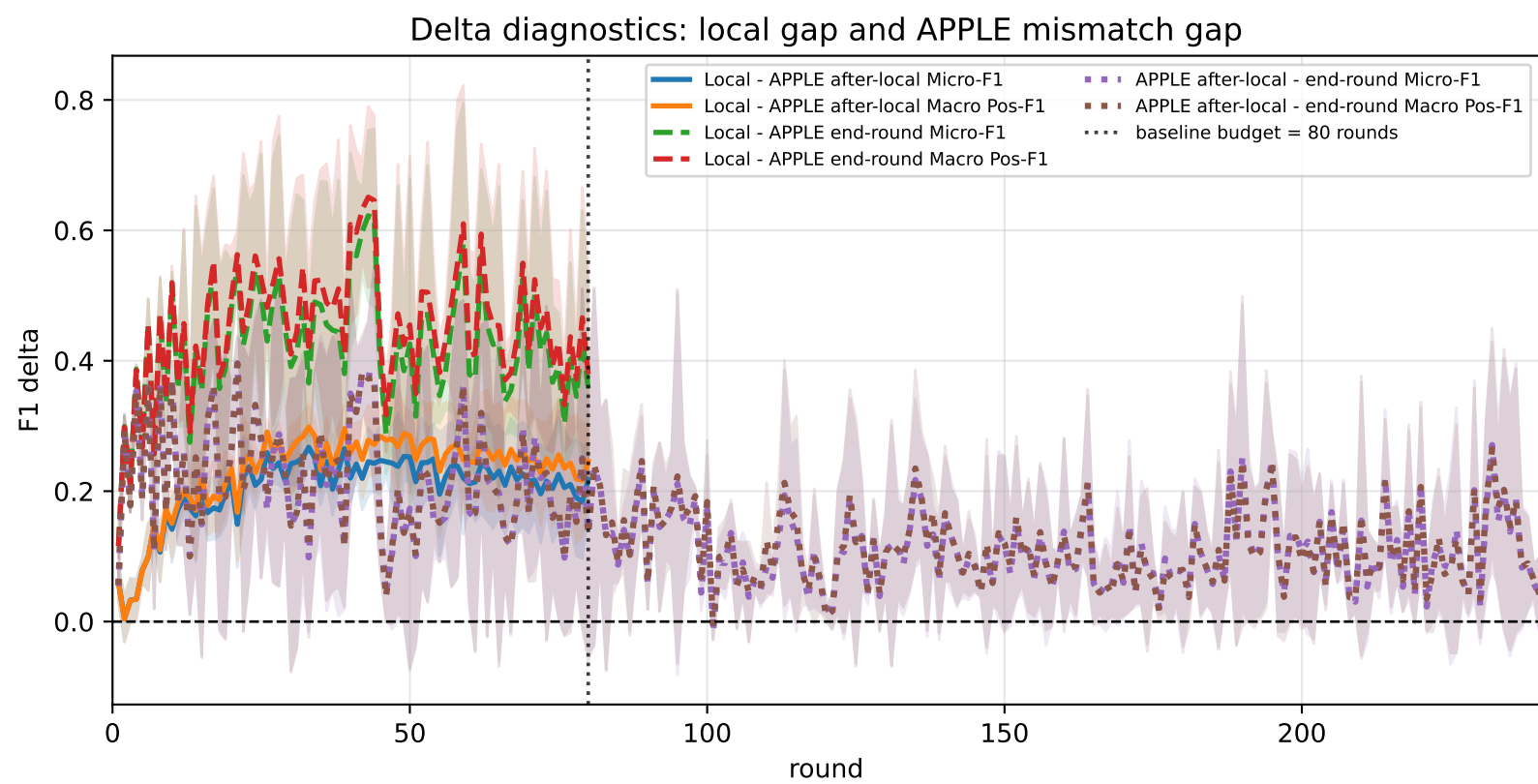
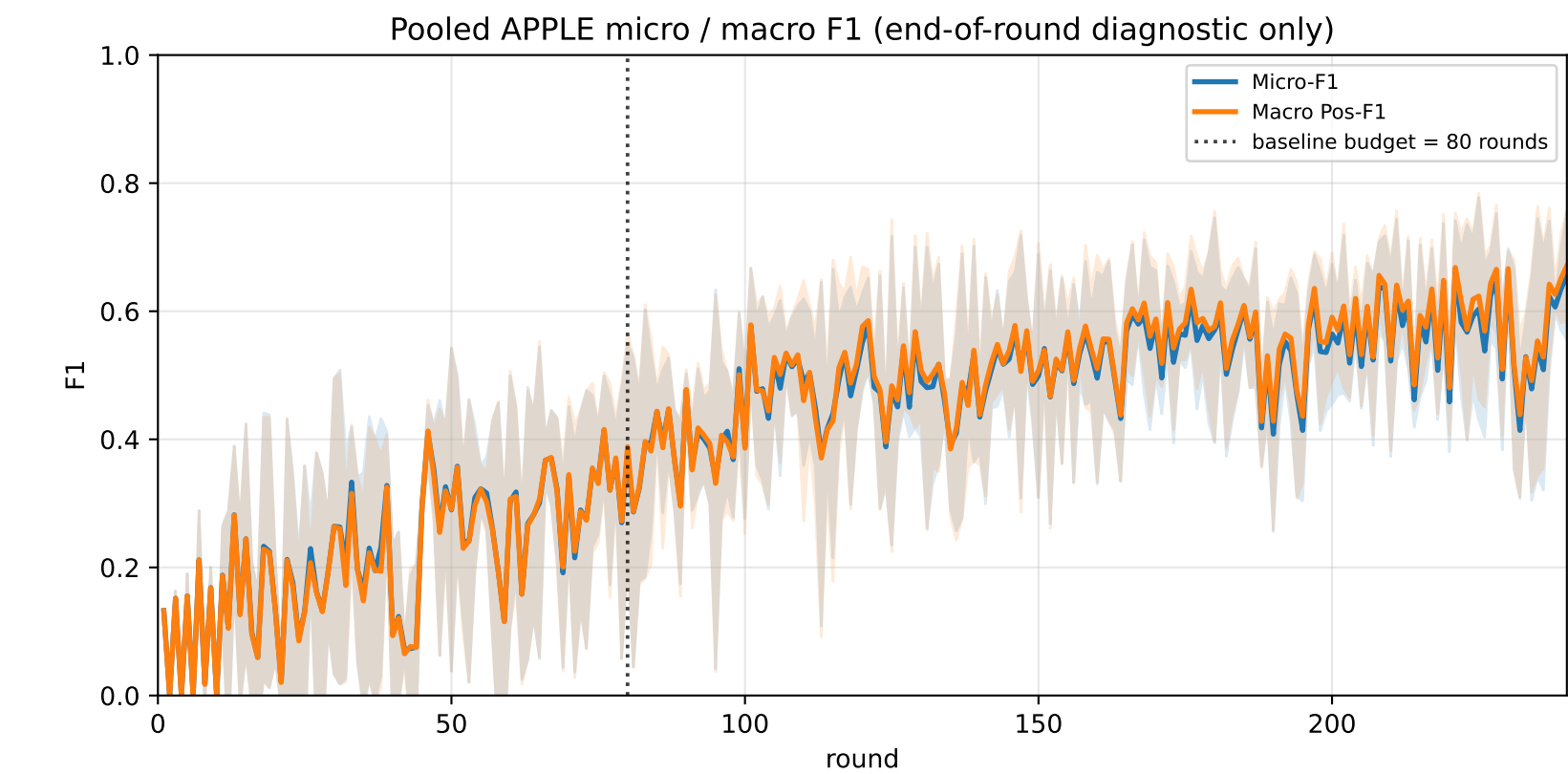
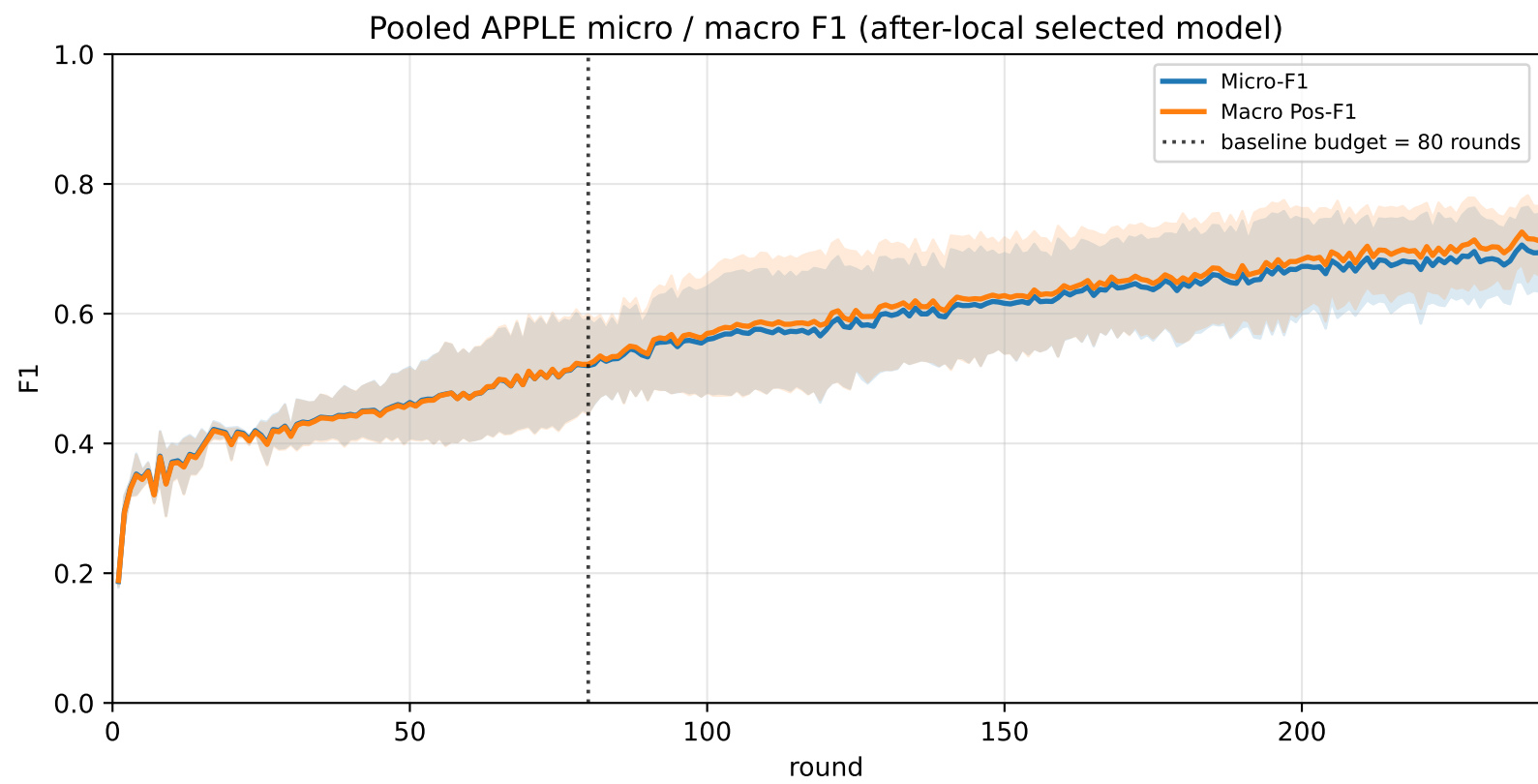
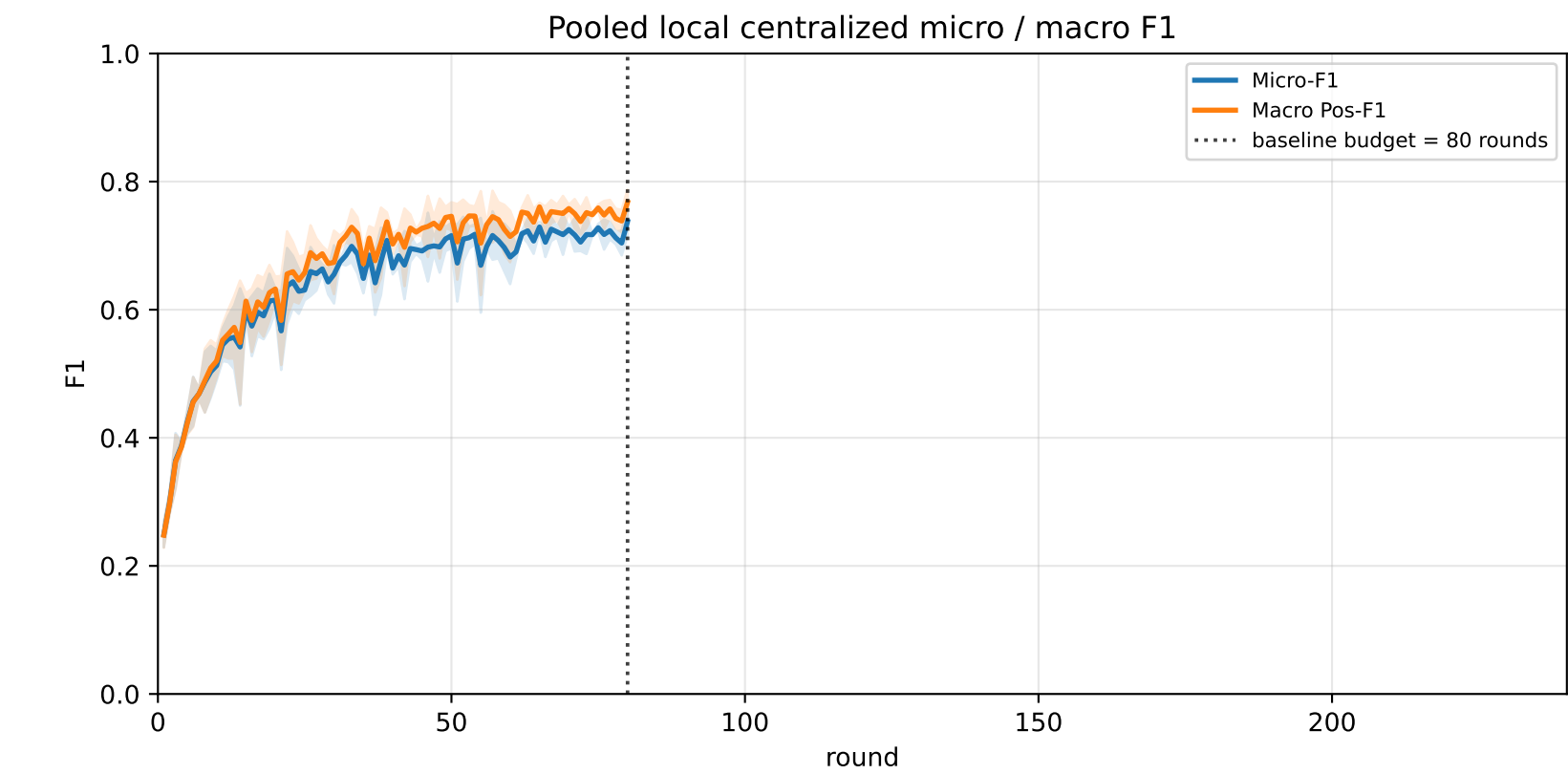
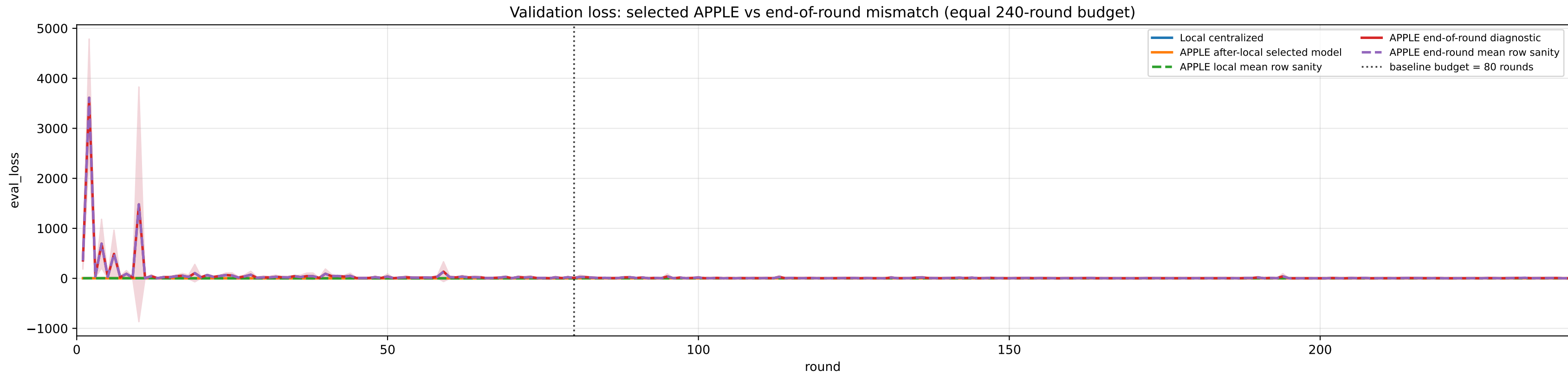


Controlled q-label heterogeneity: iid | APPLE diagnostics | aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

Best test performance (APPLE selected by after-local validation; all methods 80 rounds)

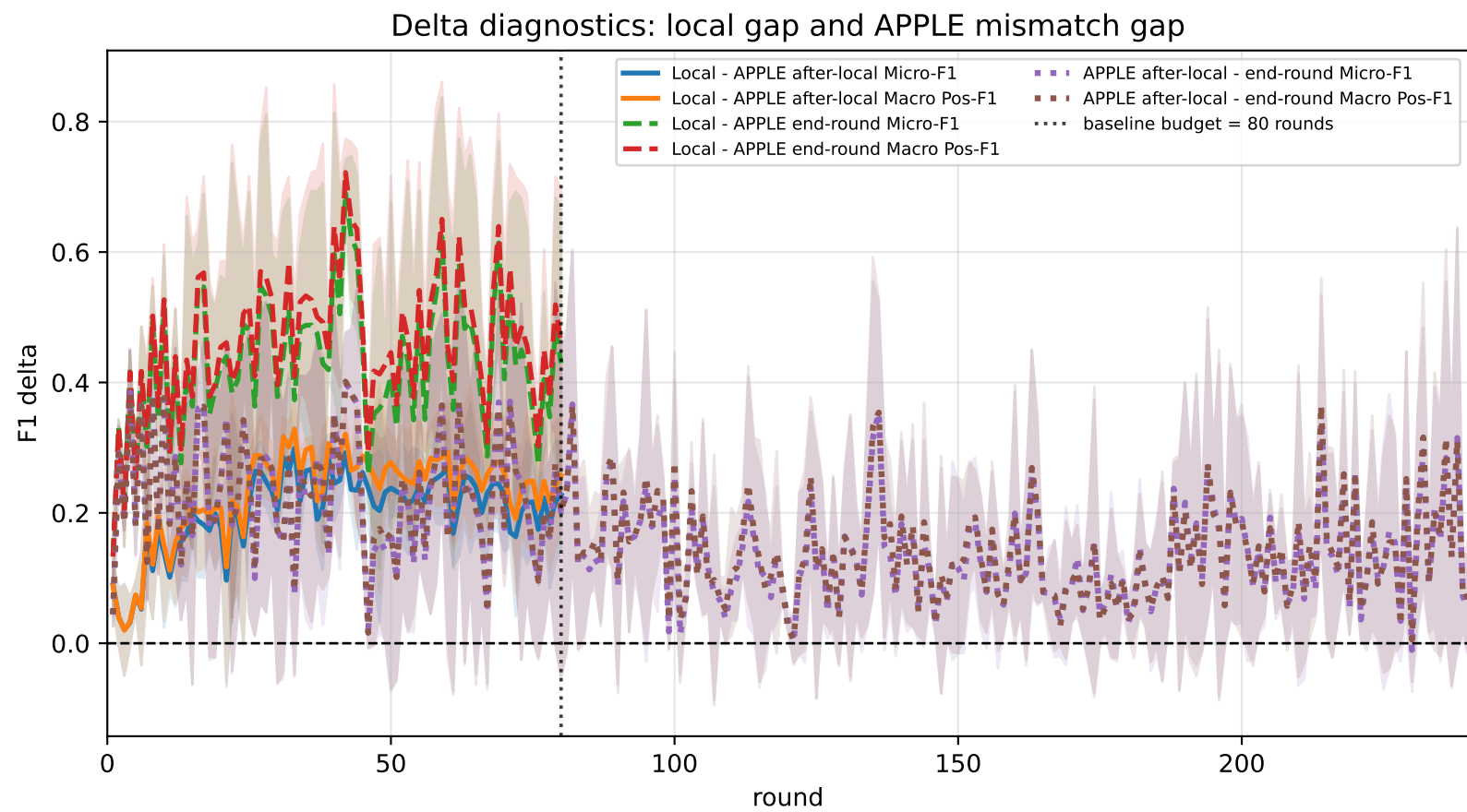
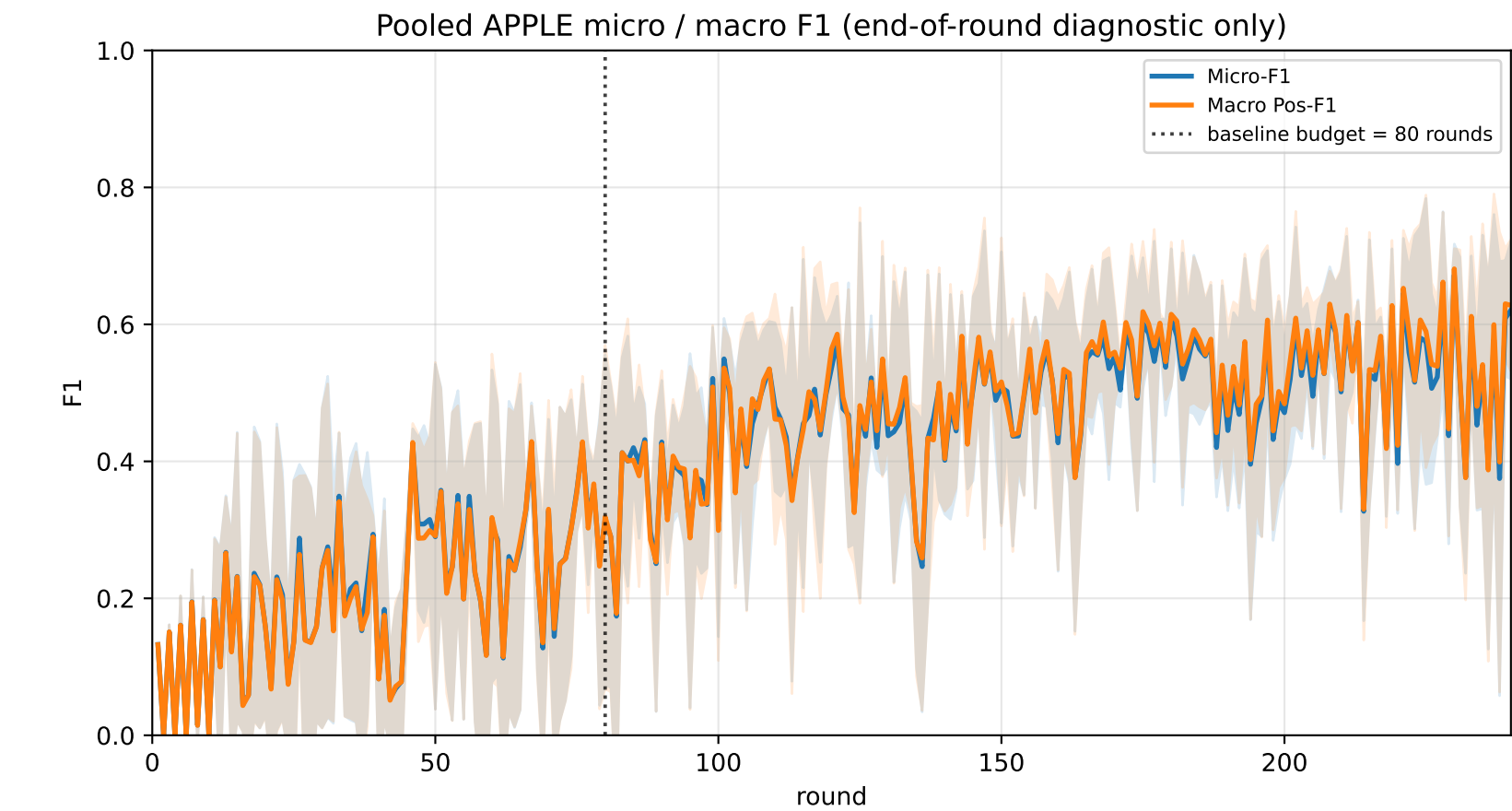
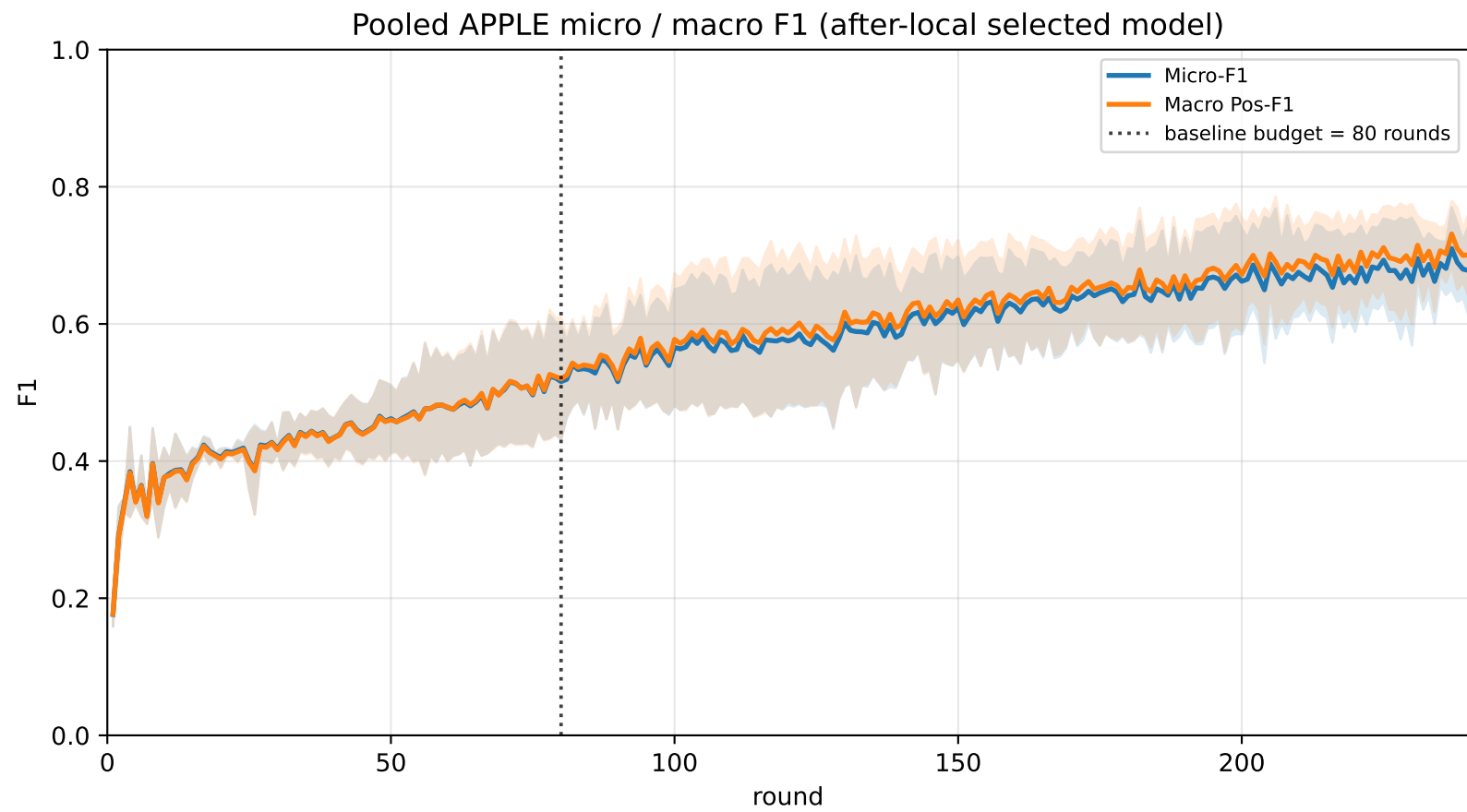
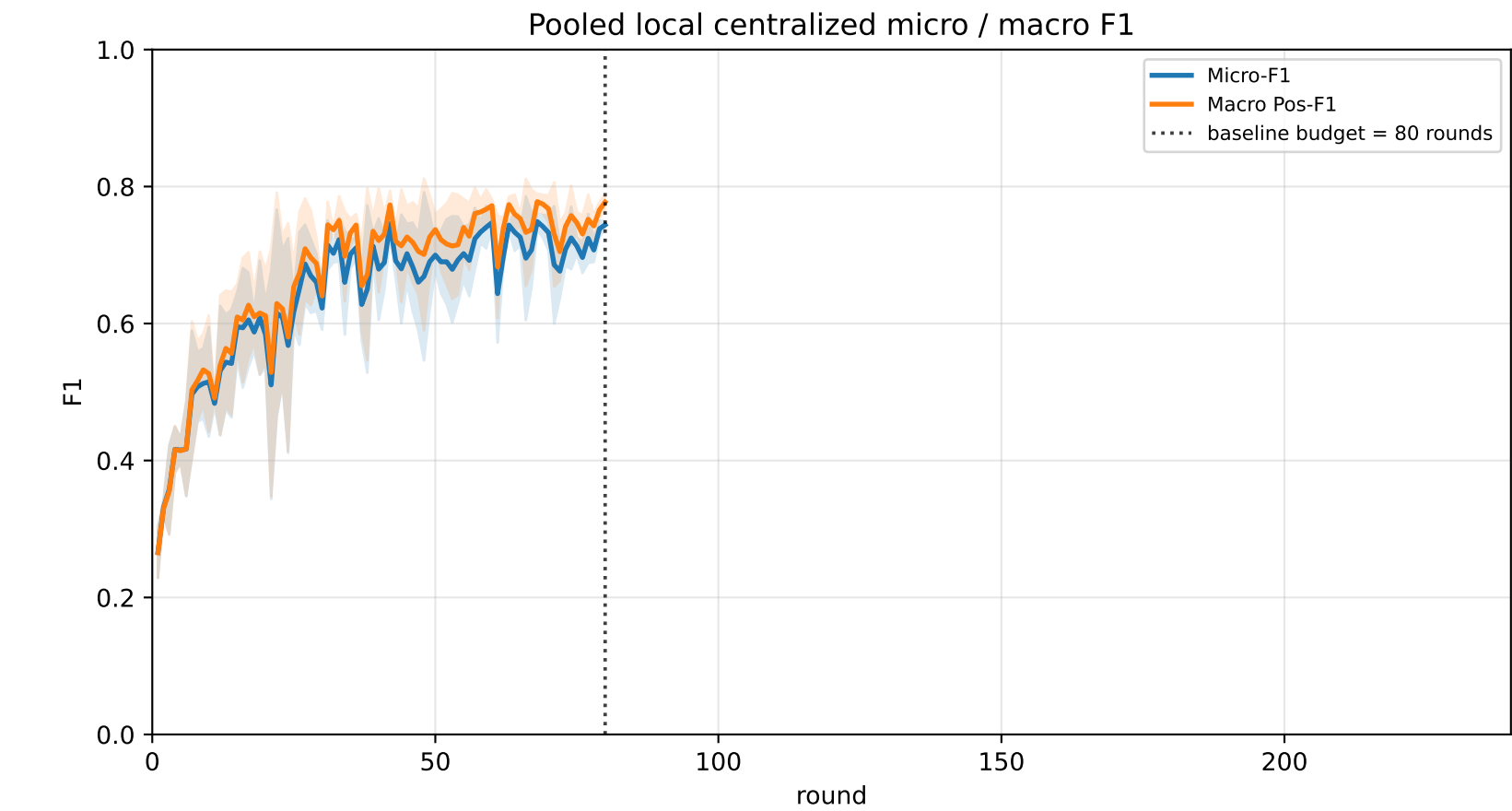
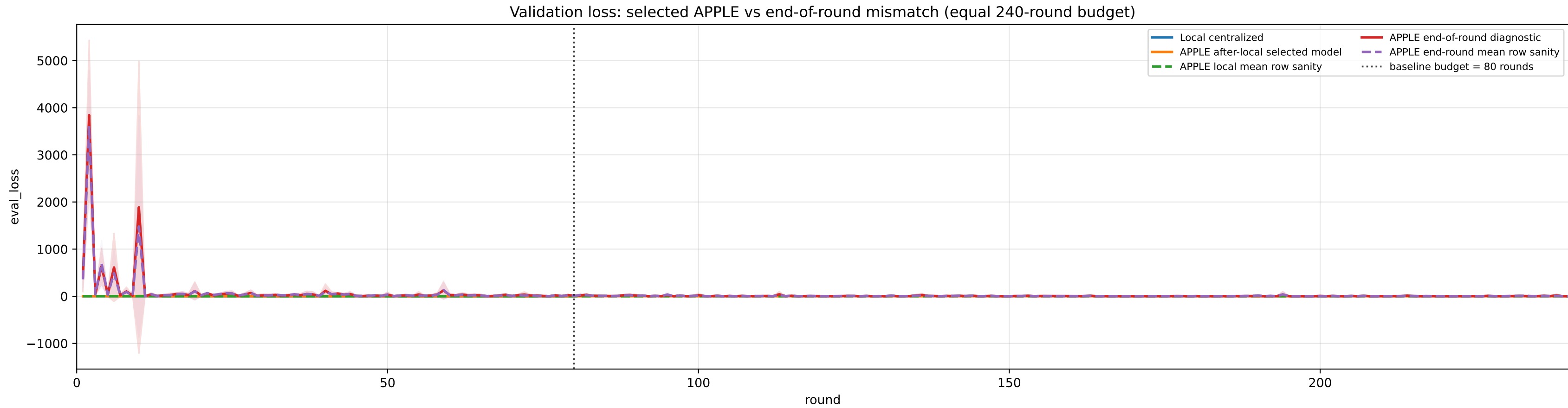
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	64.2% ± 0.6%	67.8% ± 0.9%	79.6% ± 4.0%	78.3% ± 5.1%	66.8% ± 1.9%	61.9% ± 2.0%	48.9% ± 0.2%
APPLE	68.2% ± 4.9%	70.4% ± 4.6%	82.2% ± 11.5%	74.8% ± 11.0%	71.1% ± 1.1%	65.5% ± 6.4%	58.0% ± 4.0%



family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

Best test performance (APPLE selected by after-local validation; all methods 80 rounds)

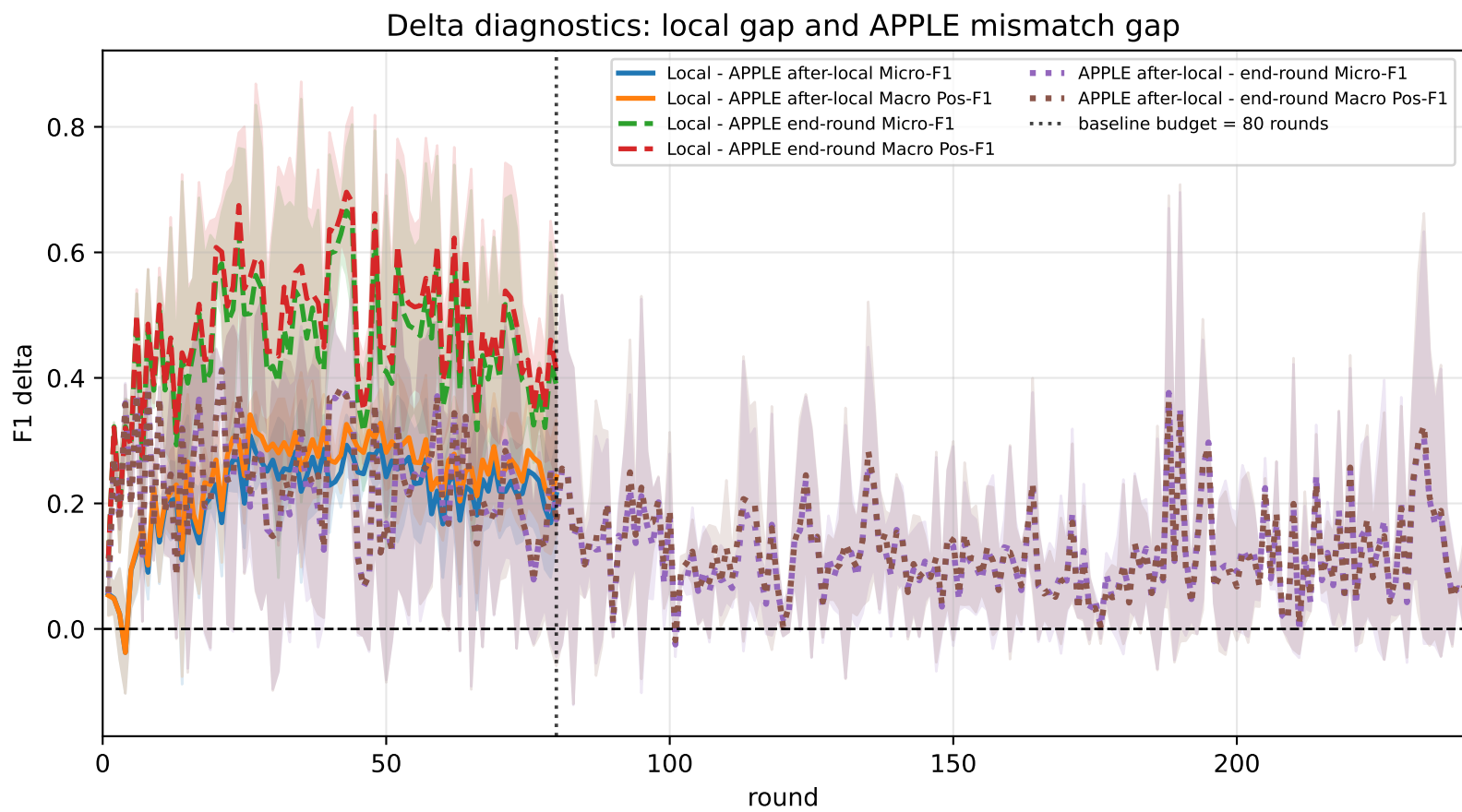
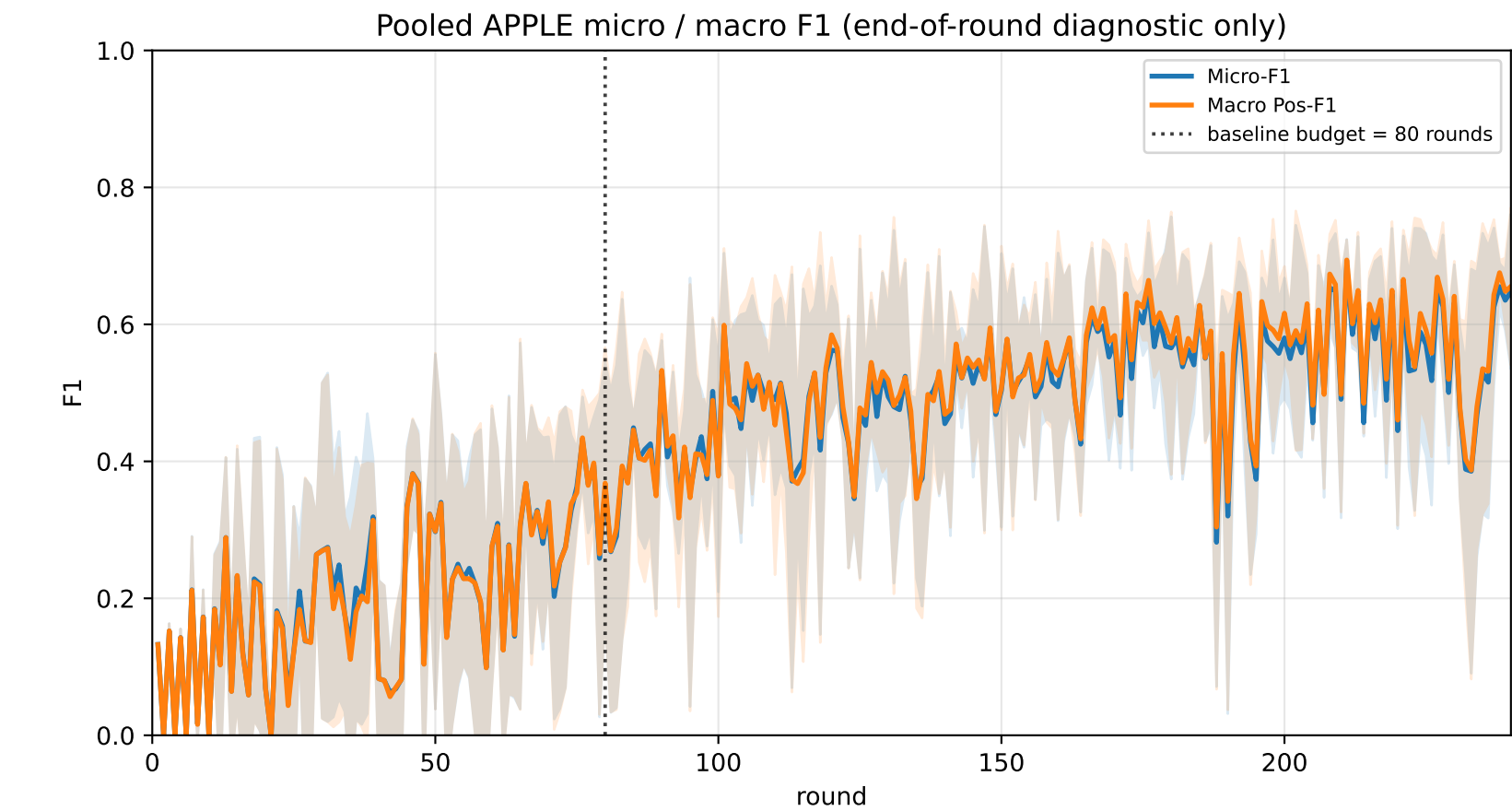
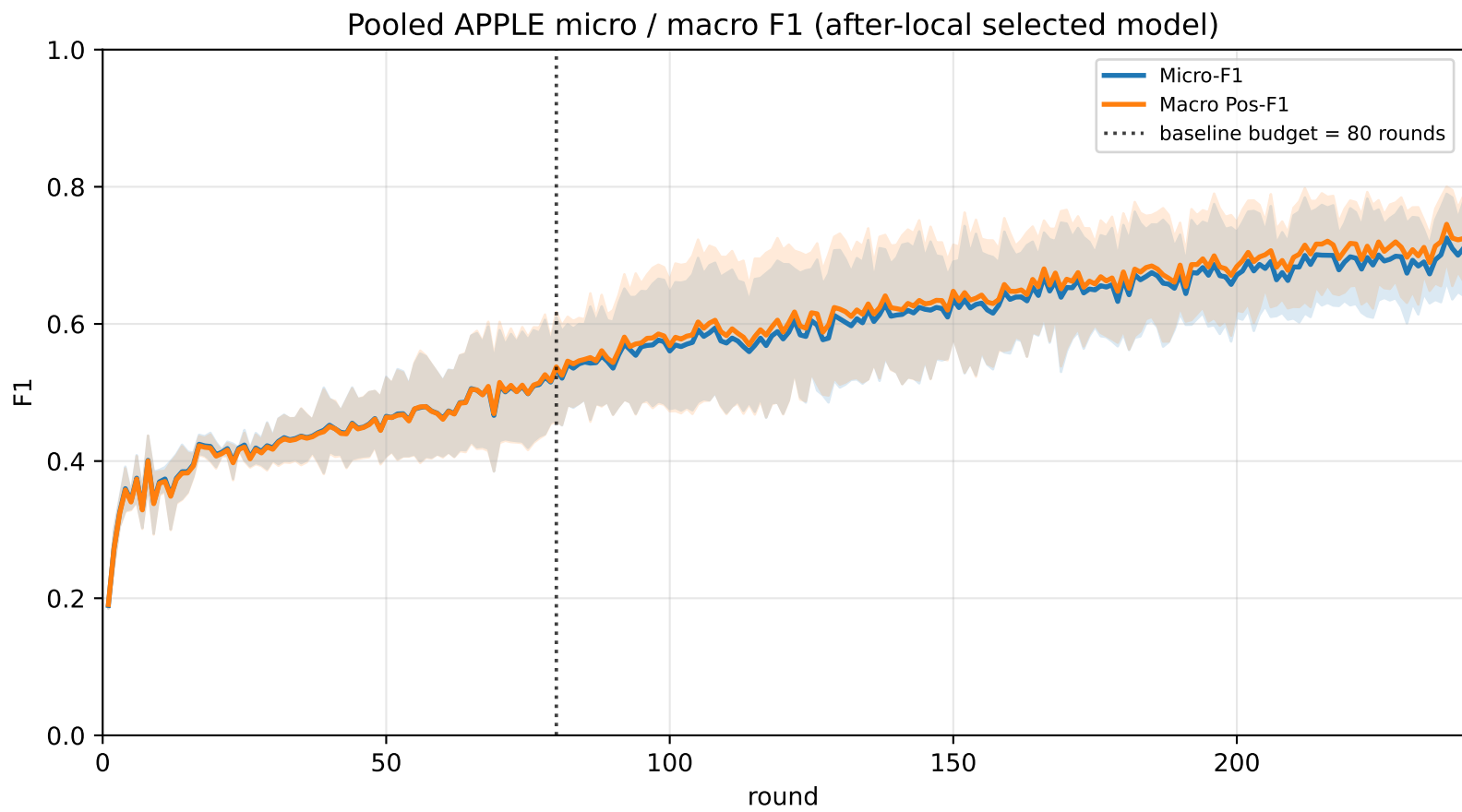
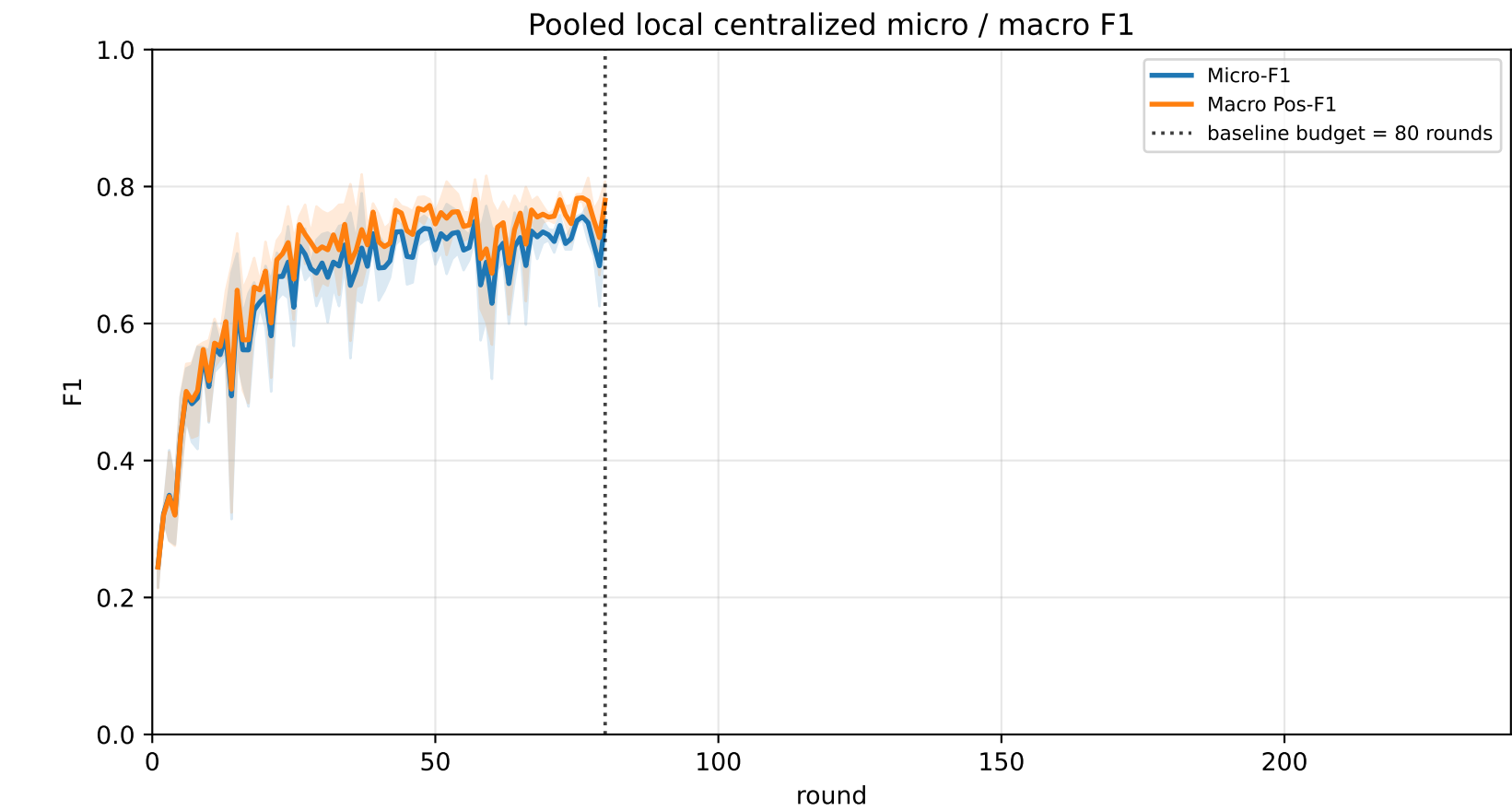
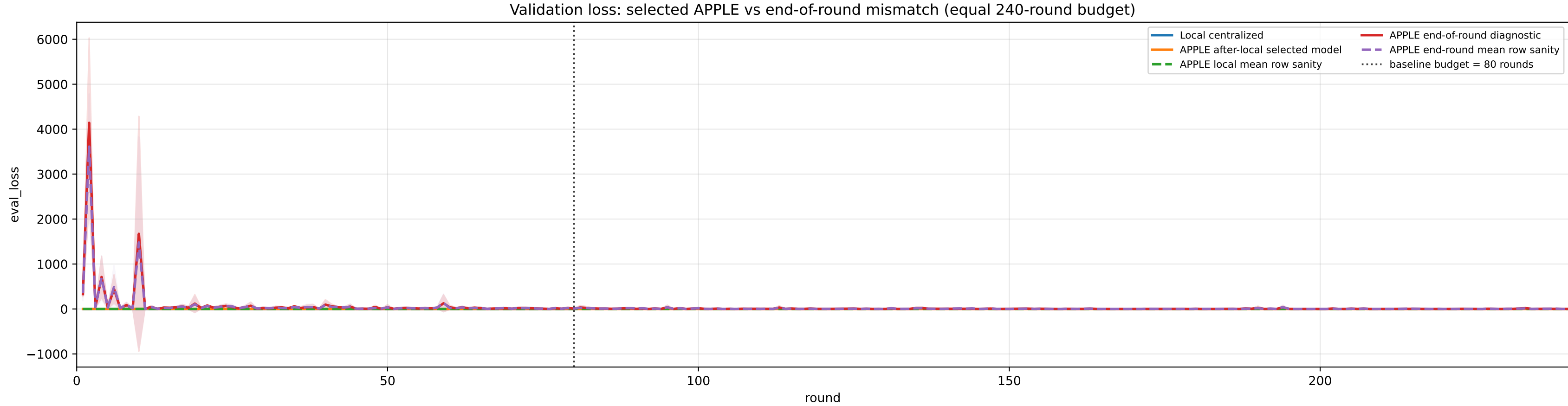
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	61.7% ± 2.9%	65.8% ± 2.5%	83.4% ± 5.2%	77.4% ± 2.3%	64.8% ± 2.9%	56.1% ± 5.8%	47.3% ± 2.8%
APPLE	68.2% ± 4.2%	70.8% ± 3.8%	84.1% ± 12.3%	75.0% ± 11.3%	72.5% ± 0.8%	65.1% ± 6.6%	57.2% ± 3.0%



family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

Best test performance (APPLE selected by after-local validation; all methods 80 rounds)

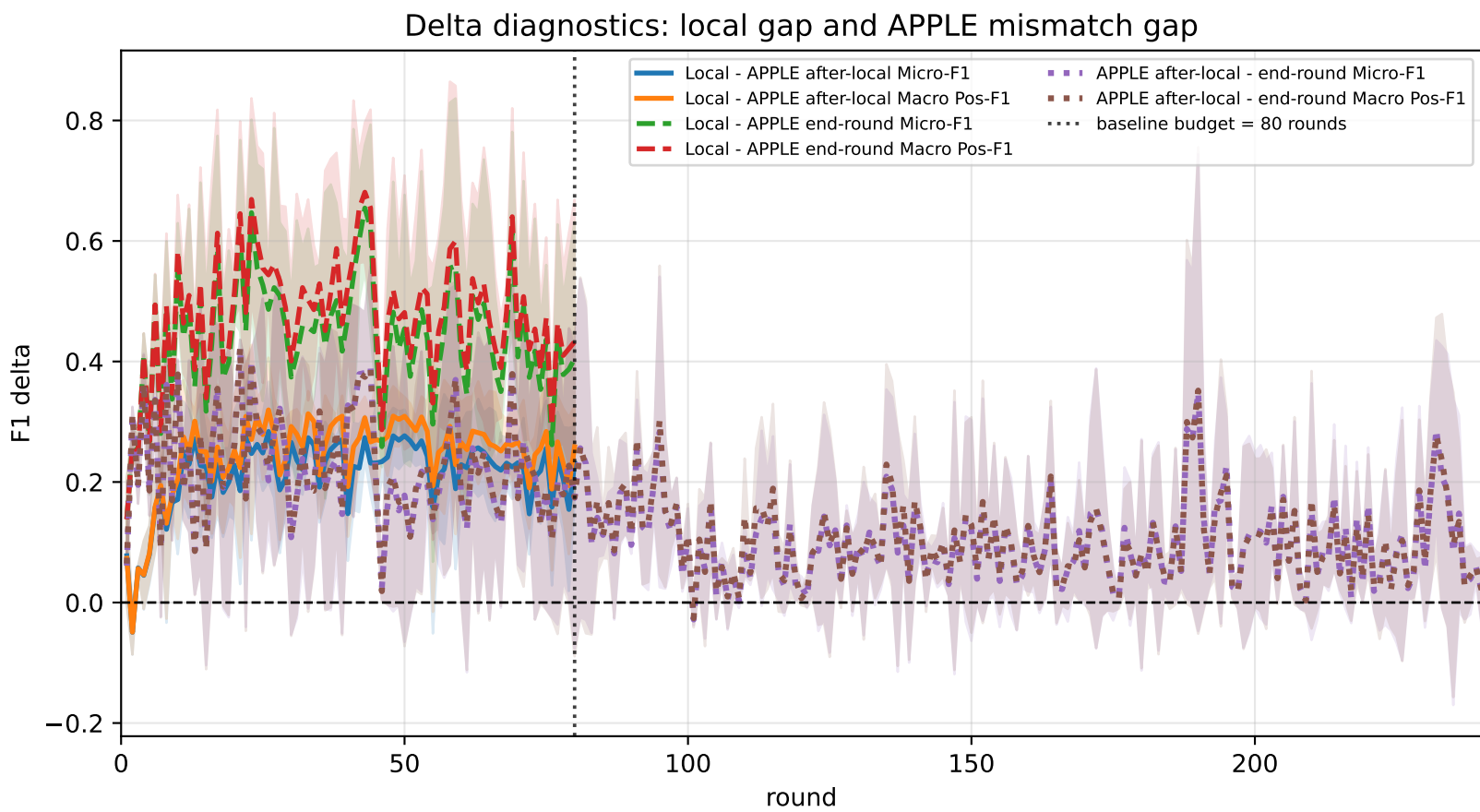
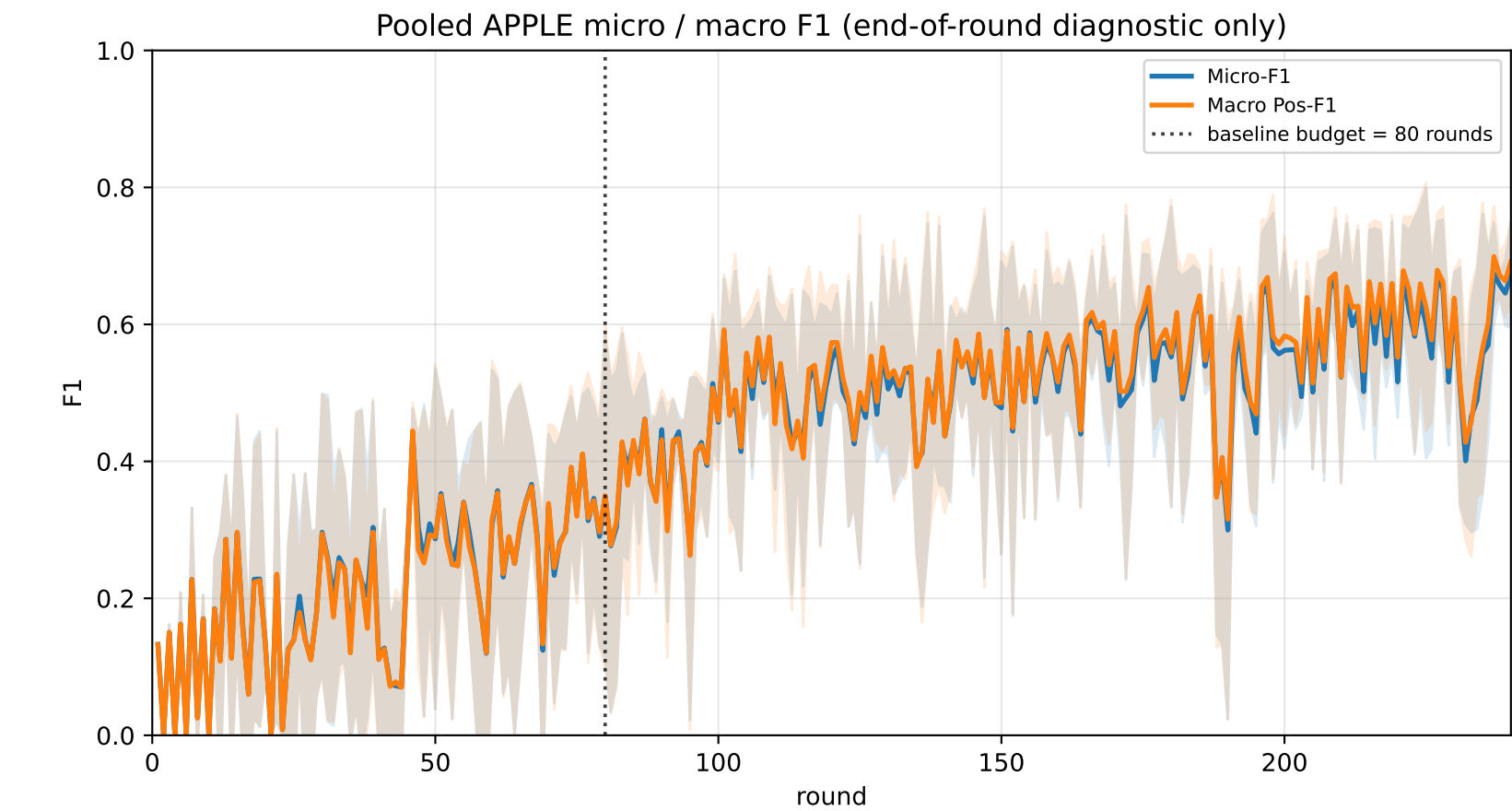
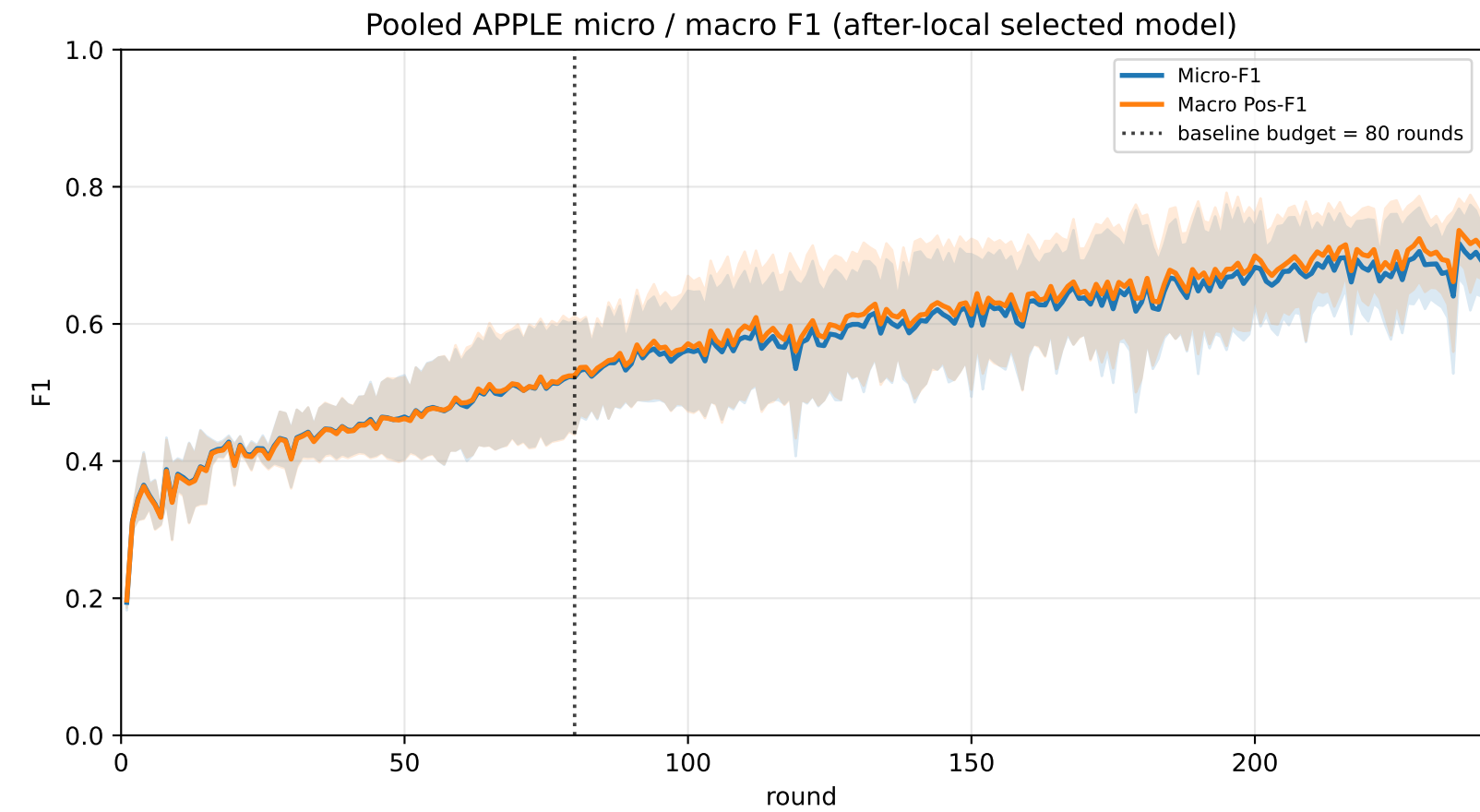
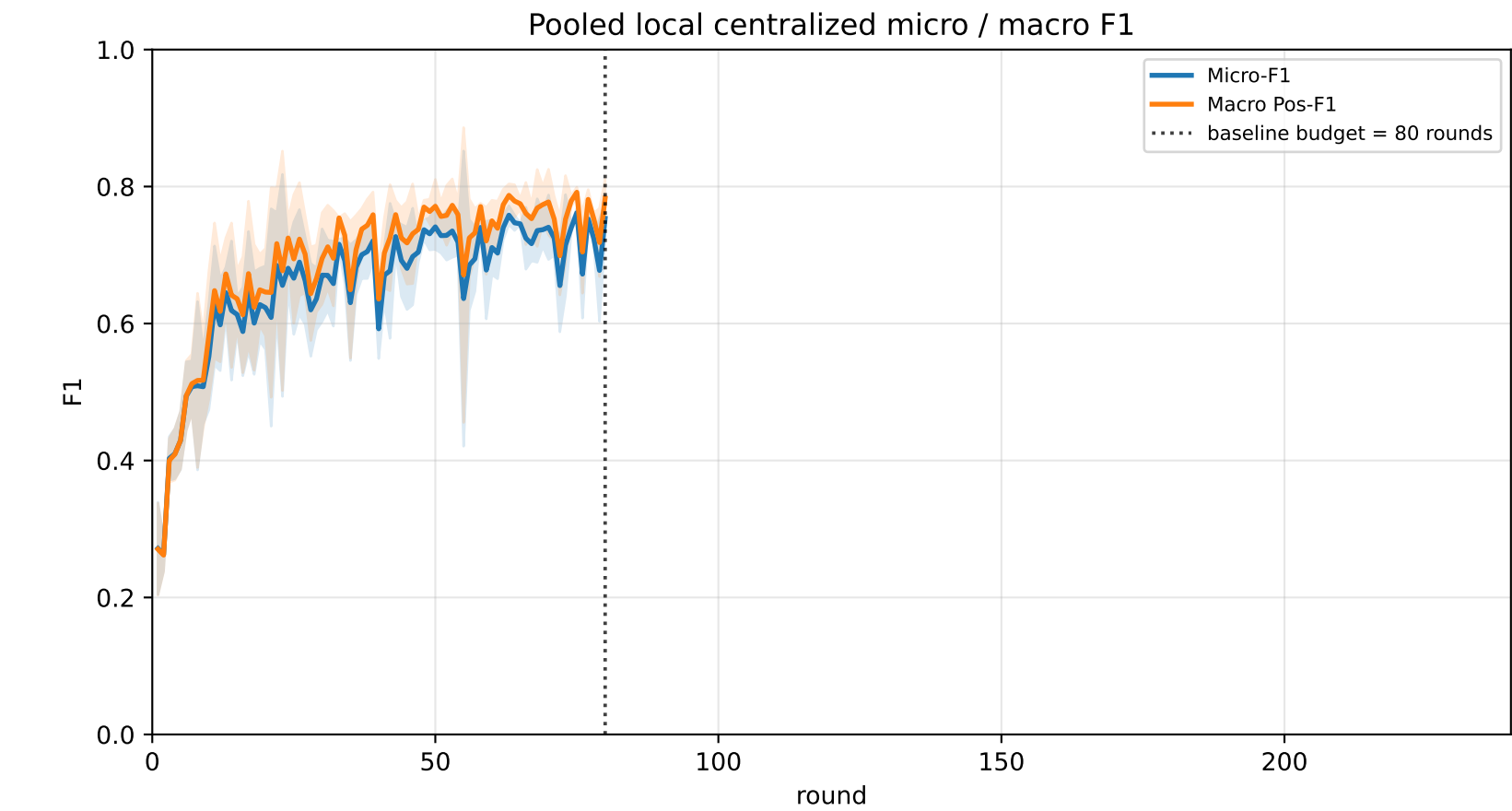
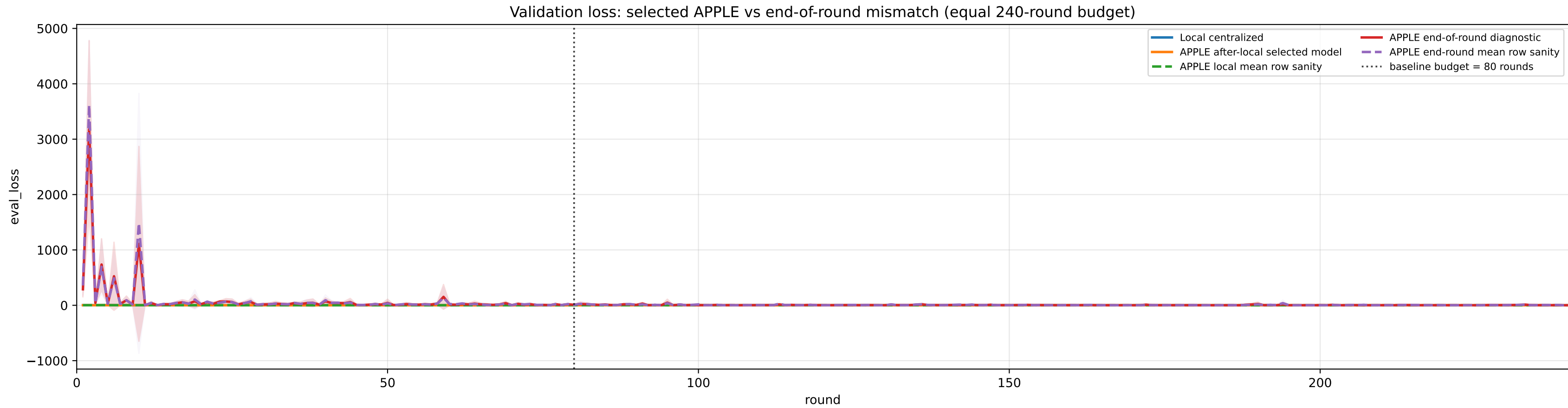
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	59.3% ± 1.8%	62.3% ± 1.2%	70.6% ± 12.2%	70.7% ± 7.8%	63.8% ± 2.1%	59.4% ± 6.7%	46.7% ± 4.1%
APPLE	69.8% ± 5.2%	71.7% ± 4.5%	82.3% ± 9.6%	75.1% ± 11.3%	72.5% ± 1.1%	68.6% ± 7.2%	60.0% ± 5.0%



family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

Best test performance (APPLE selected by after-local validation; all methods 80 rounds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	65.1% ± 6.6%	68.5% ± 6.7%	78.1% ± 11.0%	82.9% ± 5.3%	69.7% ± 5.5%	62.2% ± 8.5%	49.8% ± 5.3%
APPLE	67.2% ± 5.2%	69.2% ± 4.9%	80.7% ± 11.3%	73.4% ± 10.5%	69.3% ± 1.6%	64.6% ± 6.7%	57.9% ± 4.6%

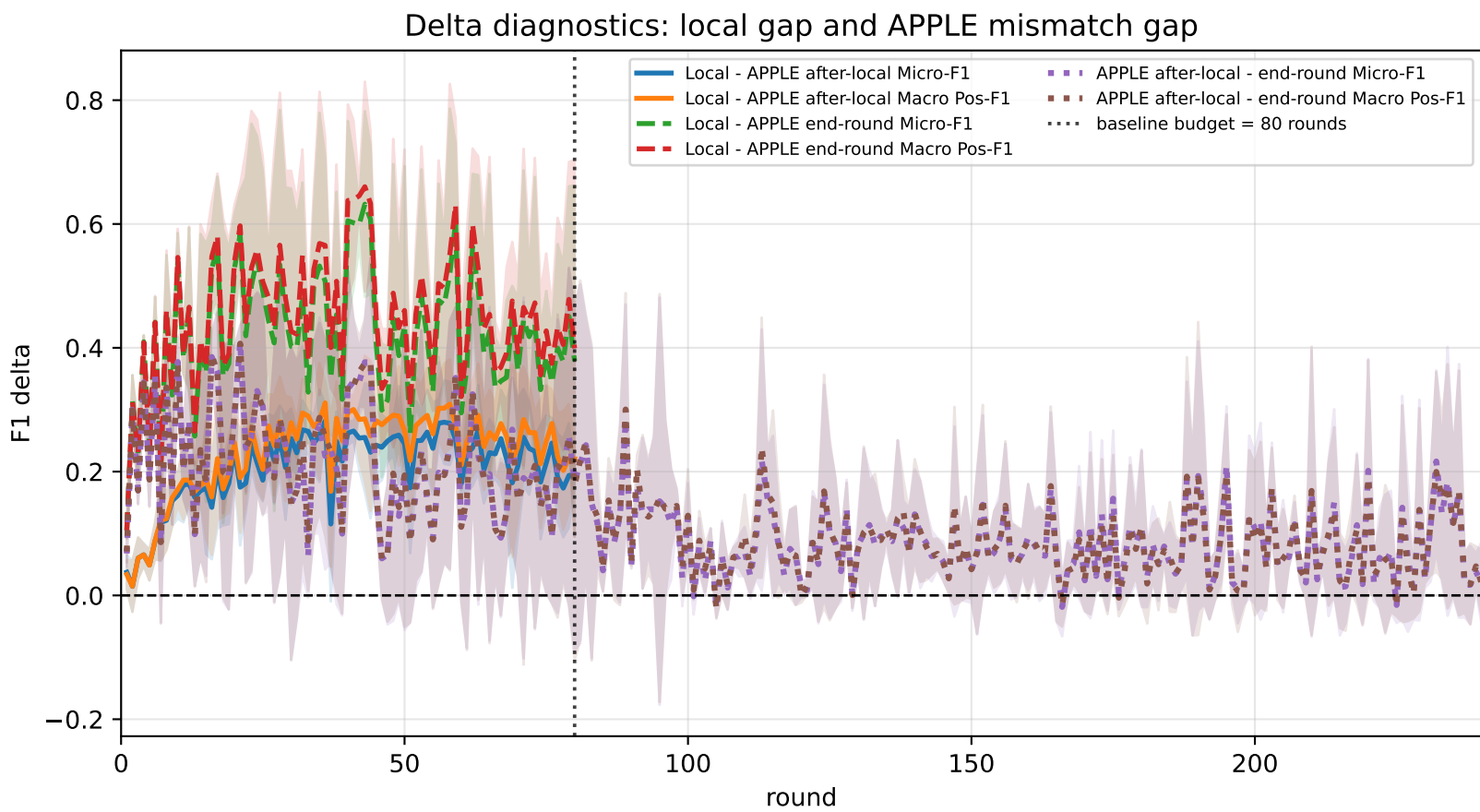
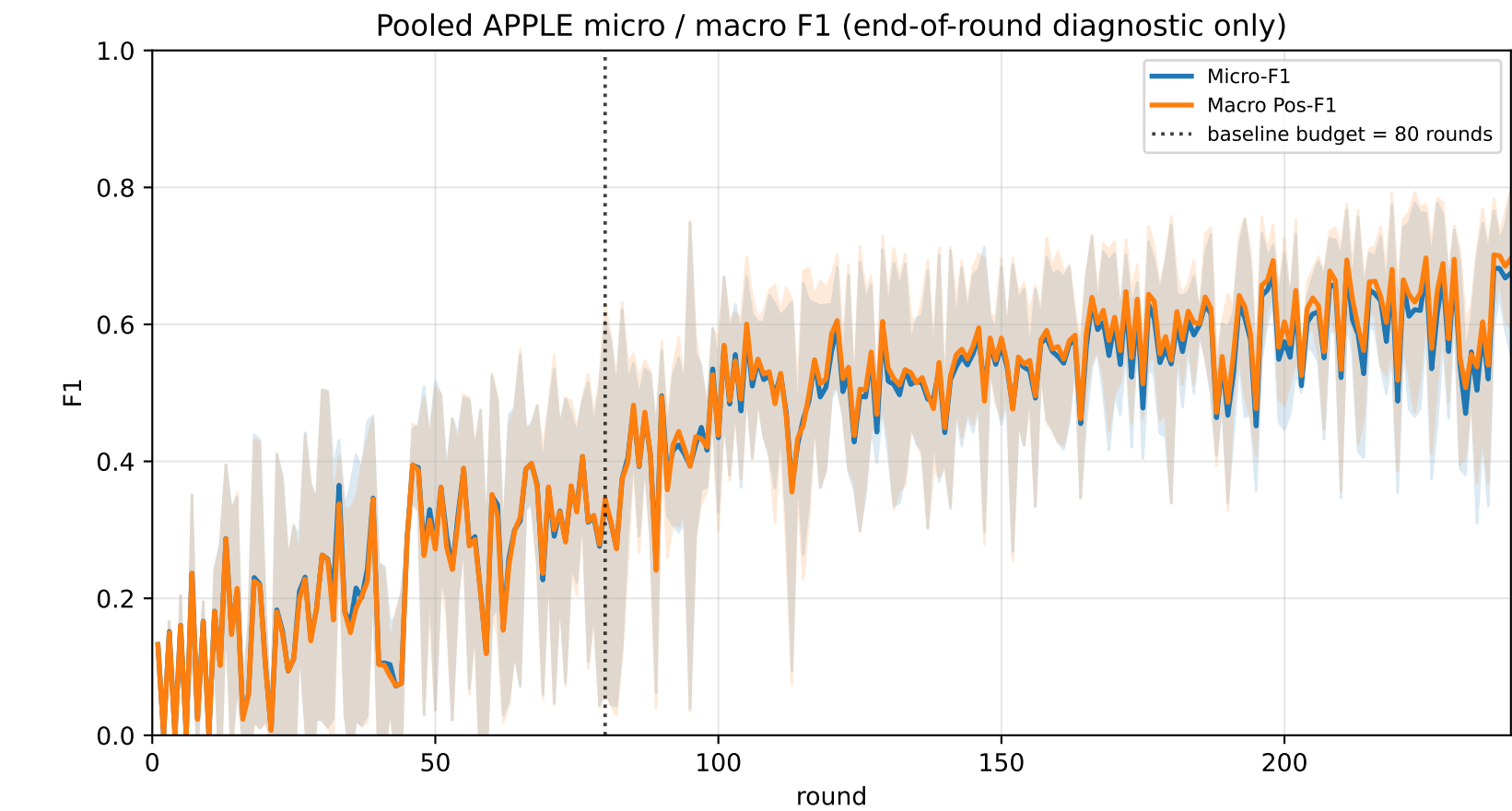
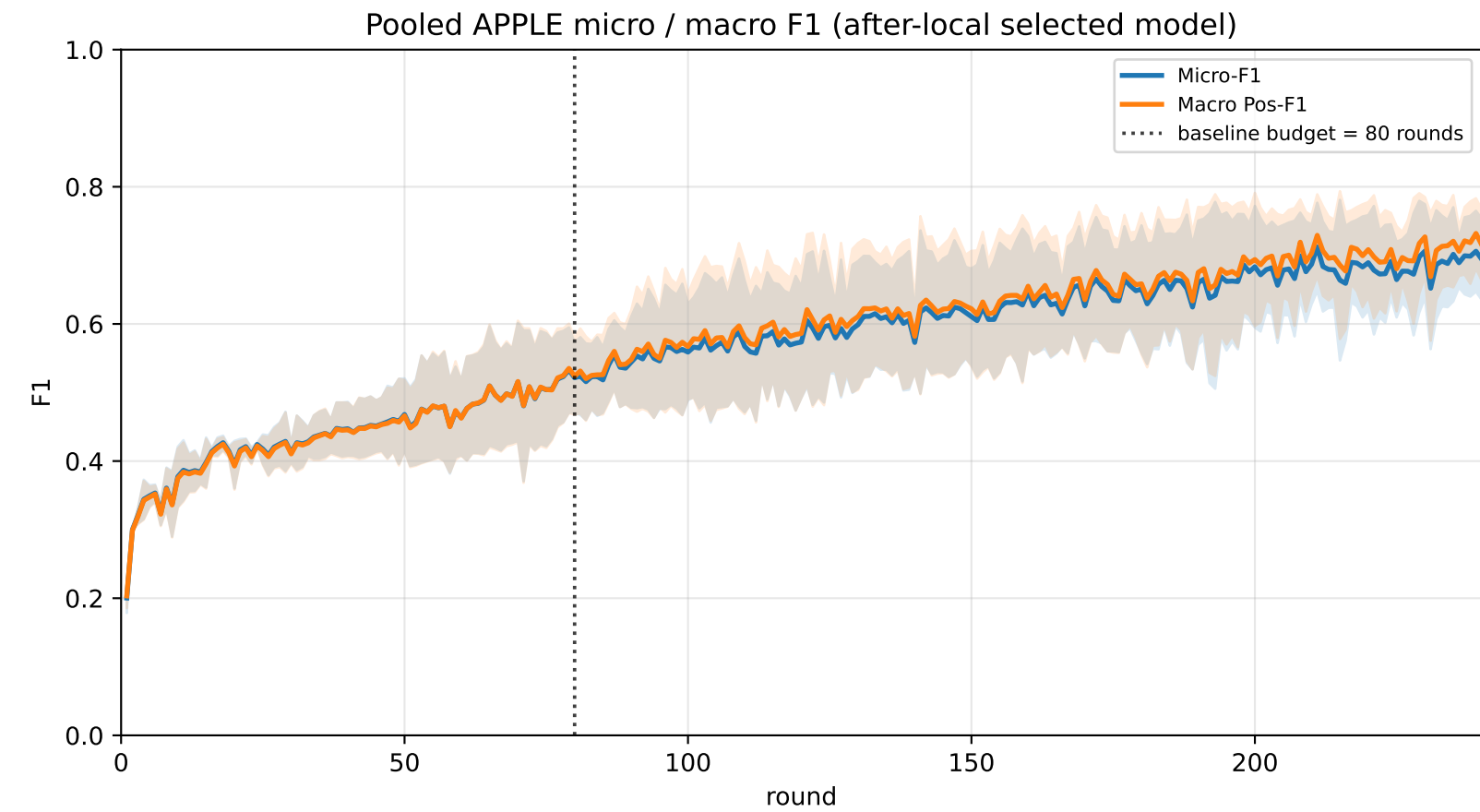
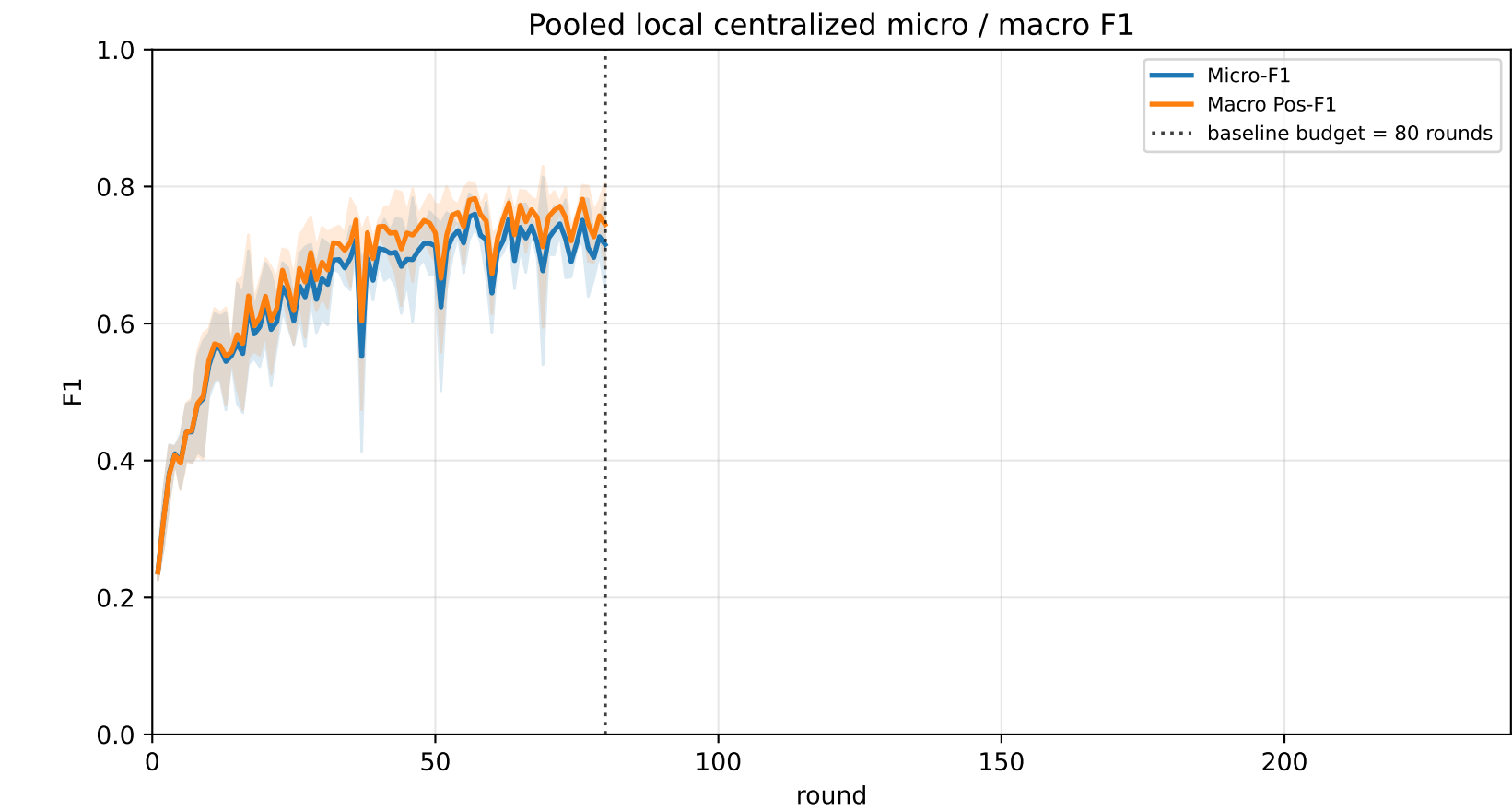
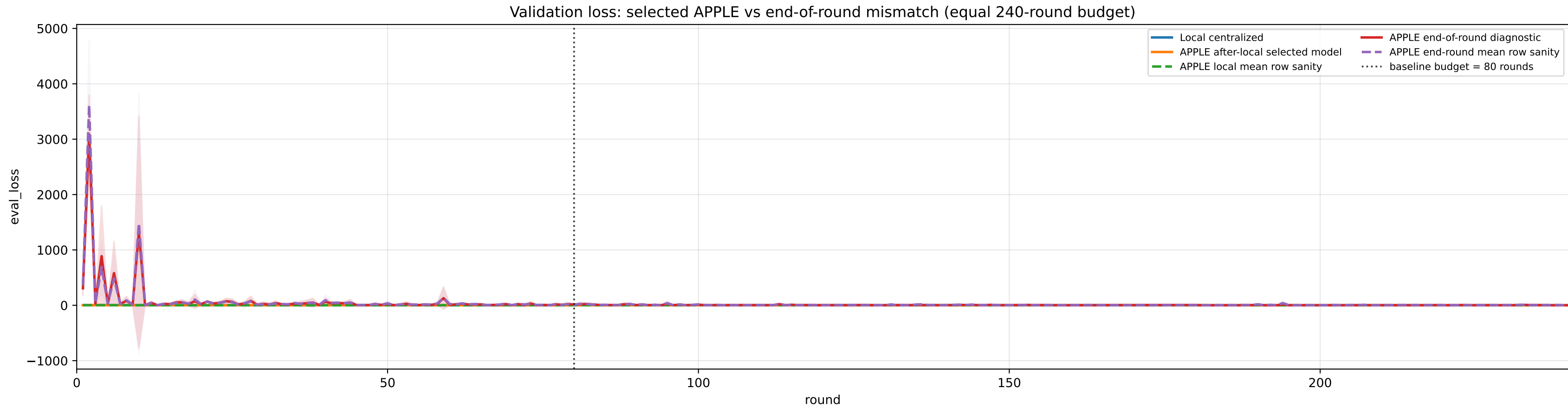


Controlled q-label heterogeneity: iid | APPLE diagnostics | family=q_task_cycle5 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

Best test performance (APPLE selected by after-local validation; all methods 80 rounds)

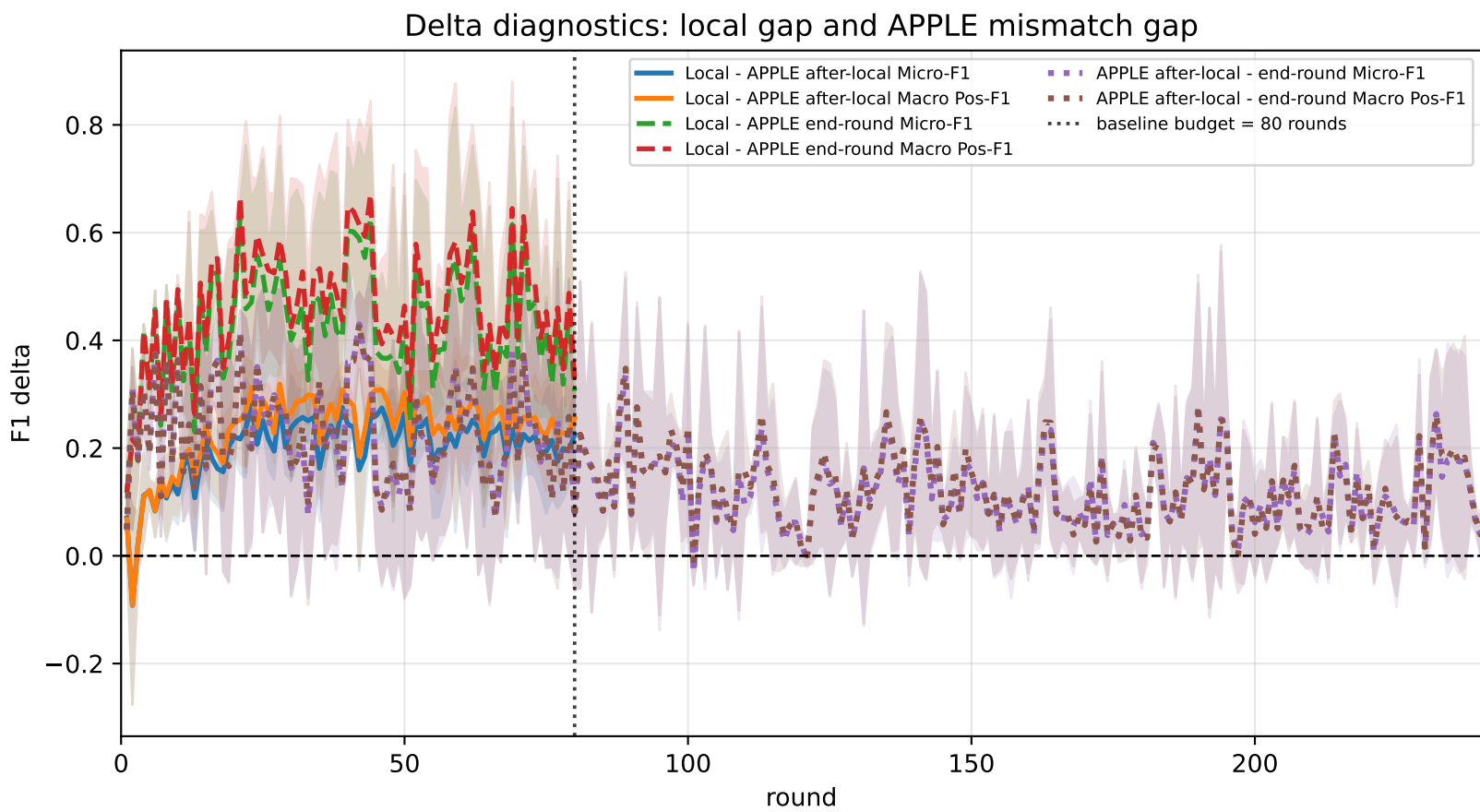
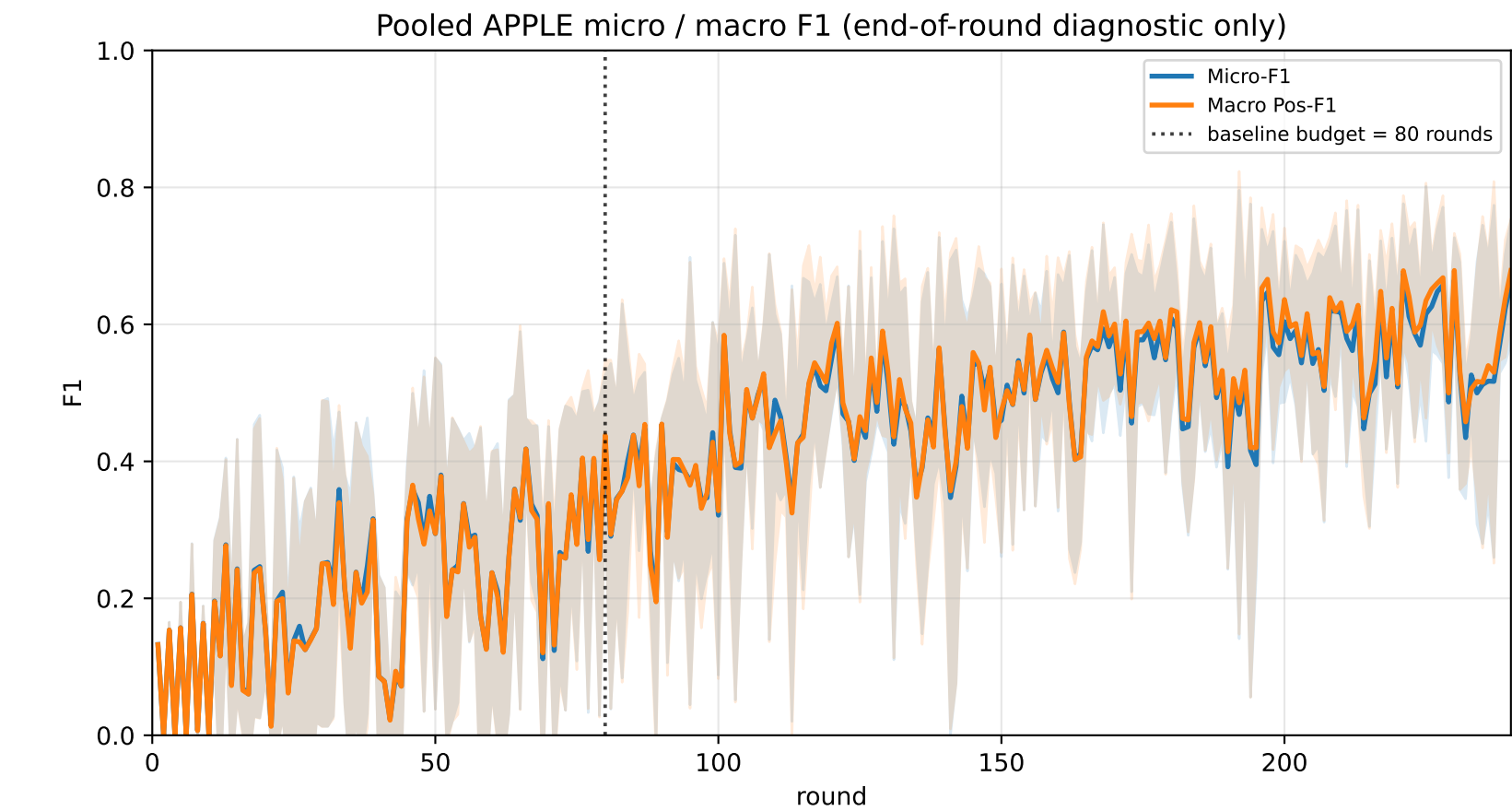
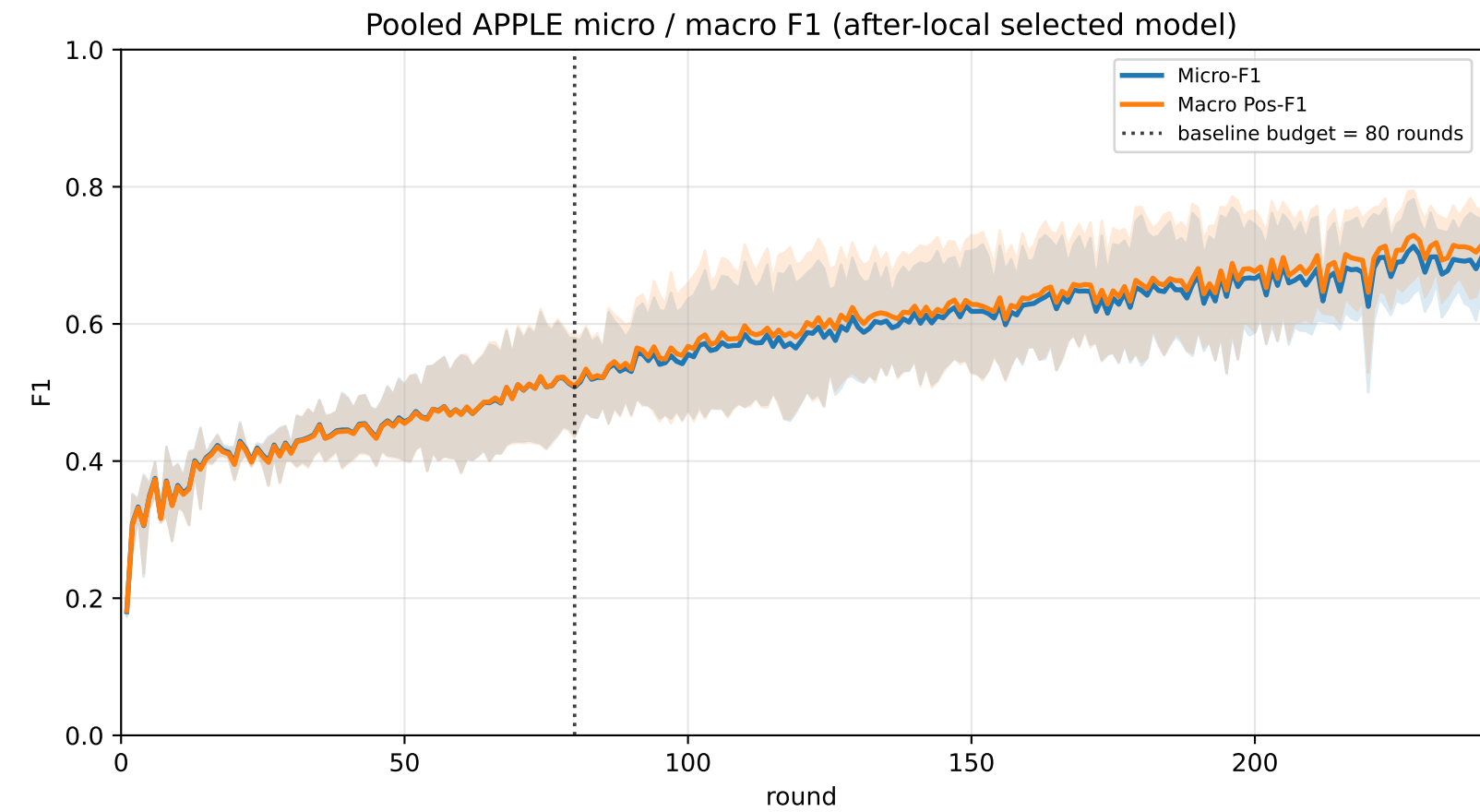
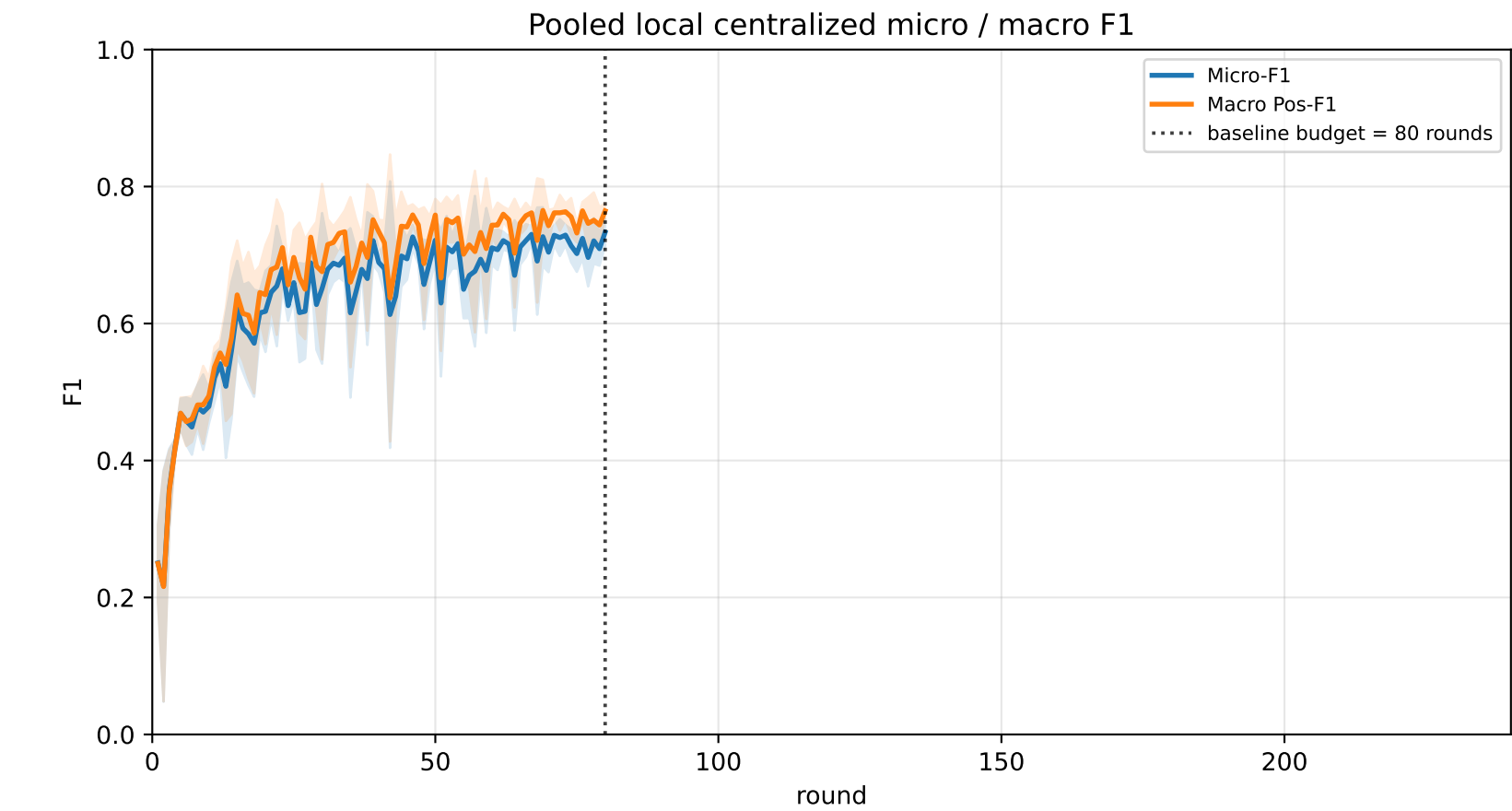
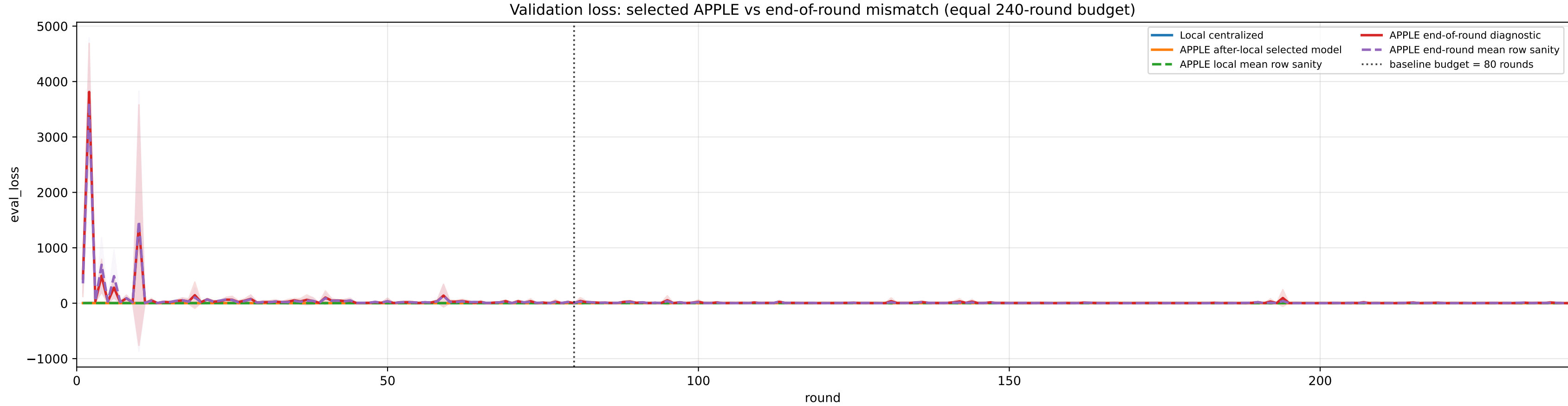
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	66.7% ± 3.2%	69.8% ± 3.5%	83.1% ± 3.5%	76.9% ± 8.5%	67.2% ± 1.6%	69.7% ± 4.1%	52.1% ± 4.2%
APPLE	67.8% ± 5.5%	70.4% ± 5.6%	84.3% ± 13.0%	74.3% ± 10.4%	72.9% ± 4.7%	64.2% ± 6.3%	56.4% ± 4.8%



family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

Best test performance (APPLE selected by after-local validation; all methods 80 rounds)

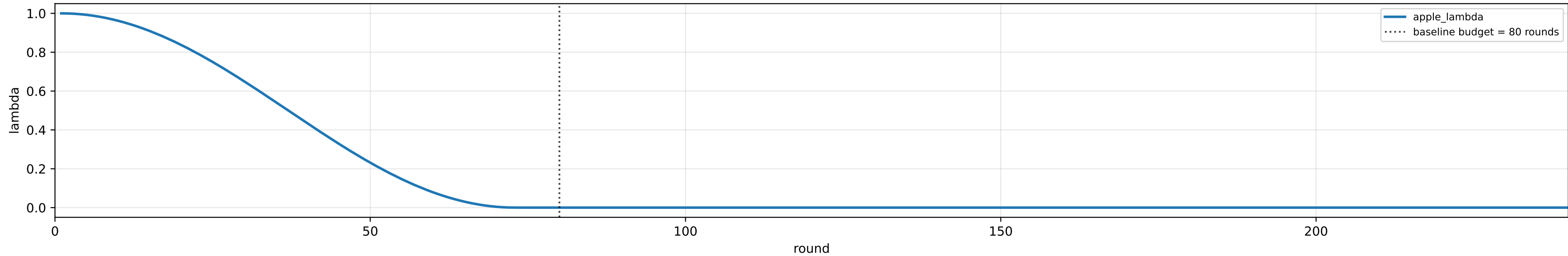
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	68.1% ± 1.5%	72.6% ± 1.2%	89.4% ± 1.7%	86.4% ± 0.7%	69.7% ± 3.6%	67.1% ± 2.2%	50.4% ± 1.9%
APPLE	68.0% ± 4.9%	70.0% ± 4.8%	80.2% ± 11.6%	76.3% ± 11.8%	69.0% ± 2.1%	65.5% ± 6.1%	58.8% ± 3.5%



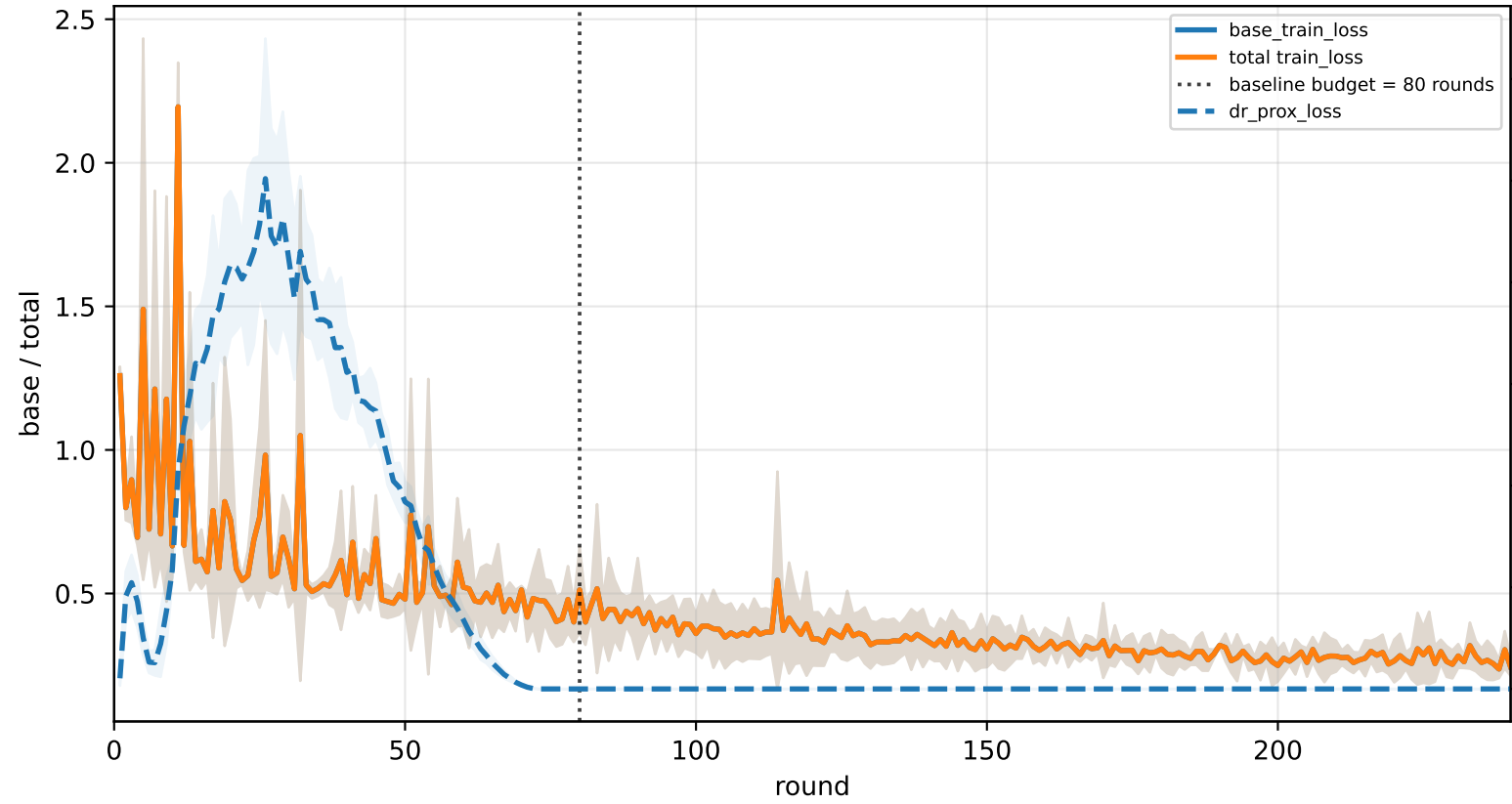
APPLE training-loss decomposition | lid heterogeneity

family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

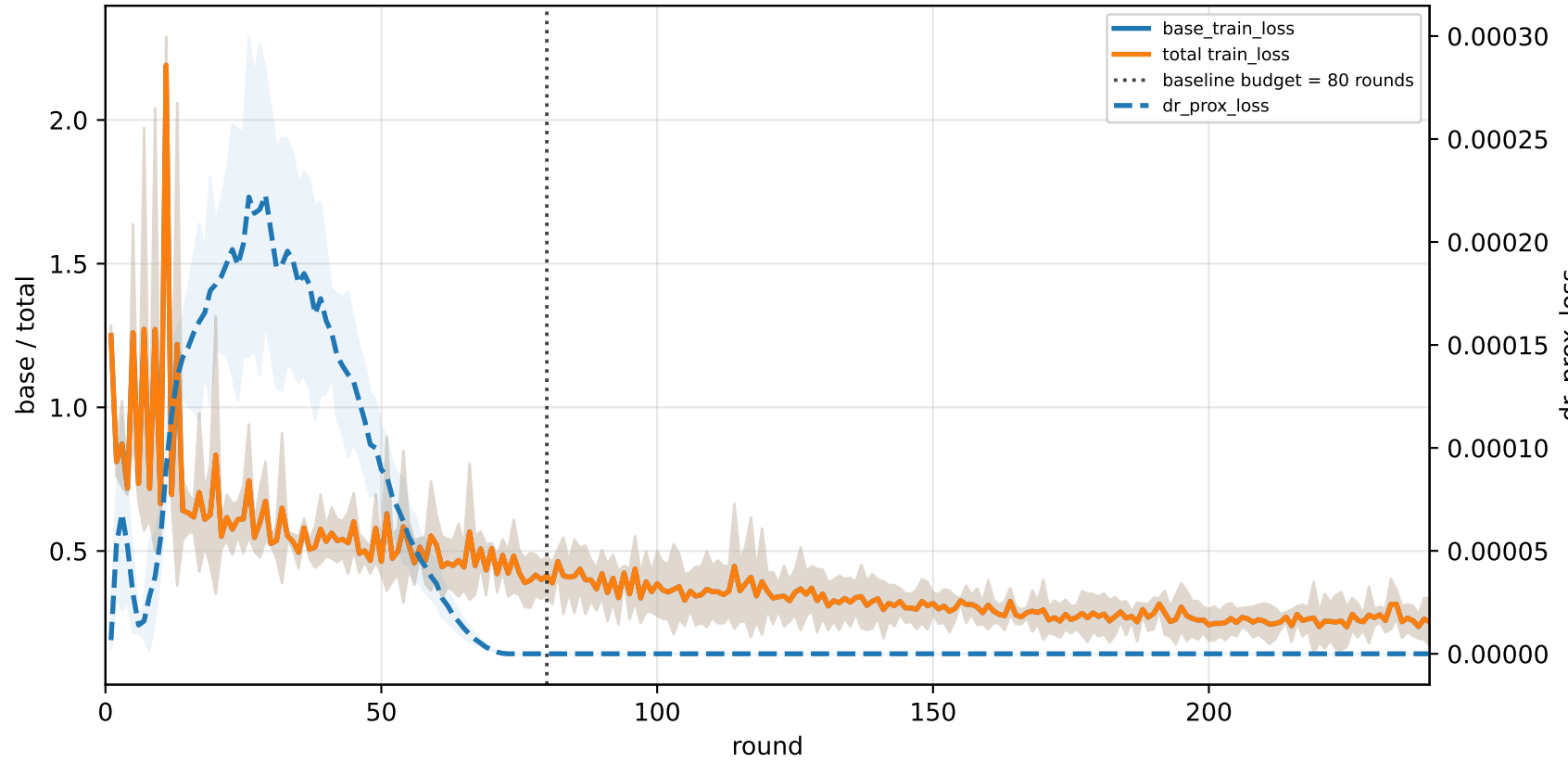
APPLE lambda scheduler | apple_mu=0.01



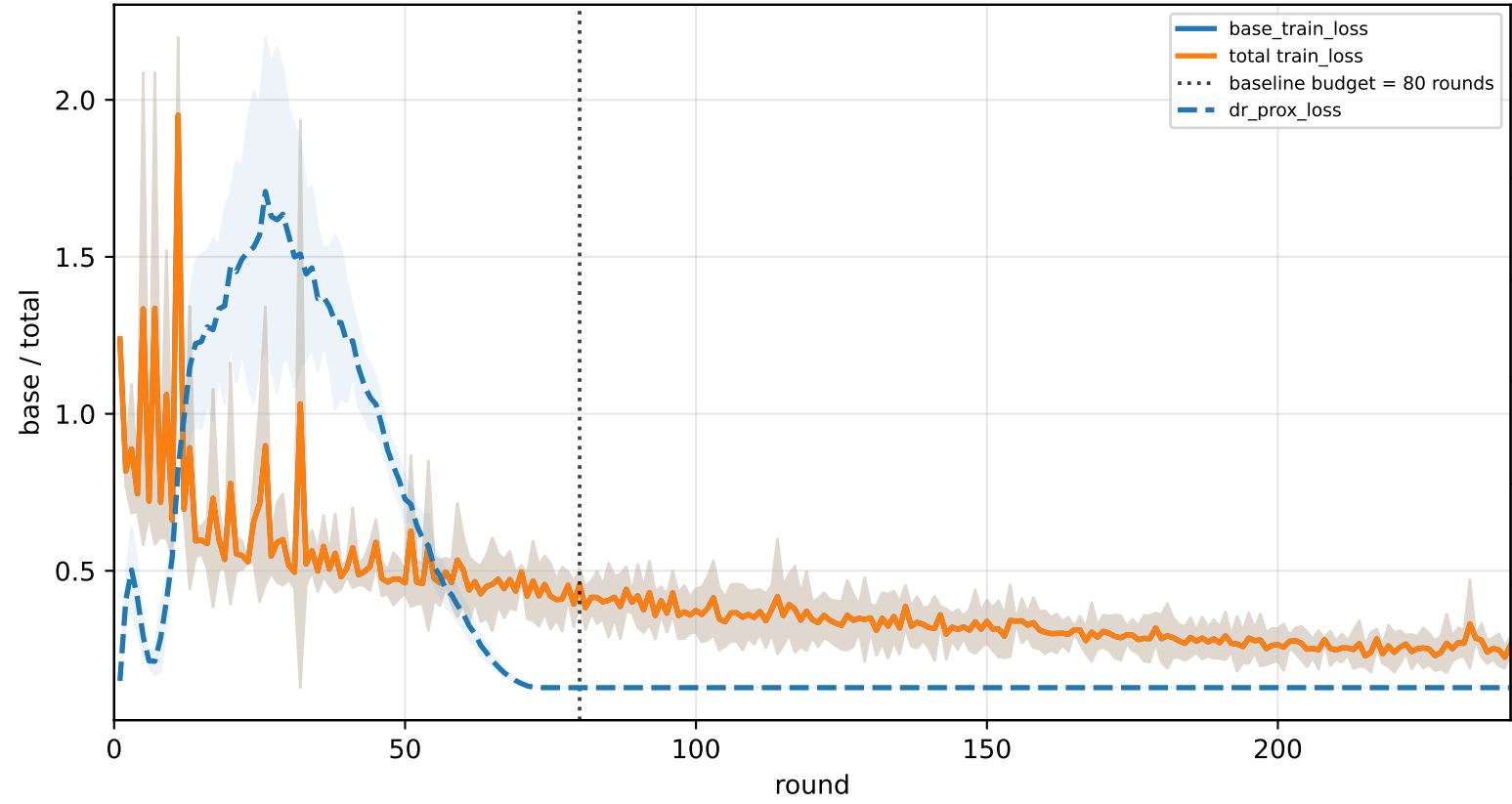
q_task_cycle2 | graph_id=3000000



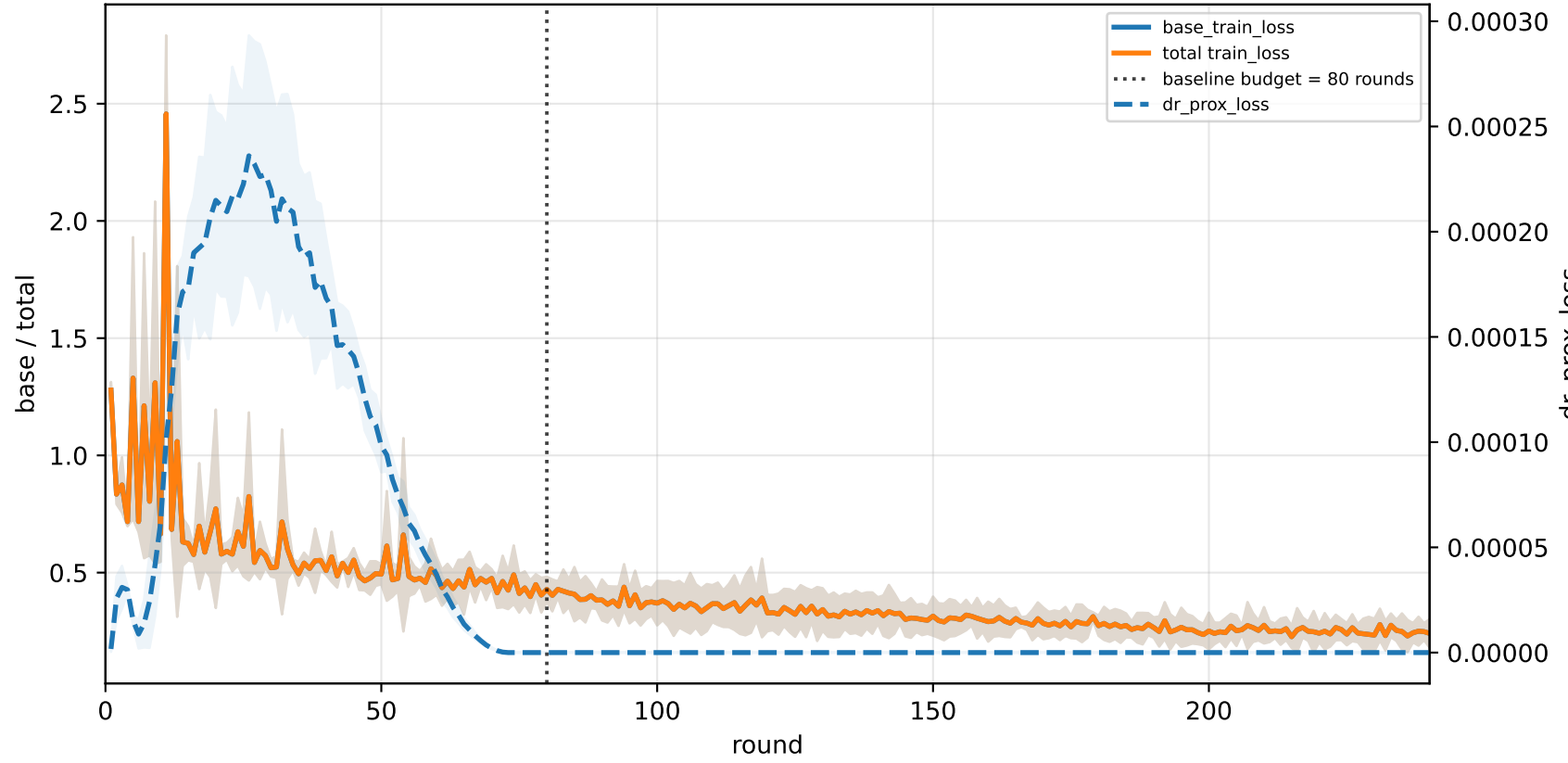
q_task_cycle3 | graph_id=3000001



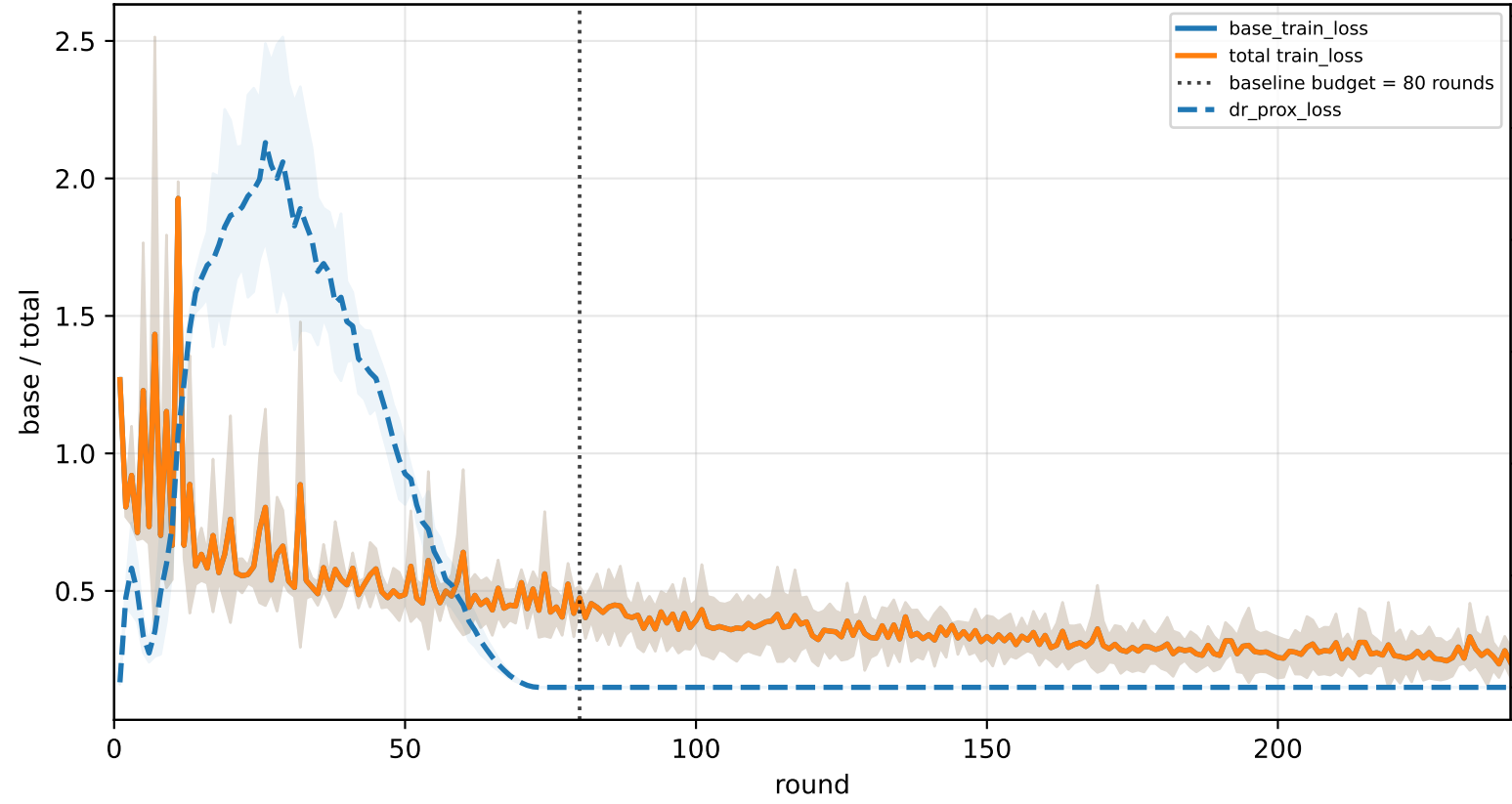
q_task_cycle4 | graph_id=3000002



q_task_cycle5 | graph_id=3000003

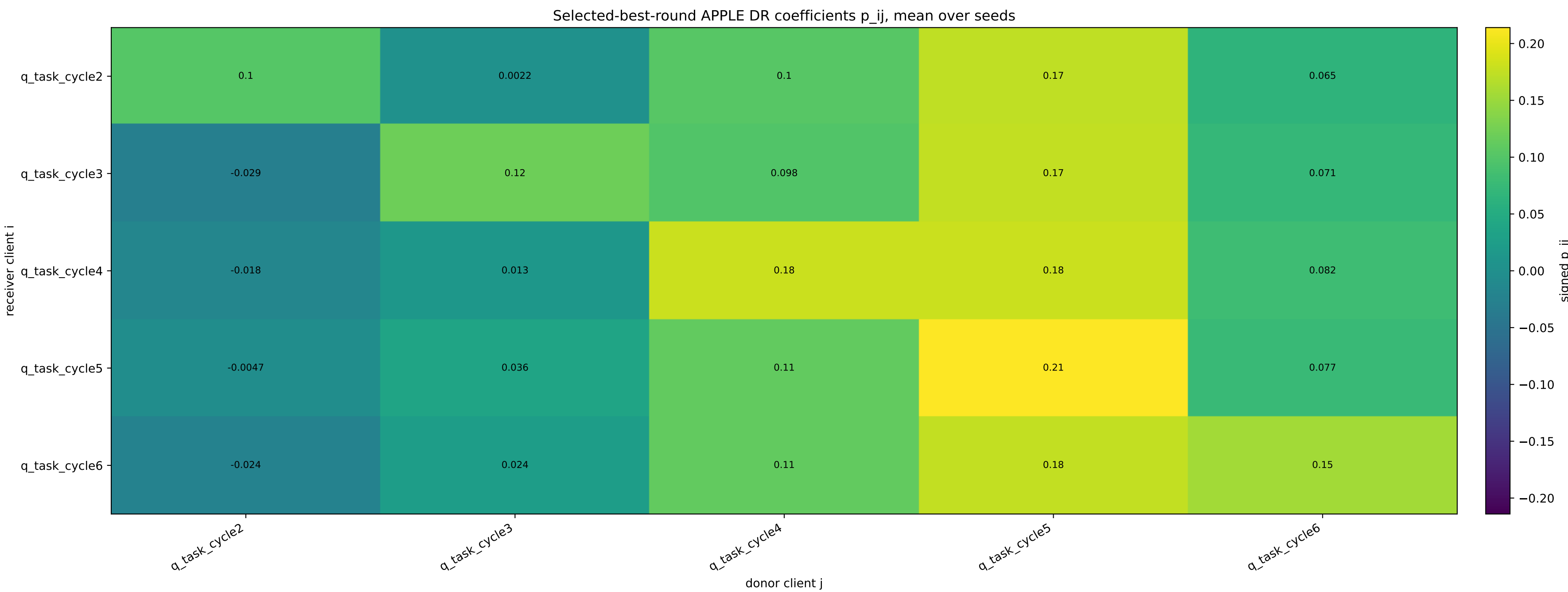
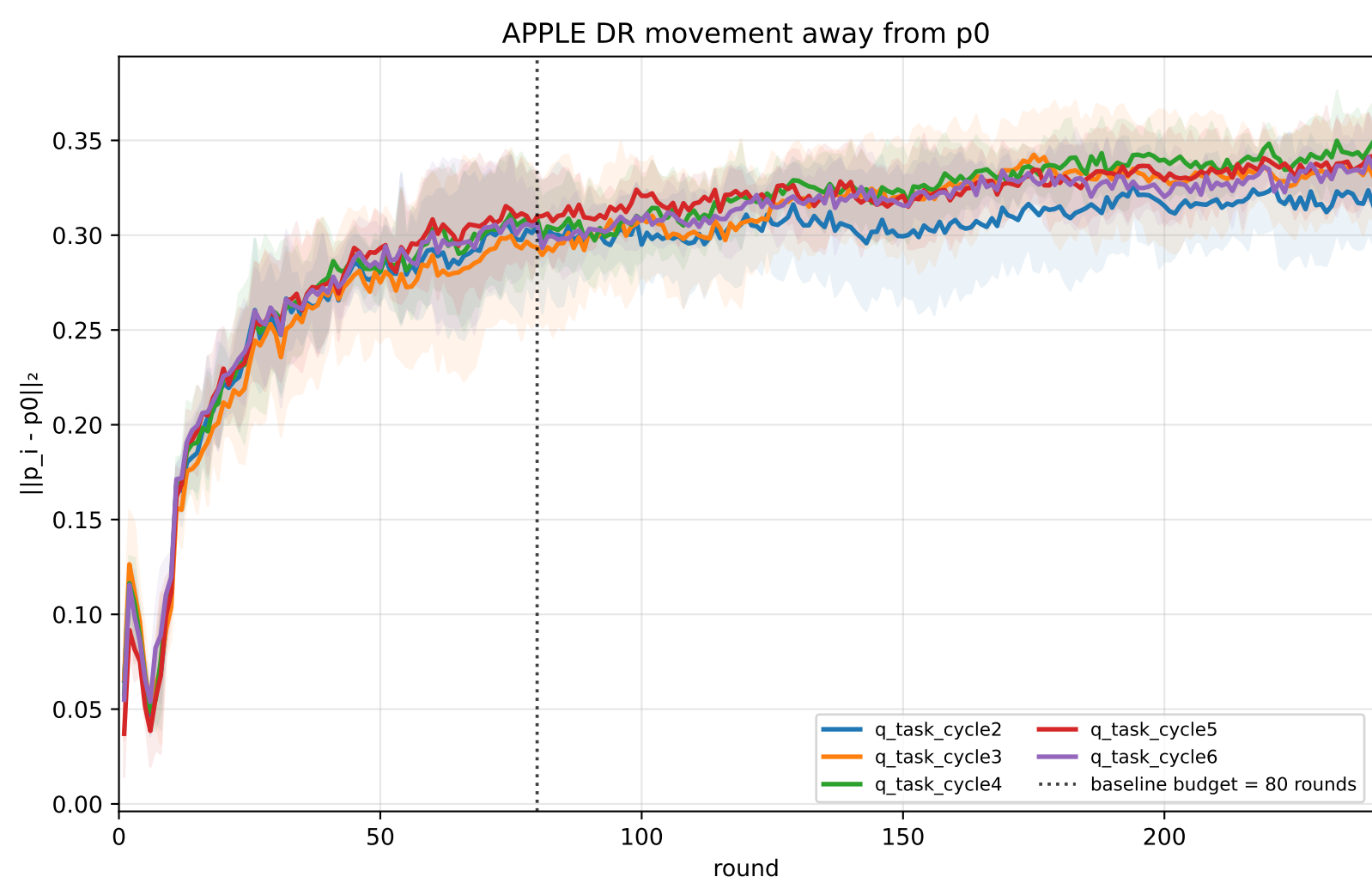
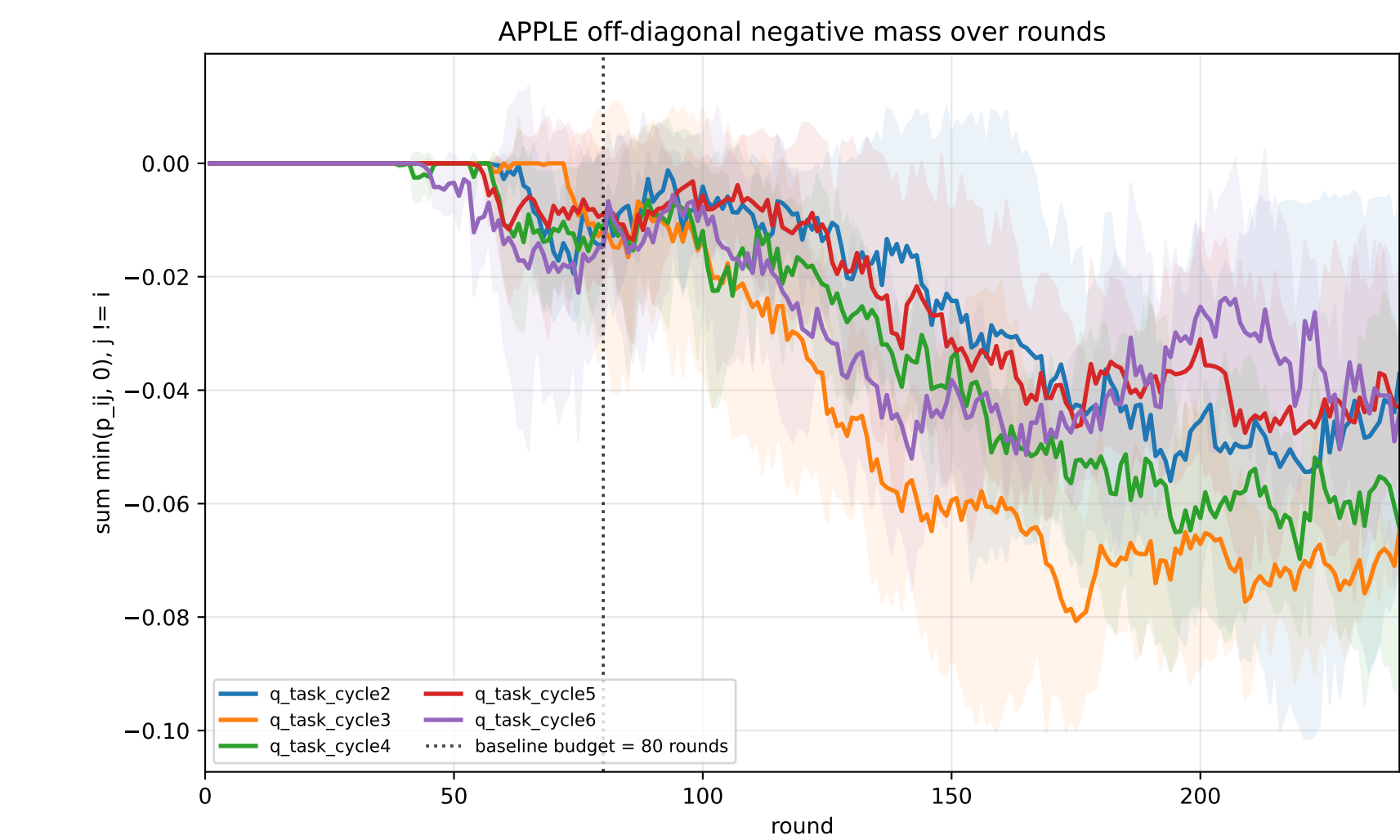
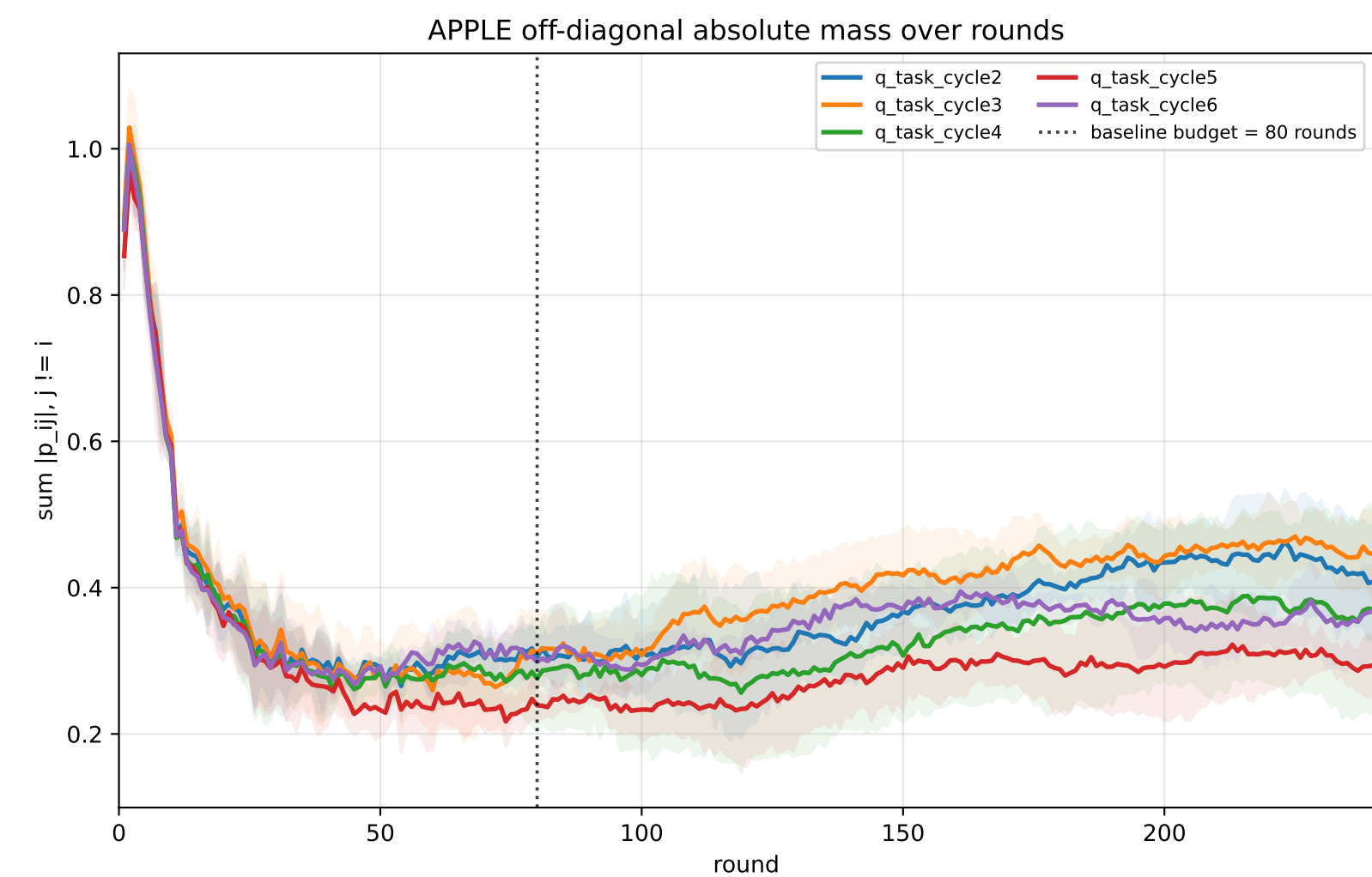
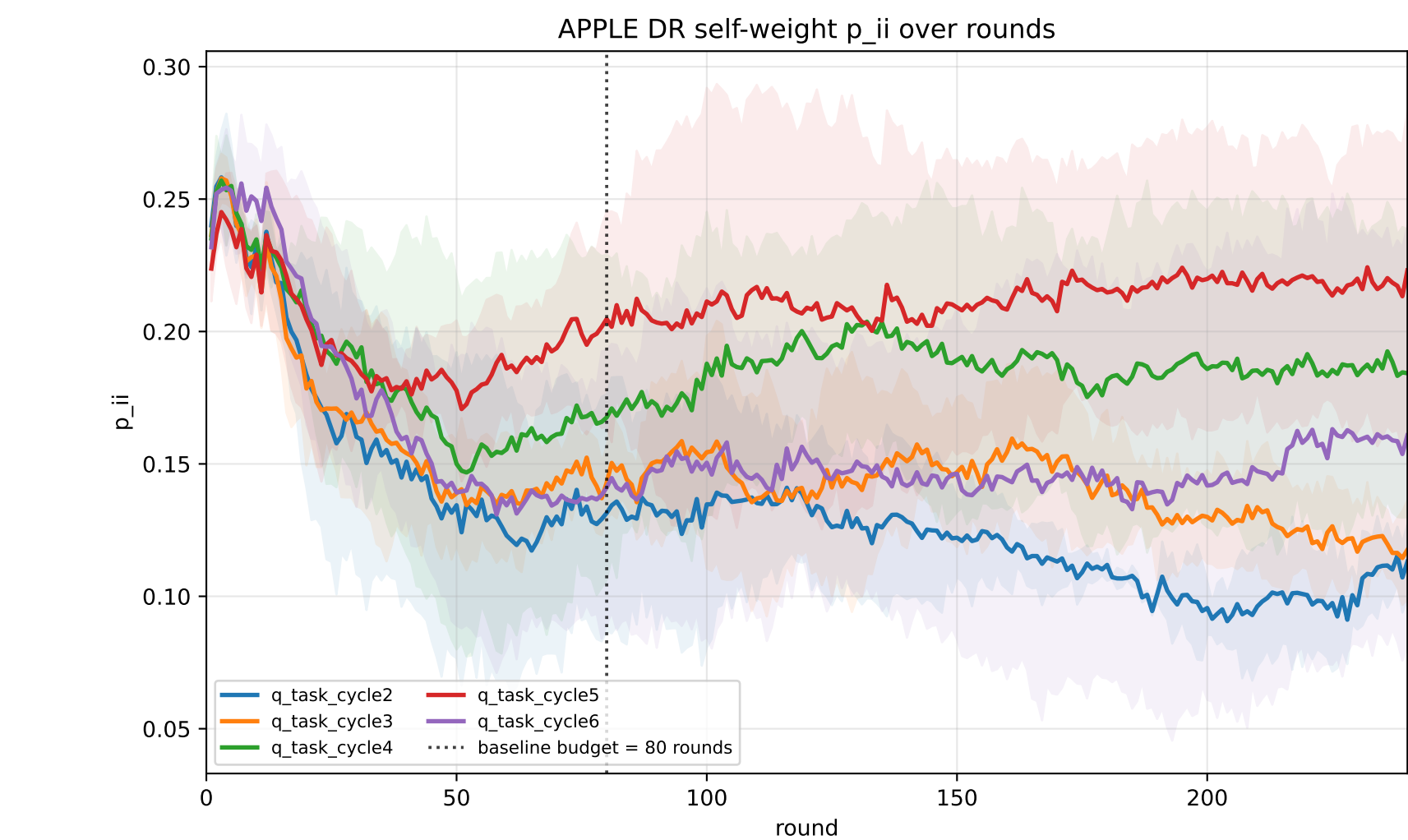


q_task_cycle6 | graph_id=3000004



APPLE DR behavior | Iid heterogeneity

family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

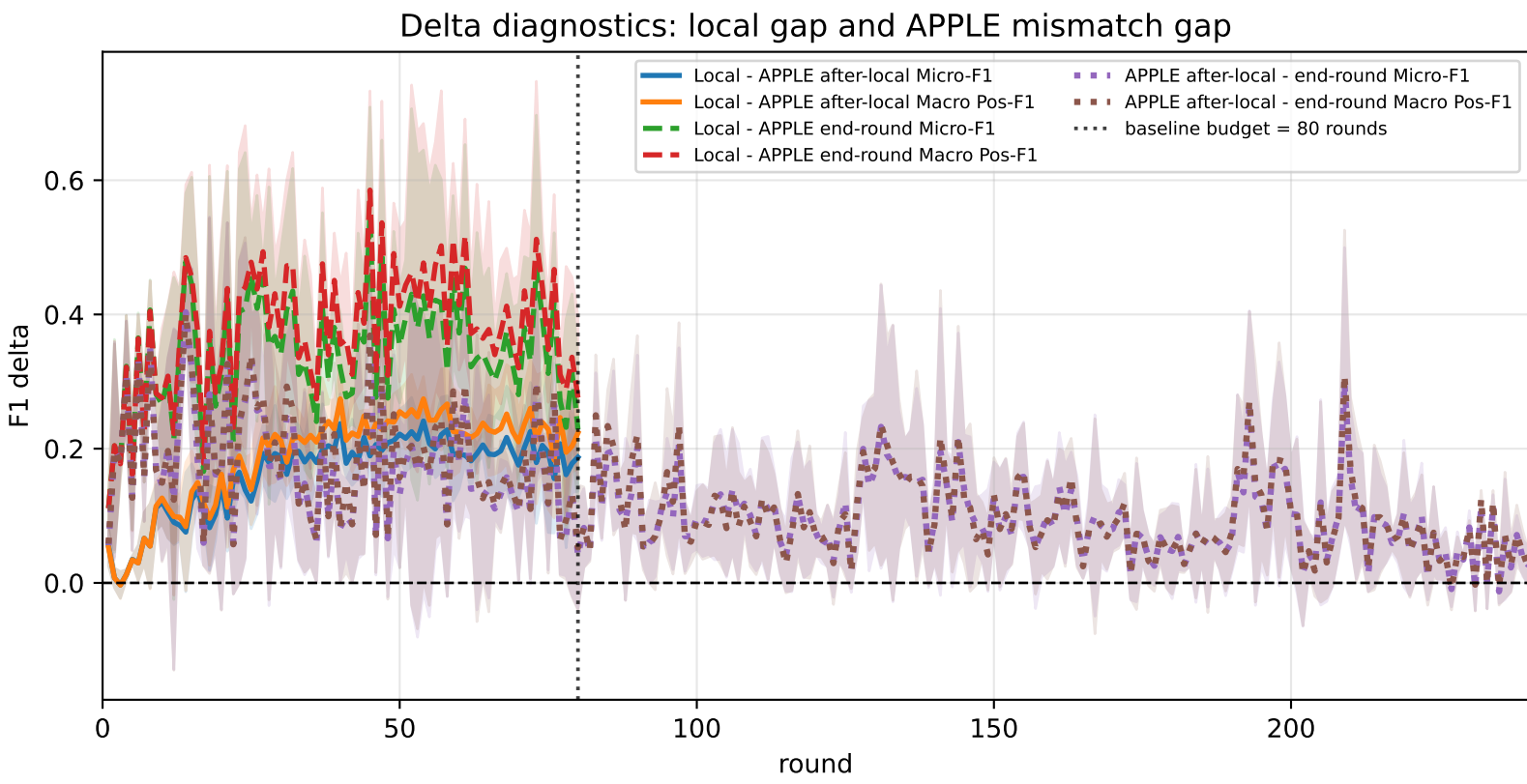
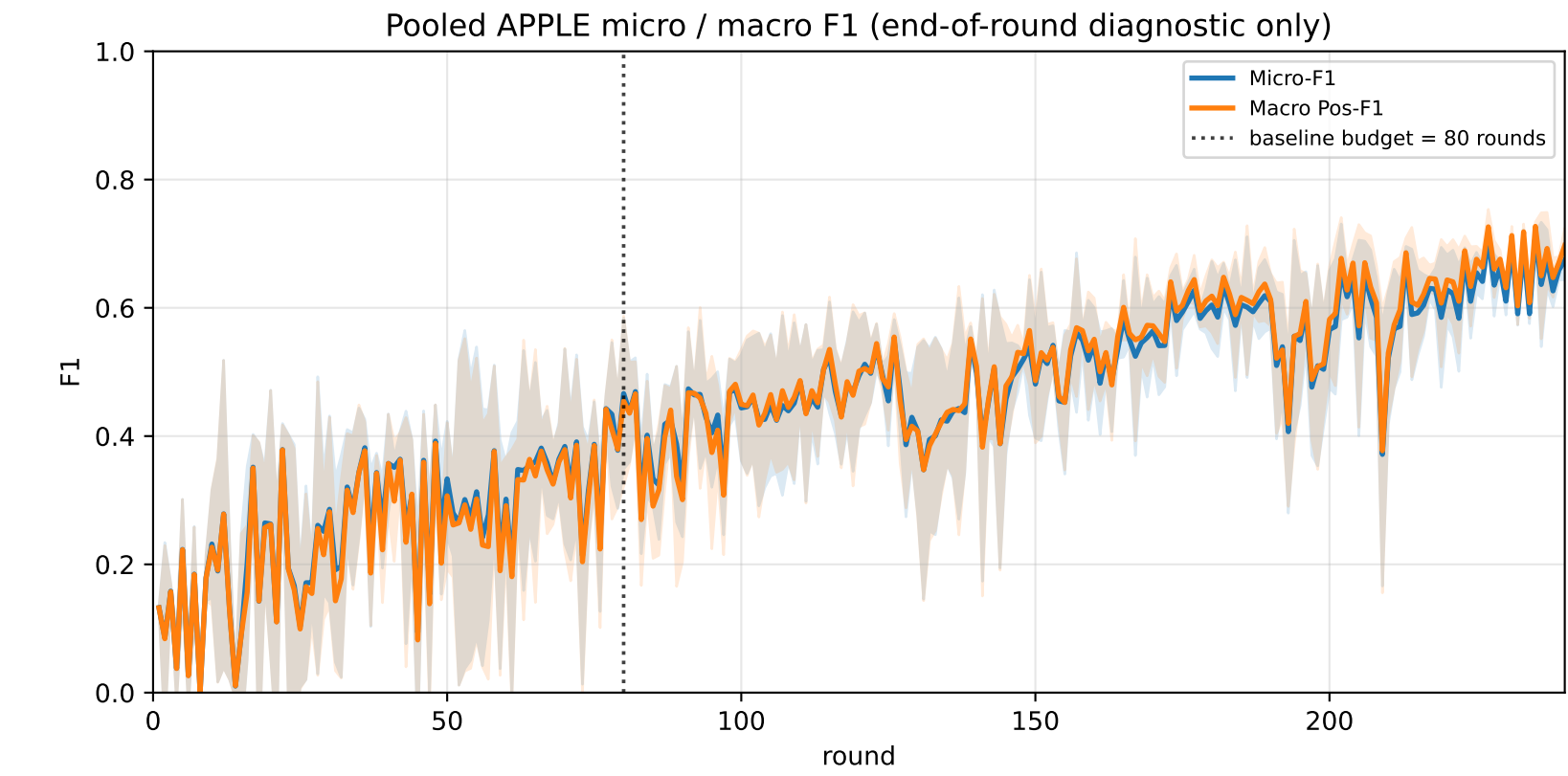
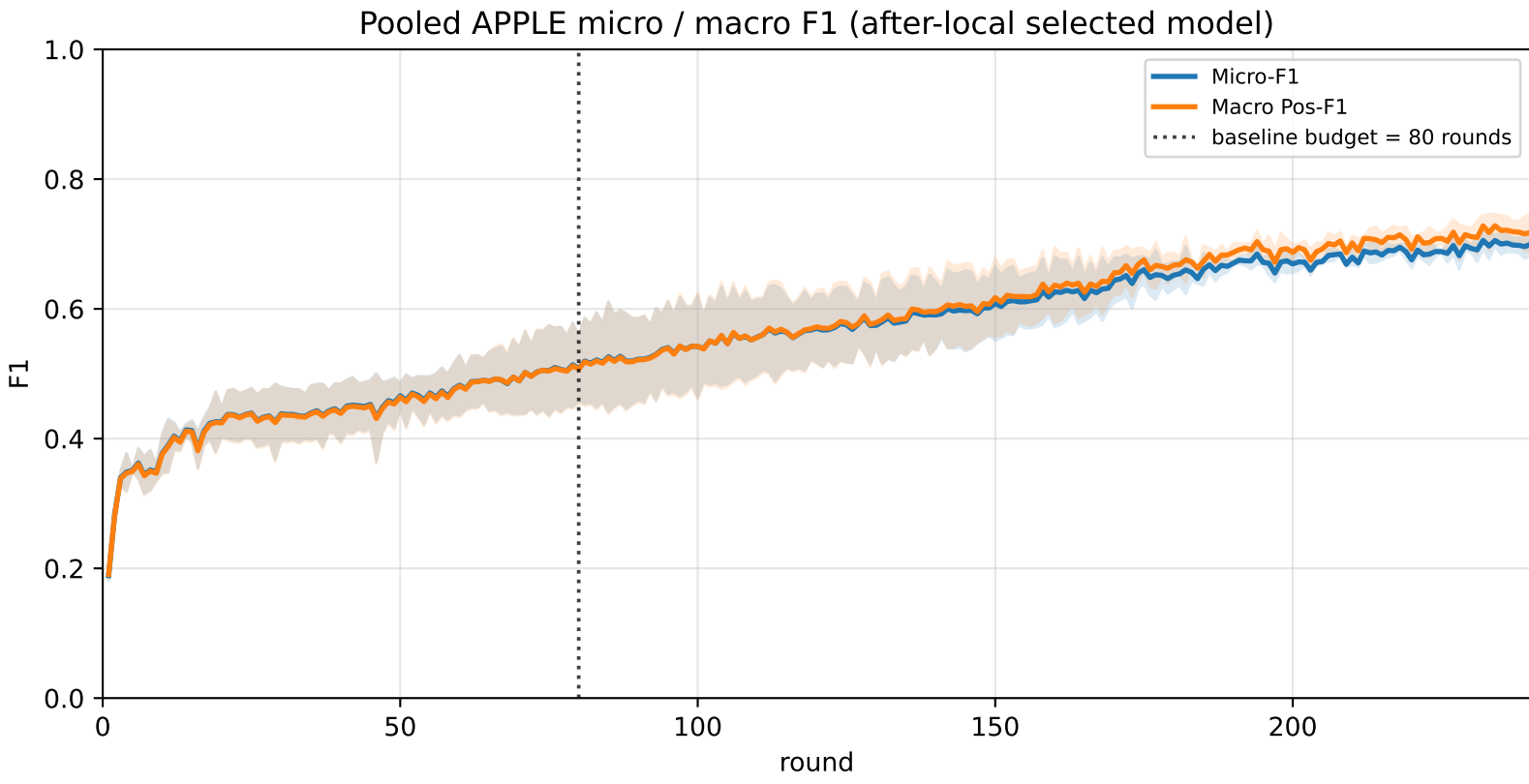
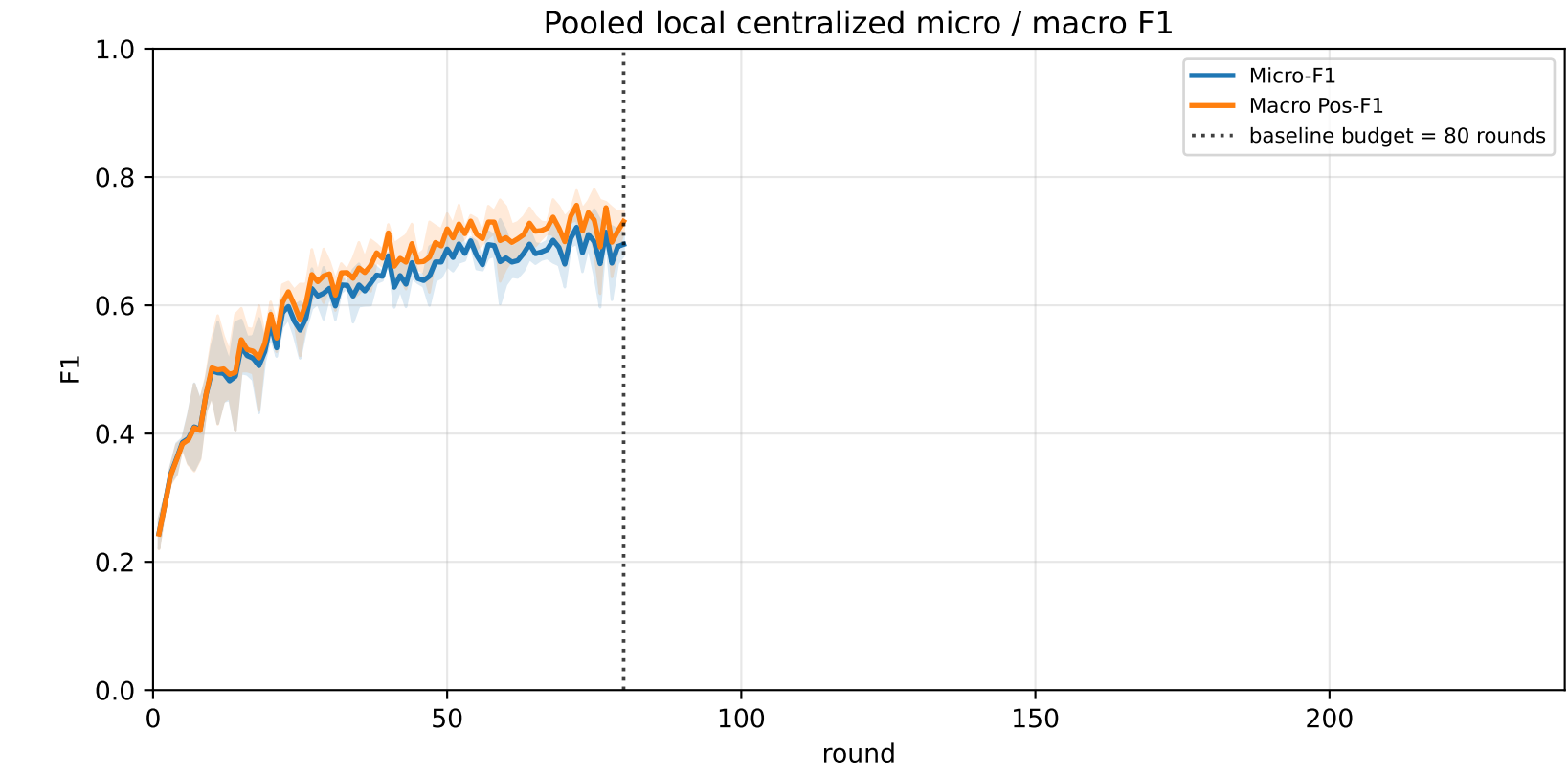
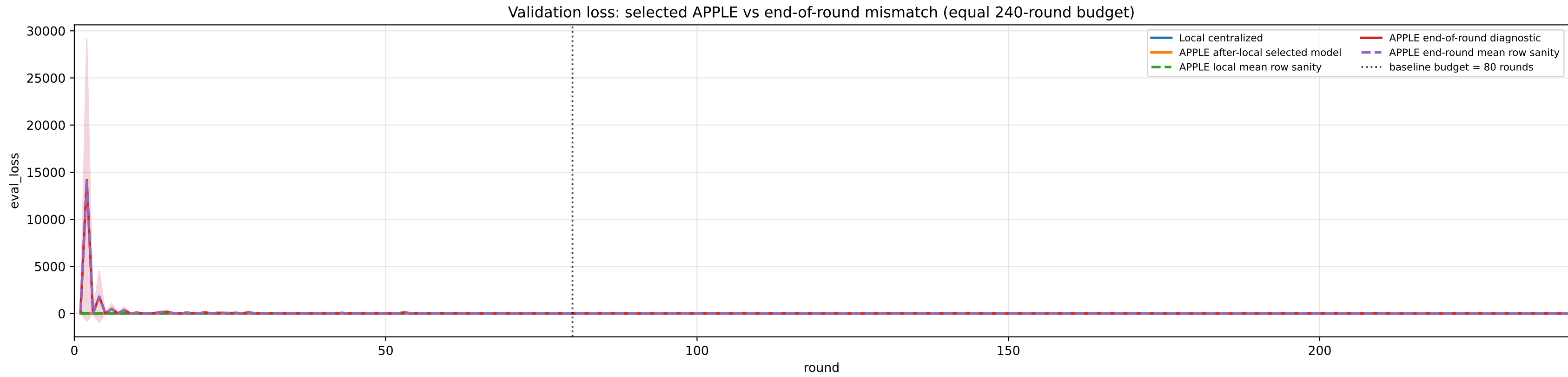


Controlled q-label heterogeneity: Mid | APPLE diagnostics | aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

Best test performance (APPLE selected by after-local validation; all methods 80 rounds)

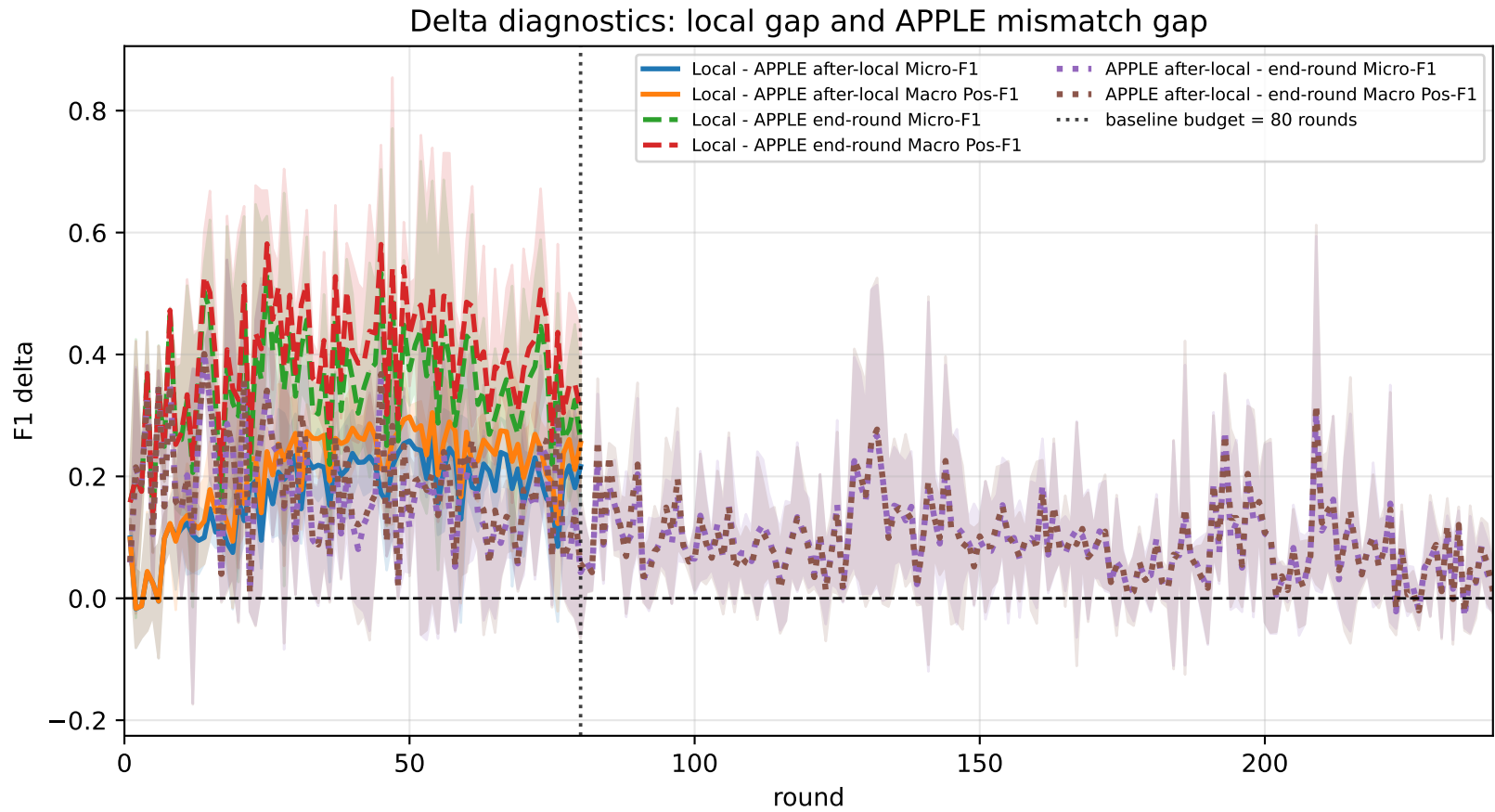
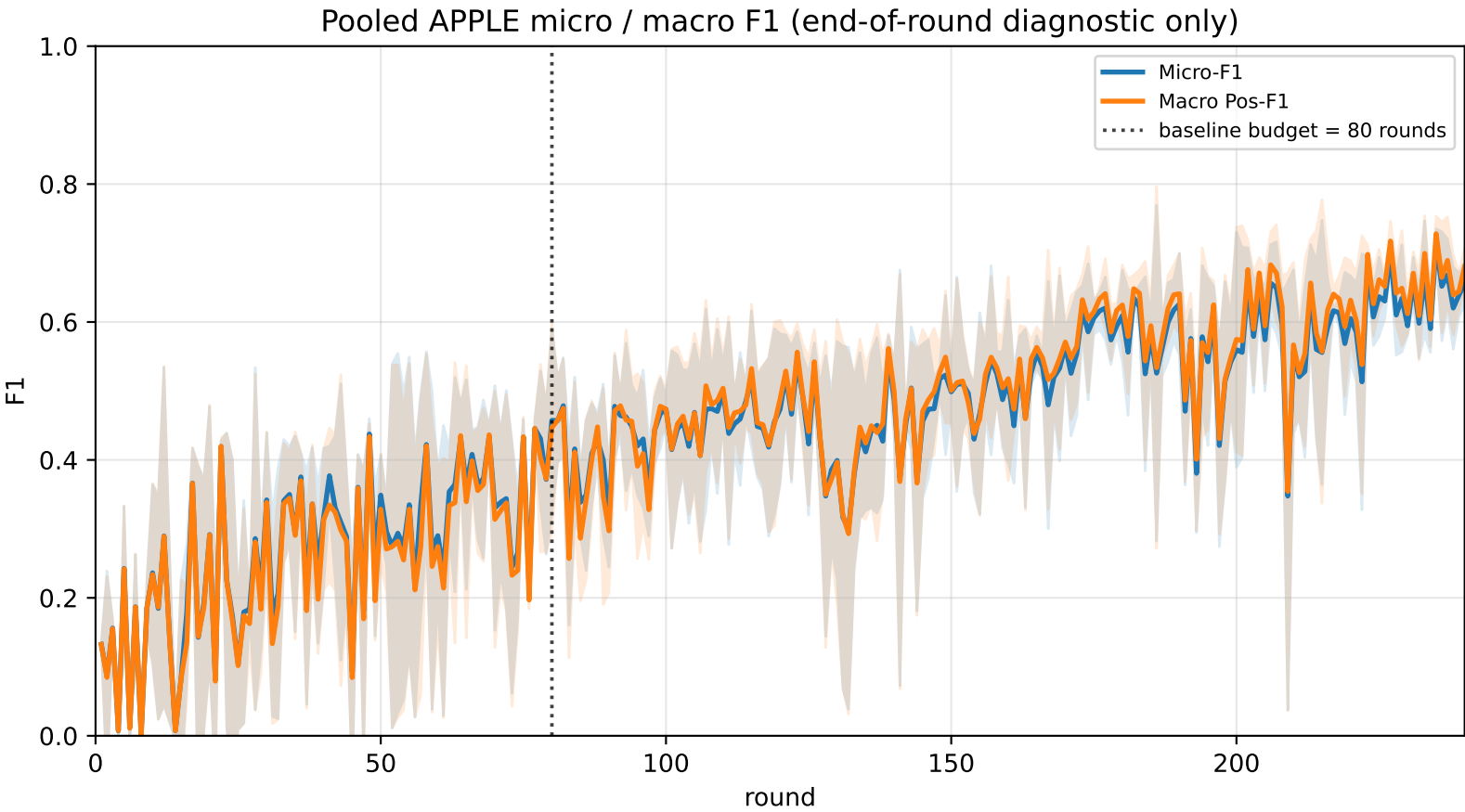
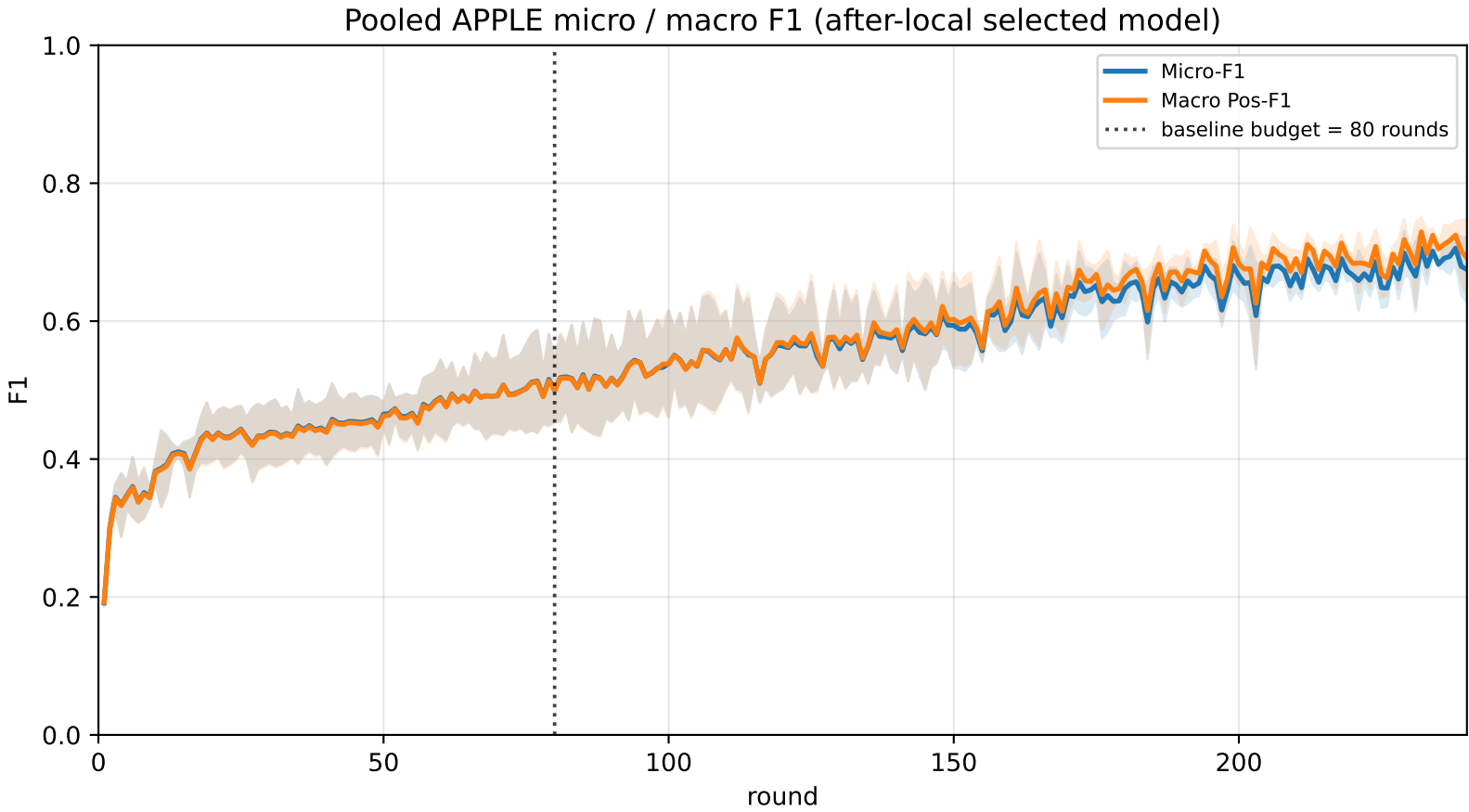
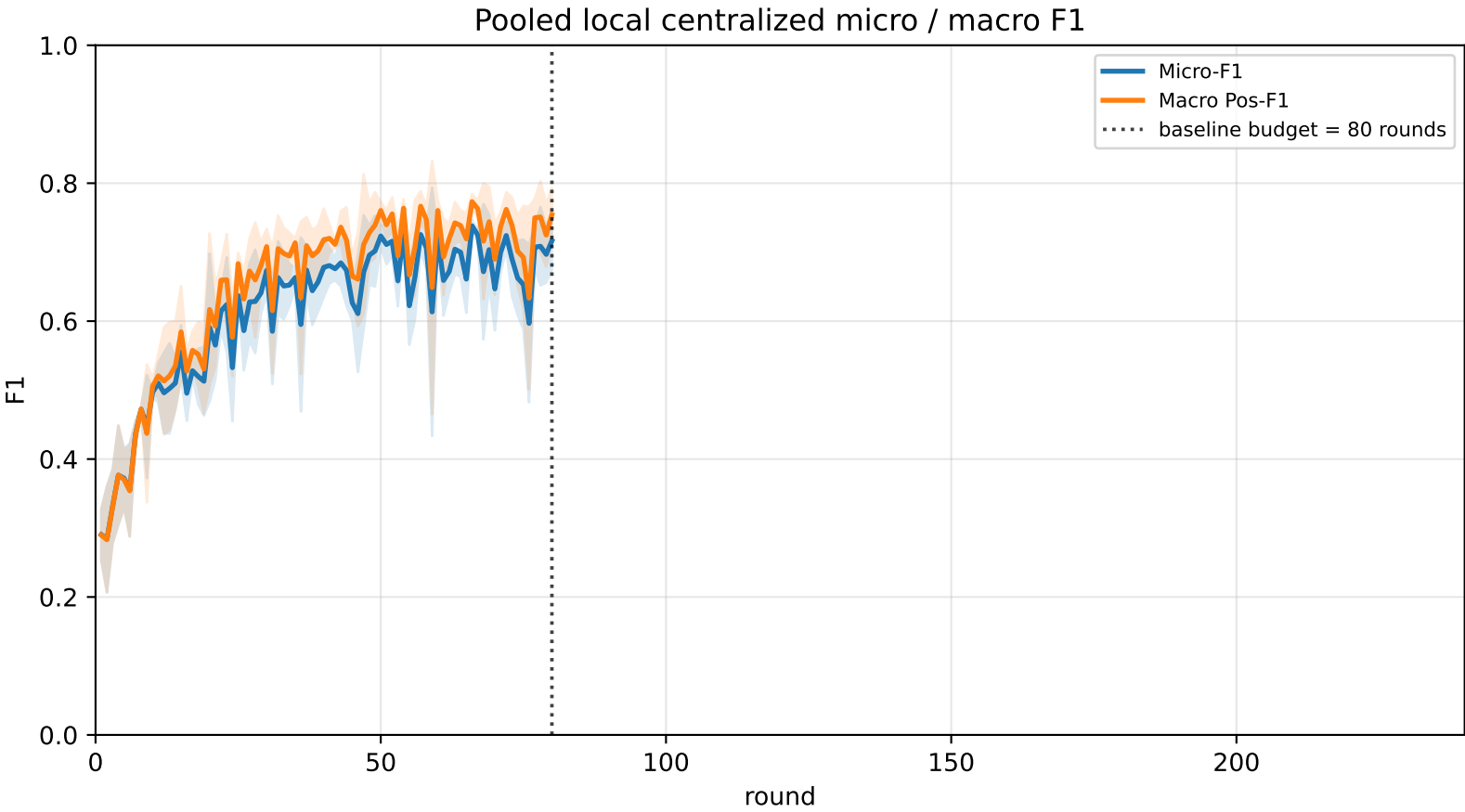
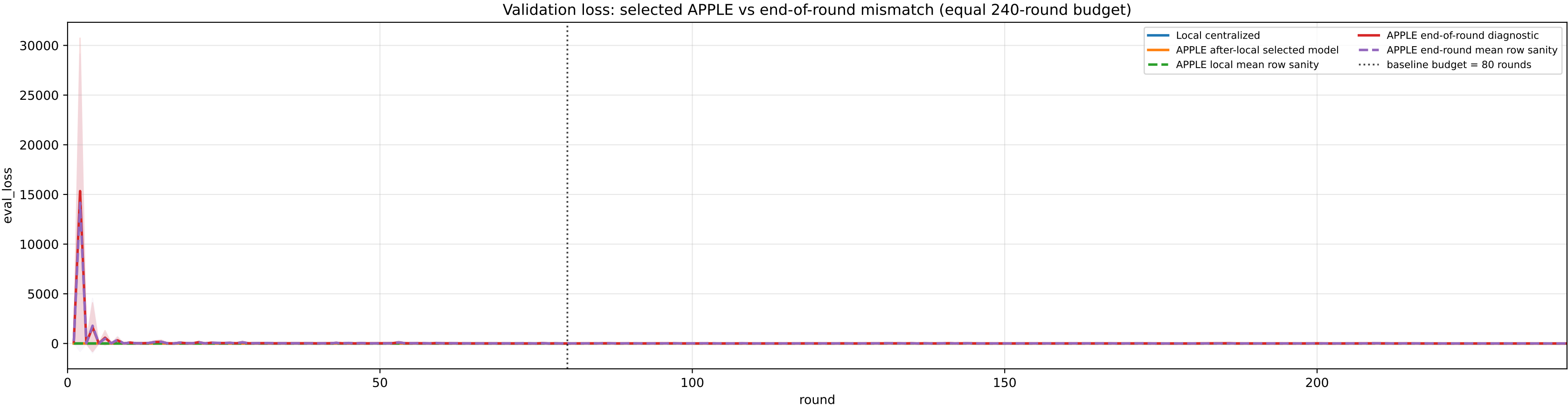
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	60.3% ± 0.4%	63.4% ± 0.4%	74.7% ± 4.2%	70.7% ± 0.1%	60.9% ± 2.0%	57.0% ± 1.1%	47.0% ± 0.4%
APPLE	67.9% ± 1.6%	70.0% ± 2.2%	81.4% ± 4.7%	77.1% ± 4.4%	68.5% ± 4.0%	65.3% ± 1.4%	56.8% ± 1.1%



family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

Best test performance (APPLE selected by after-local validation; all methods 80 rounds)

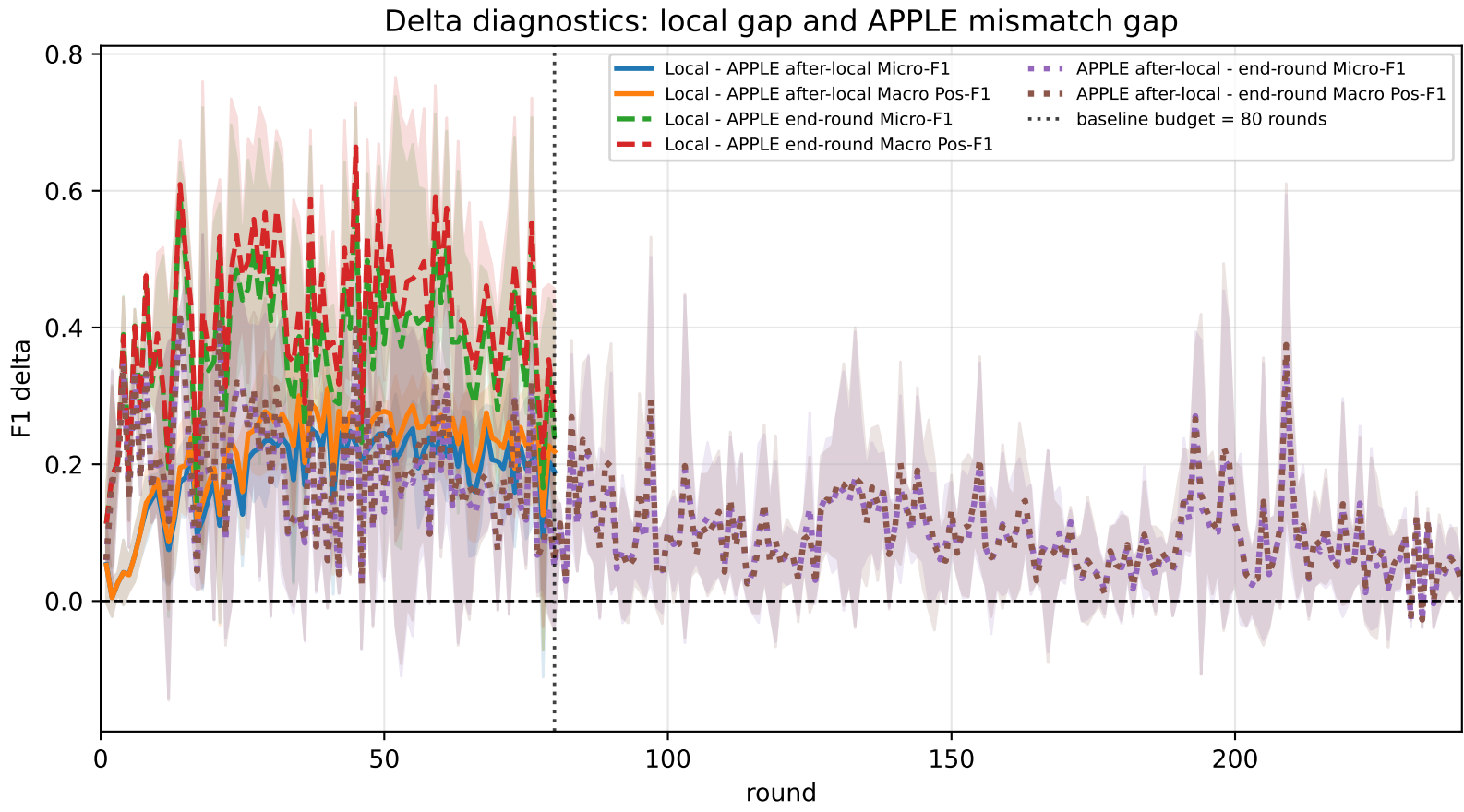
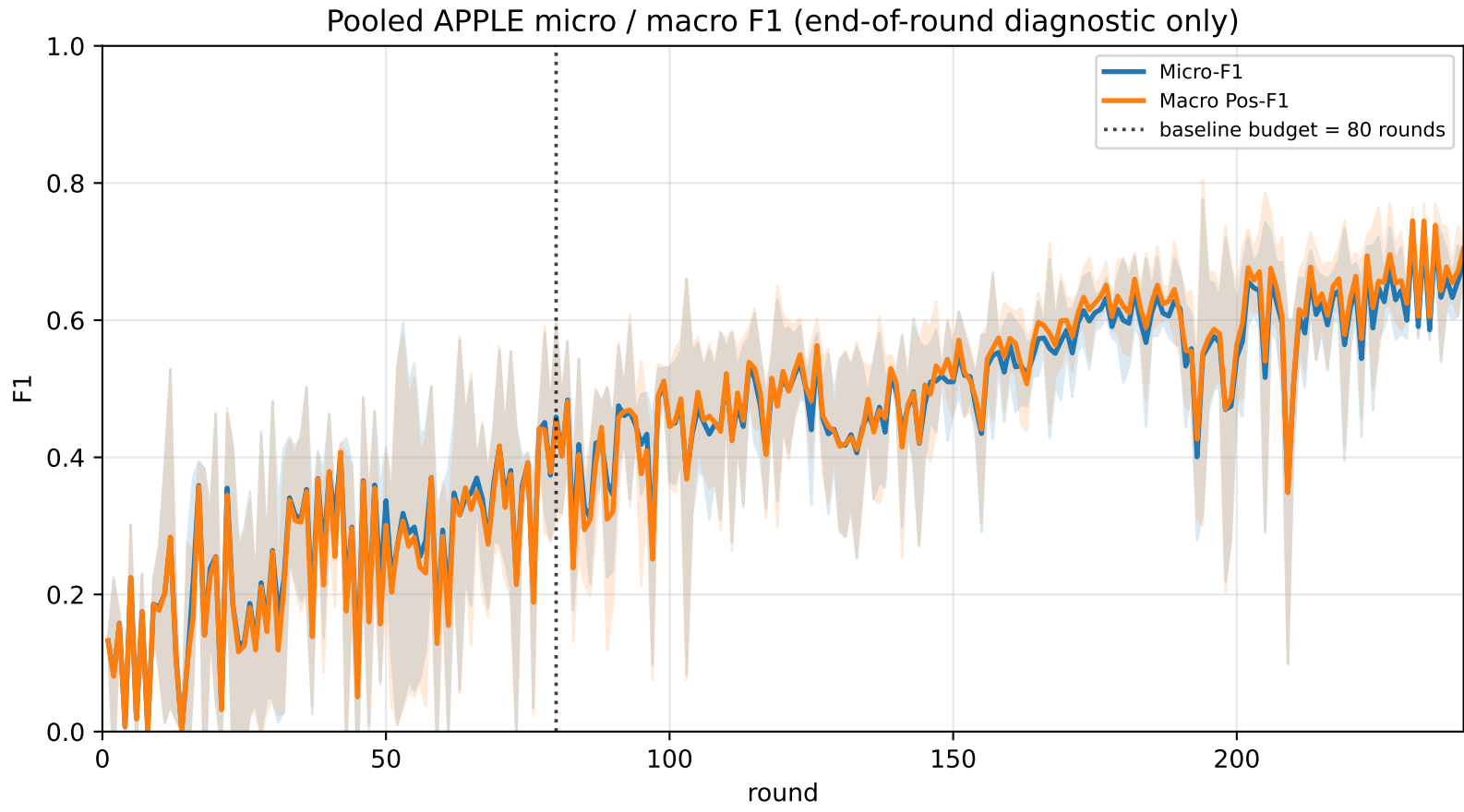
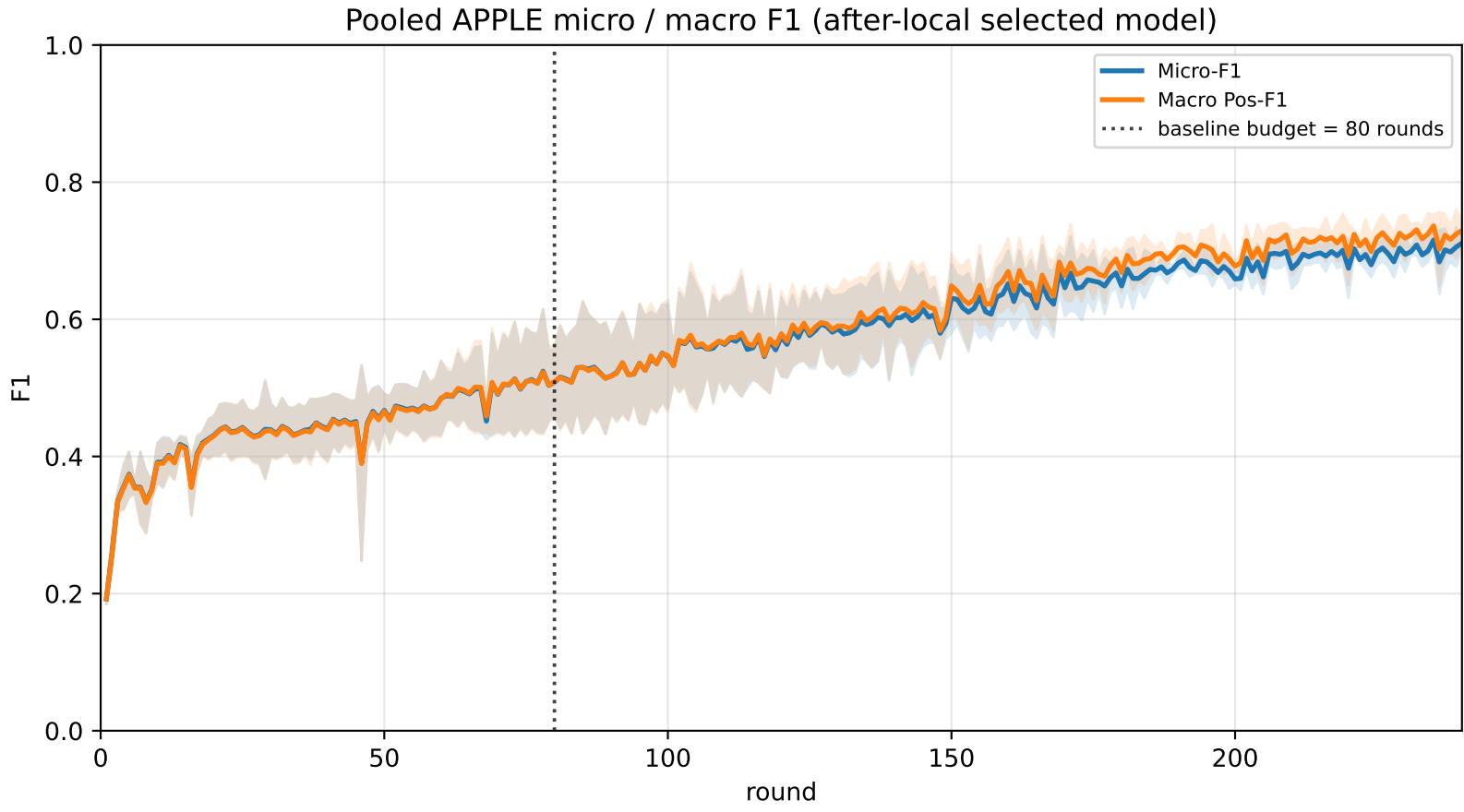
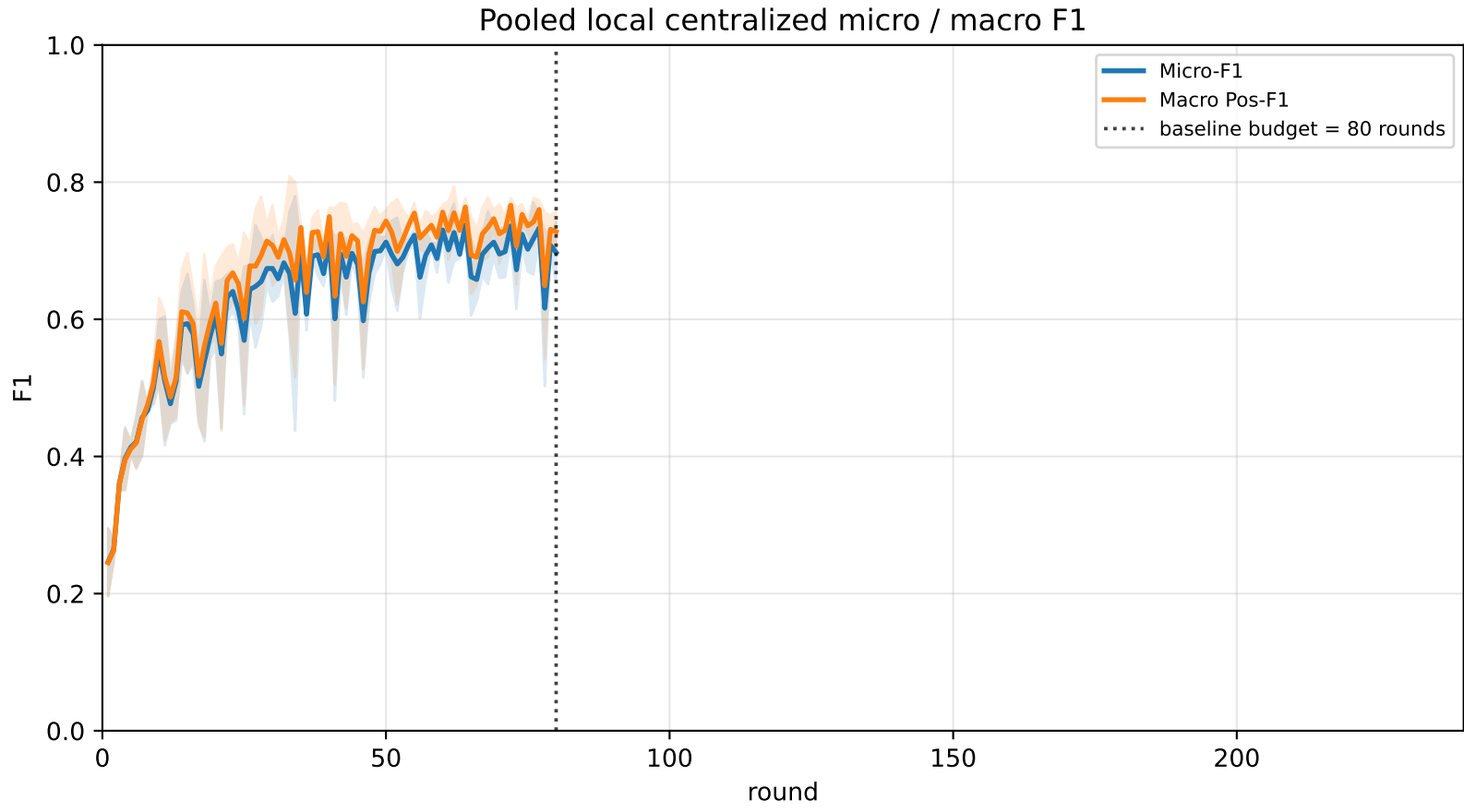
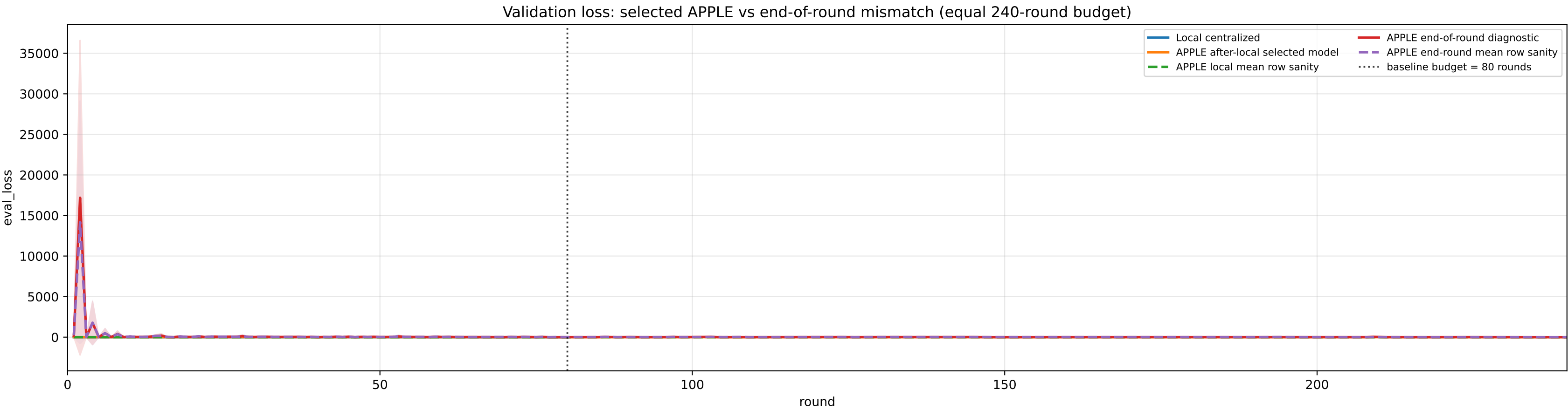
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	62.9% ± 2.7%	67.2% ± 2.6%	84.2% ± 3.3%	78.8% ± 3.0%	67.1% ± 2.5%	59.7% ± 2.9%	46.4% ± 2.7%
APPLE	65.6% ± 2.8%	67.6% ± 3.0%	79.0% ± 4.7%	75.9% ± 4.9%	66.4% ± 3.9%	61.8% ± 3.9%	55.2% ± 3.0%



family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3% | heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

Best test performance (APPLE selected by after-local validation; all methods 80 rounds)

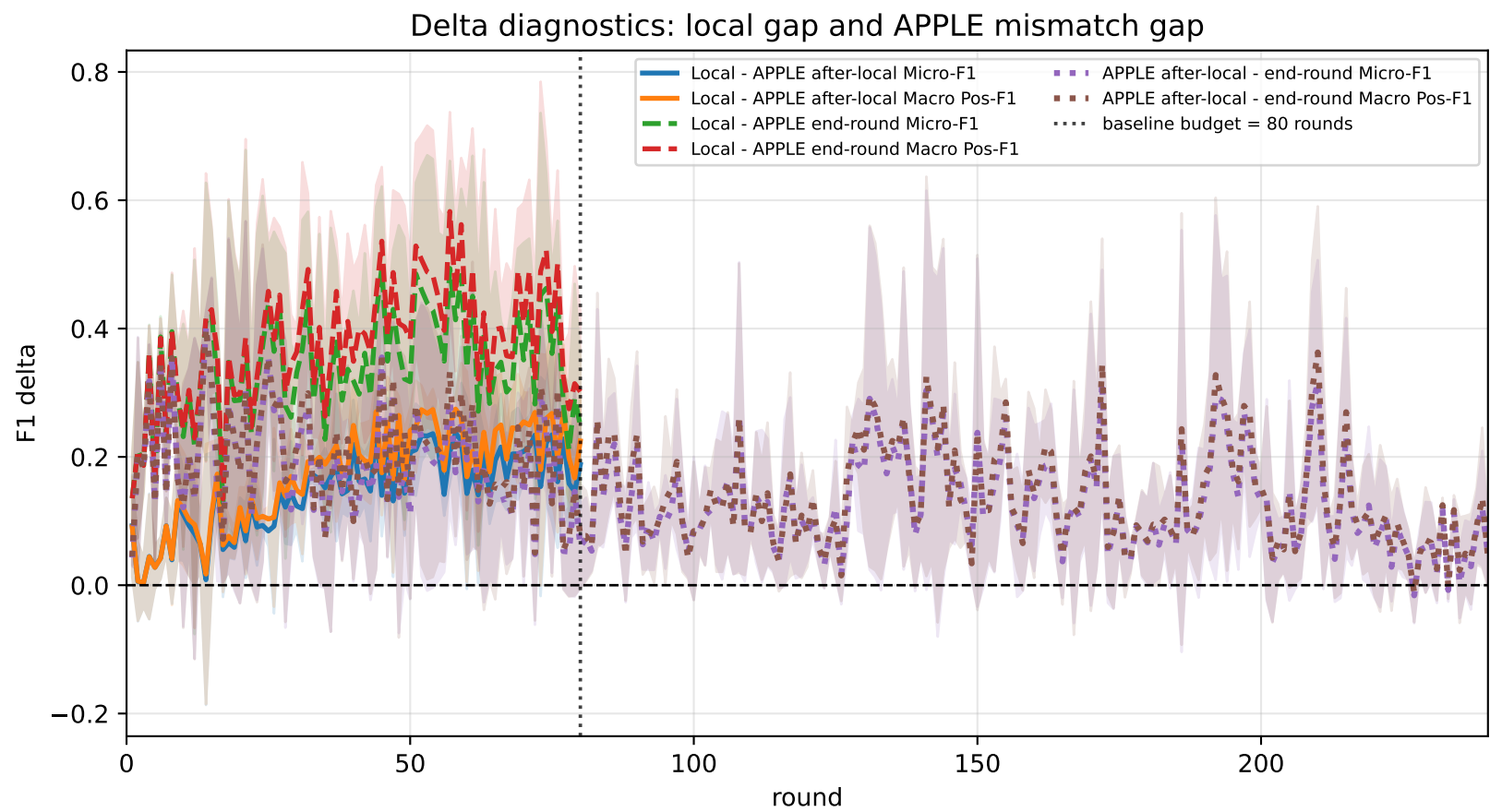
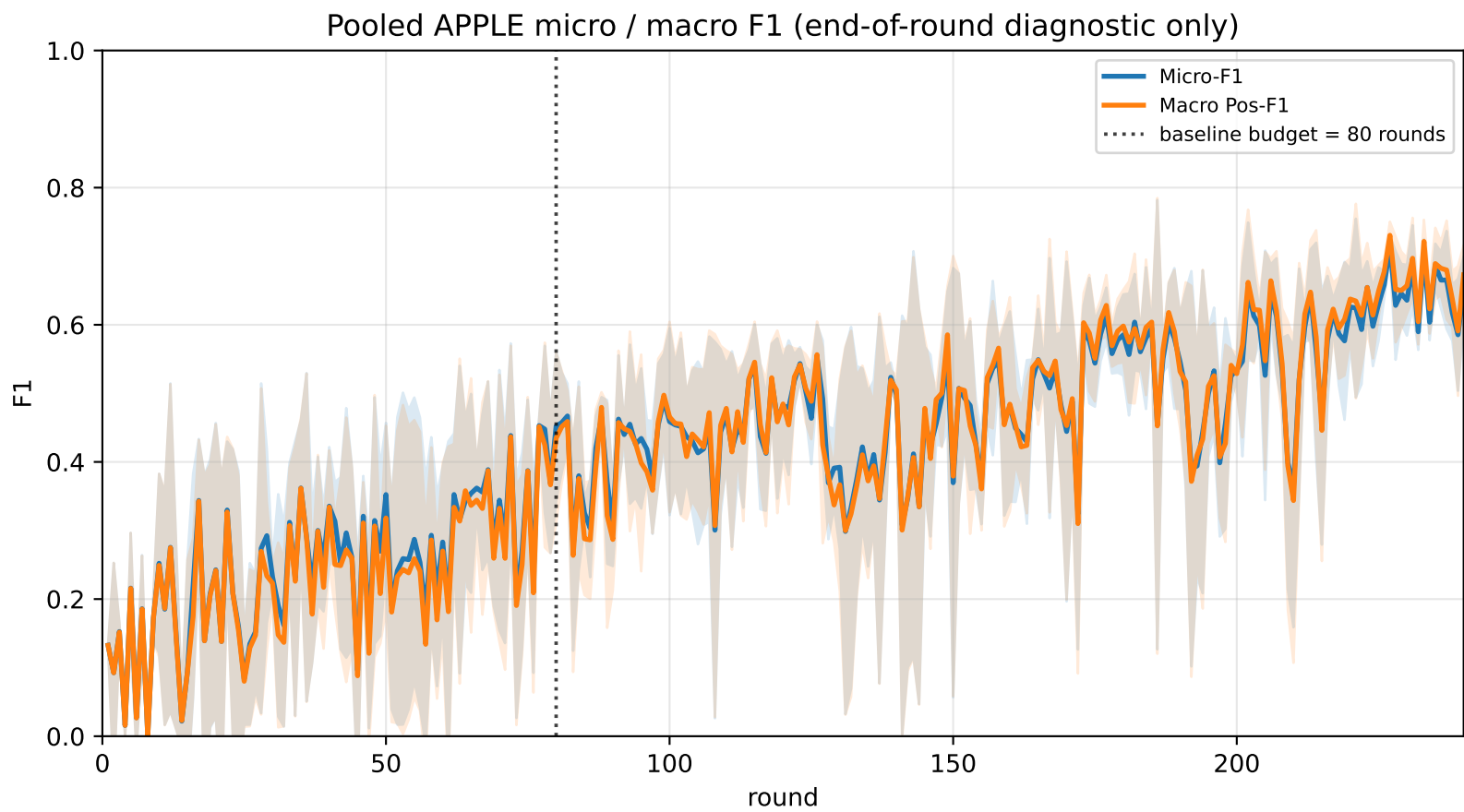
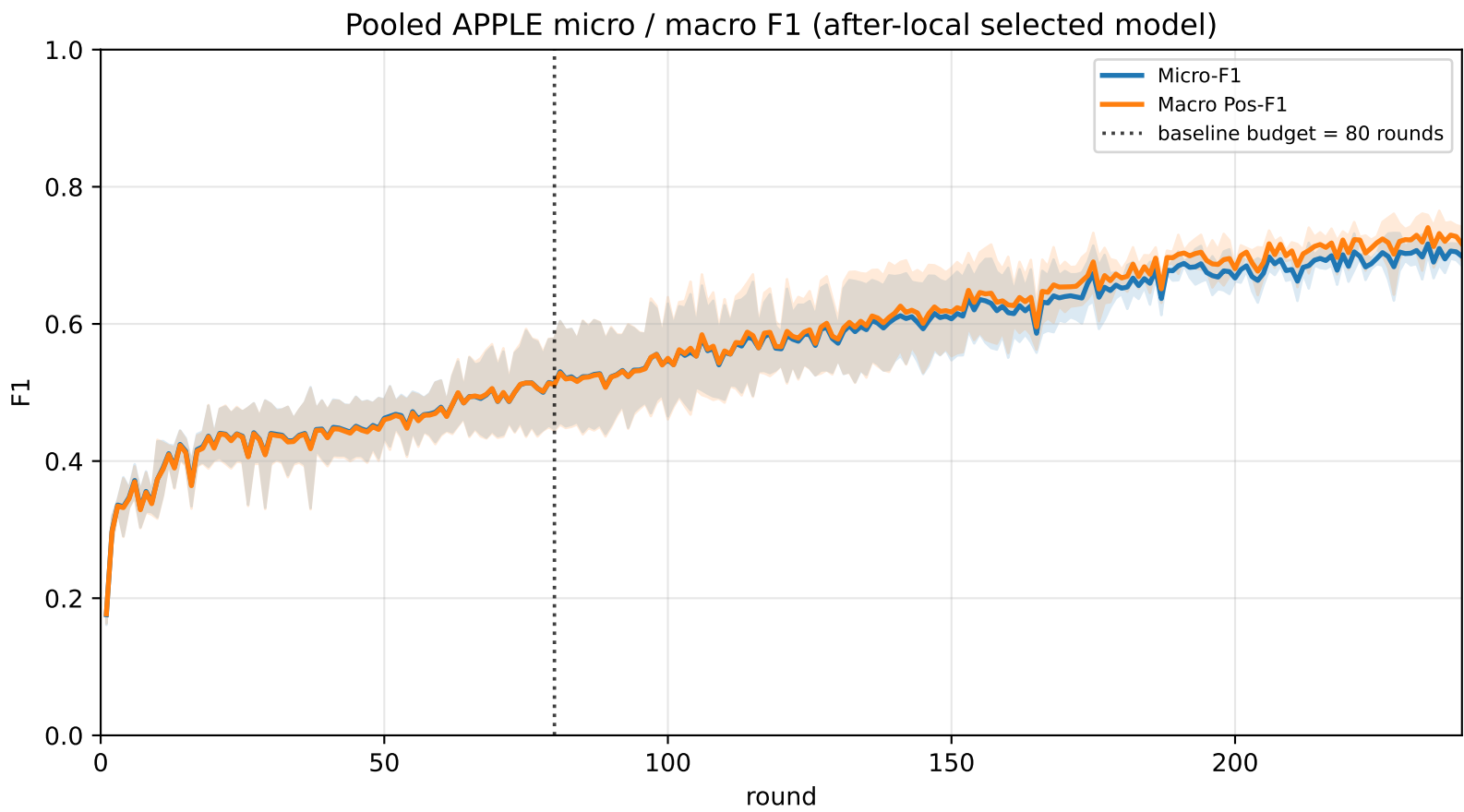
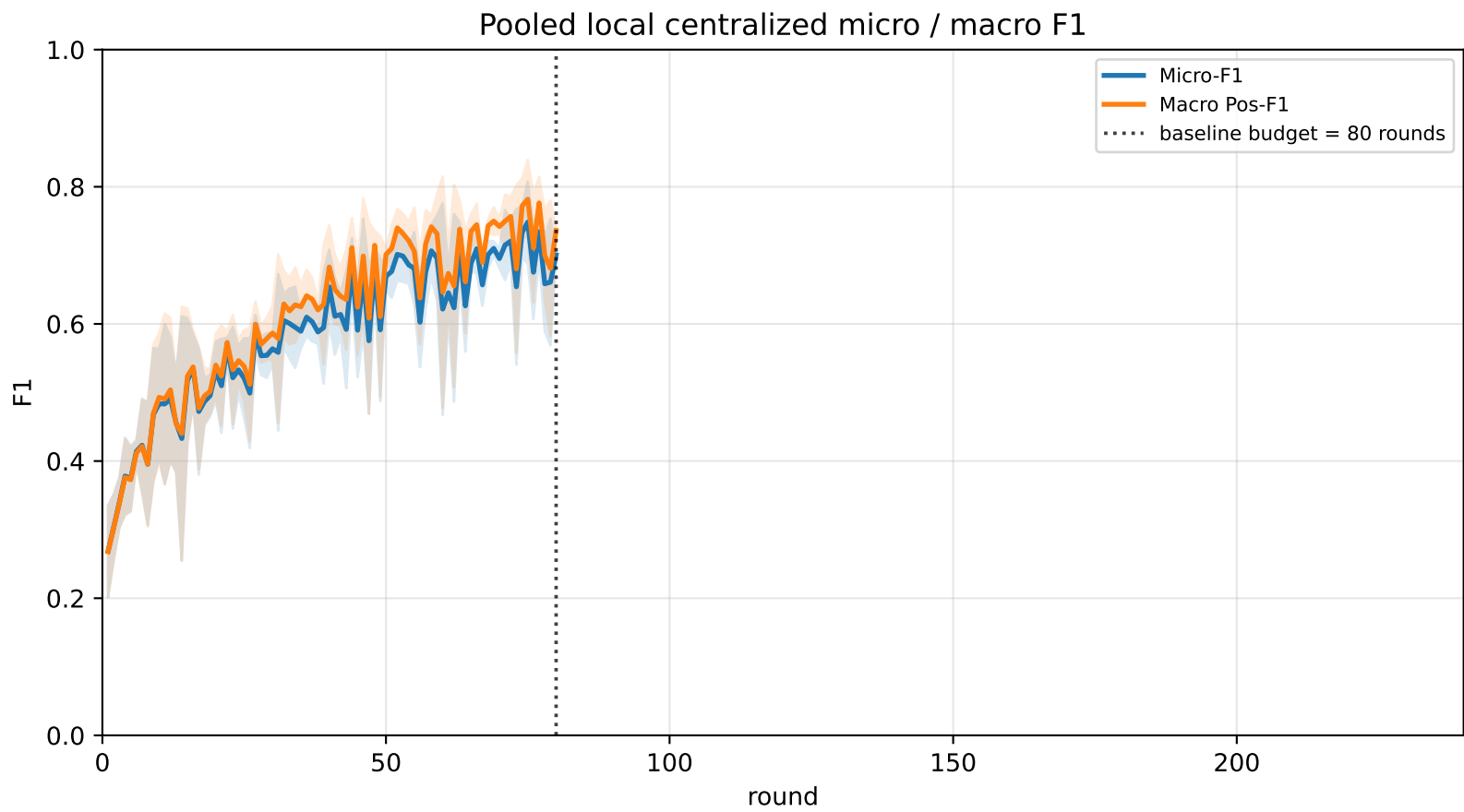
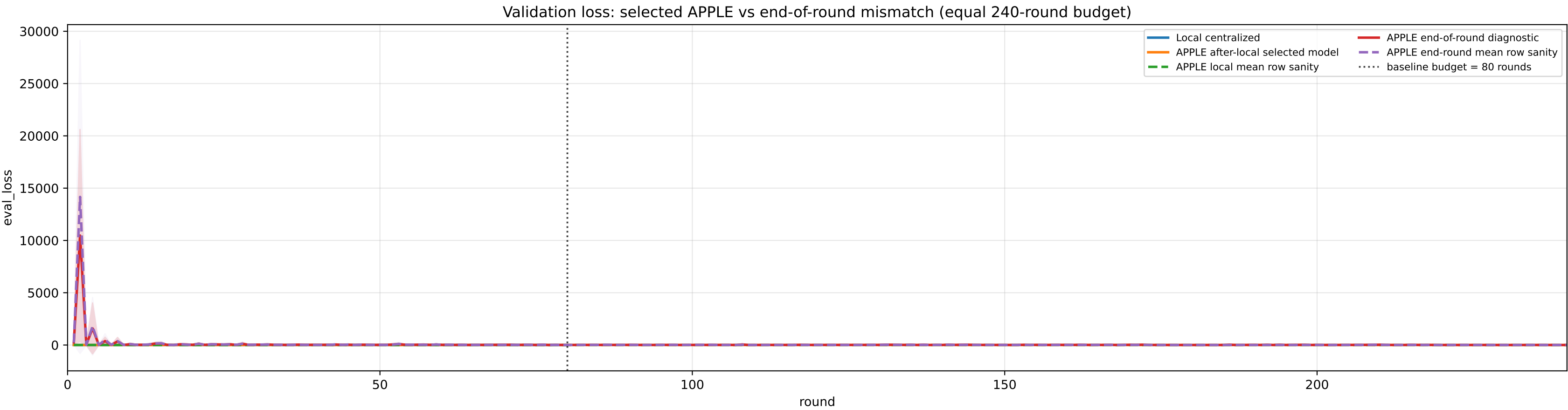
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	55.7% ± 8.8%	57.9% ± 10.2%	66.6% ± 17.9%	62.6% ± 13.2%	58.0% ± 7.4%	56.9% ± 12.7%	45.4% ± 3.0%
APPLE	67.1% ± 1.8%	69.4% ± 2.1%	83.8% ± 3.0%	75.5% ± 4.5%	68.1% ± 4.1%	65.1% ± 1.4%	54.7% ± 0.9%



family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

Best test performance (APPLE selected by after-local validation; all methods 80 rounds)

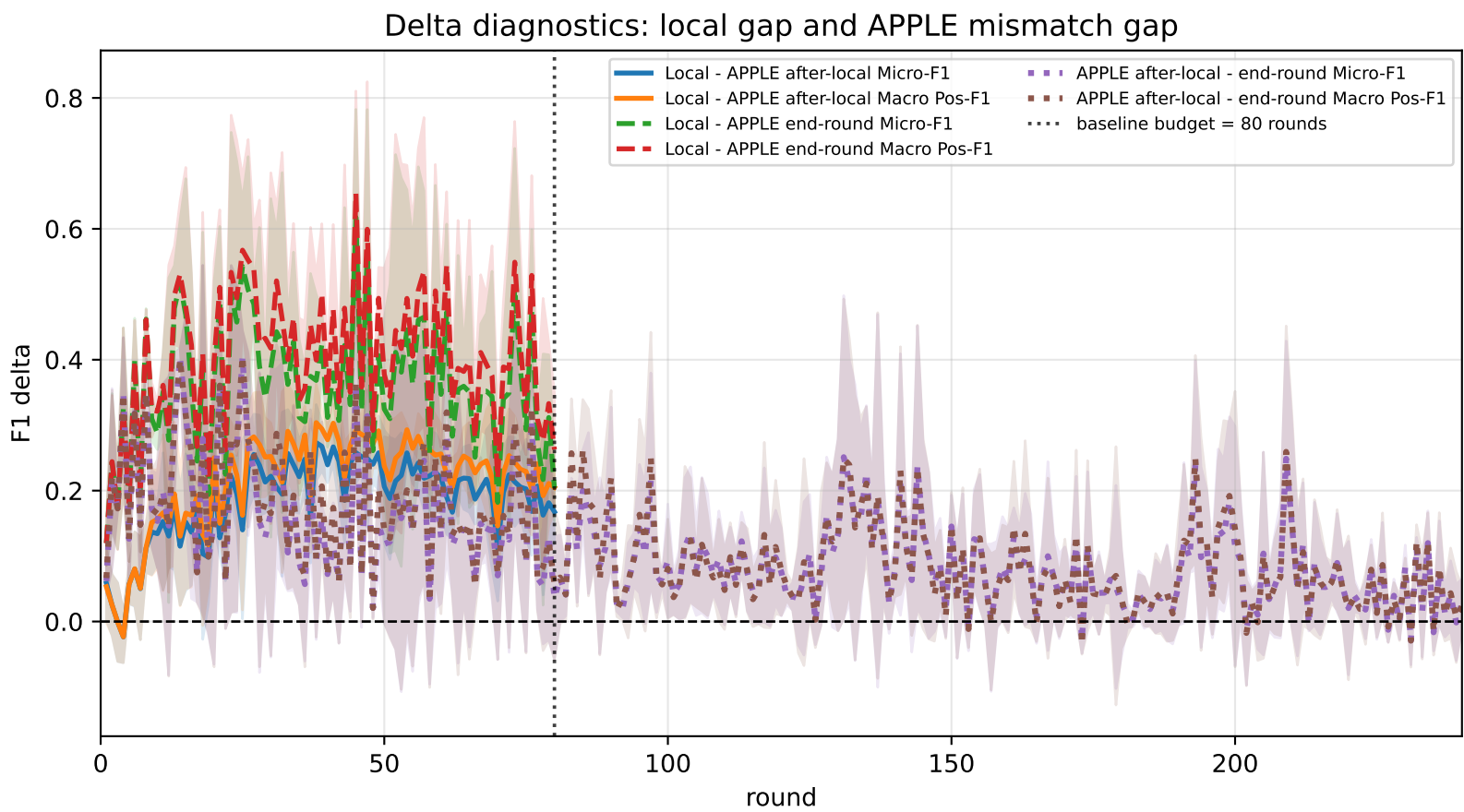
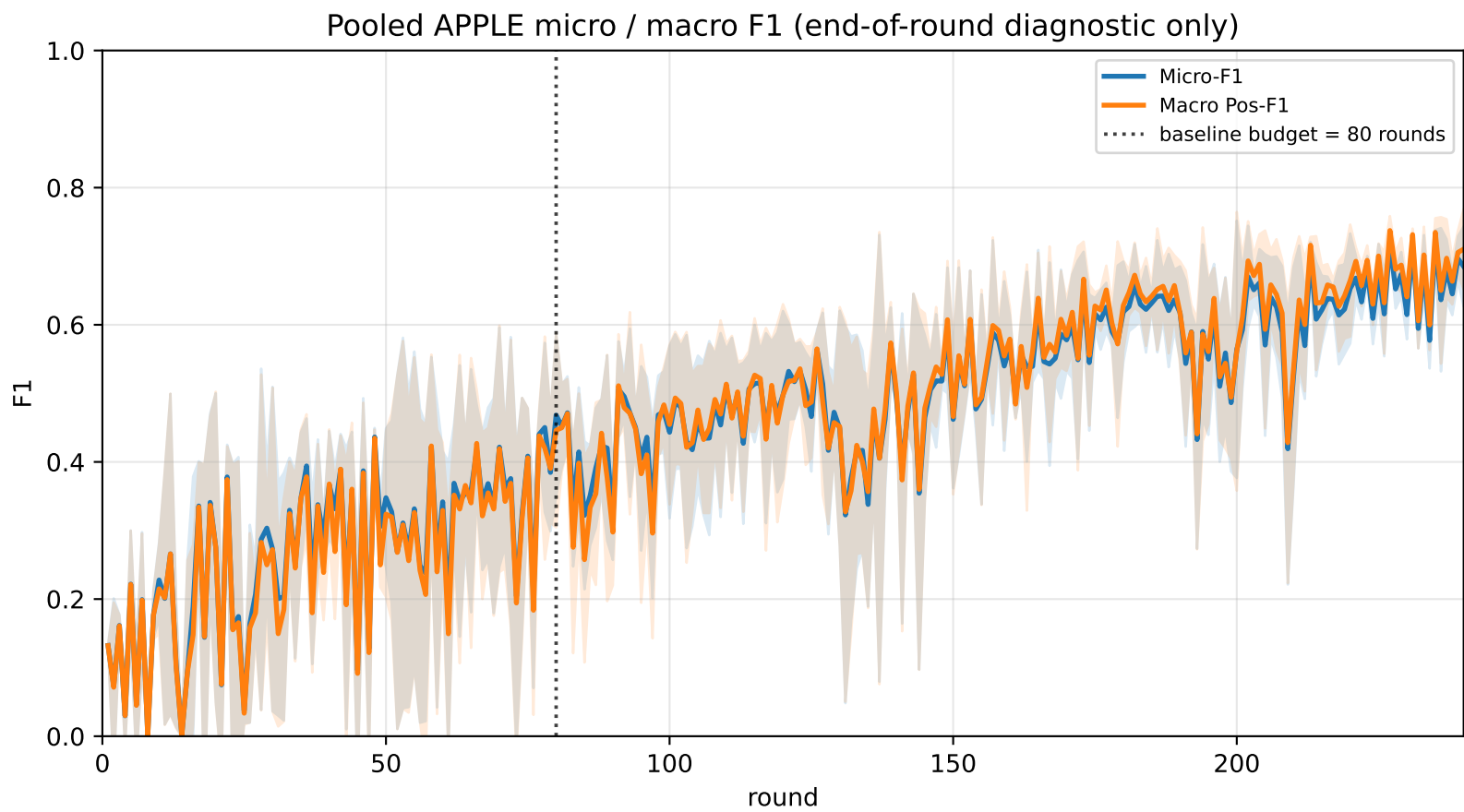
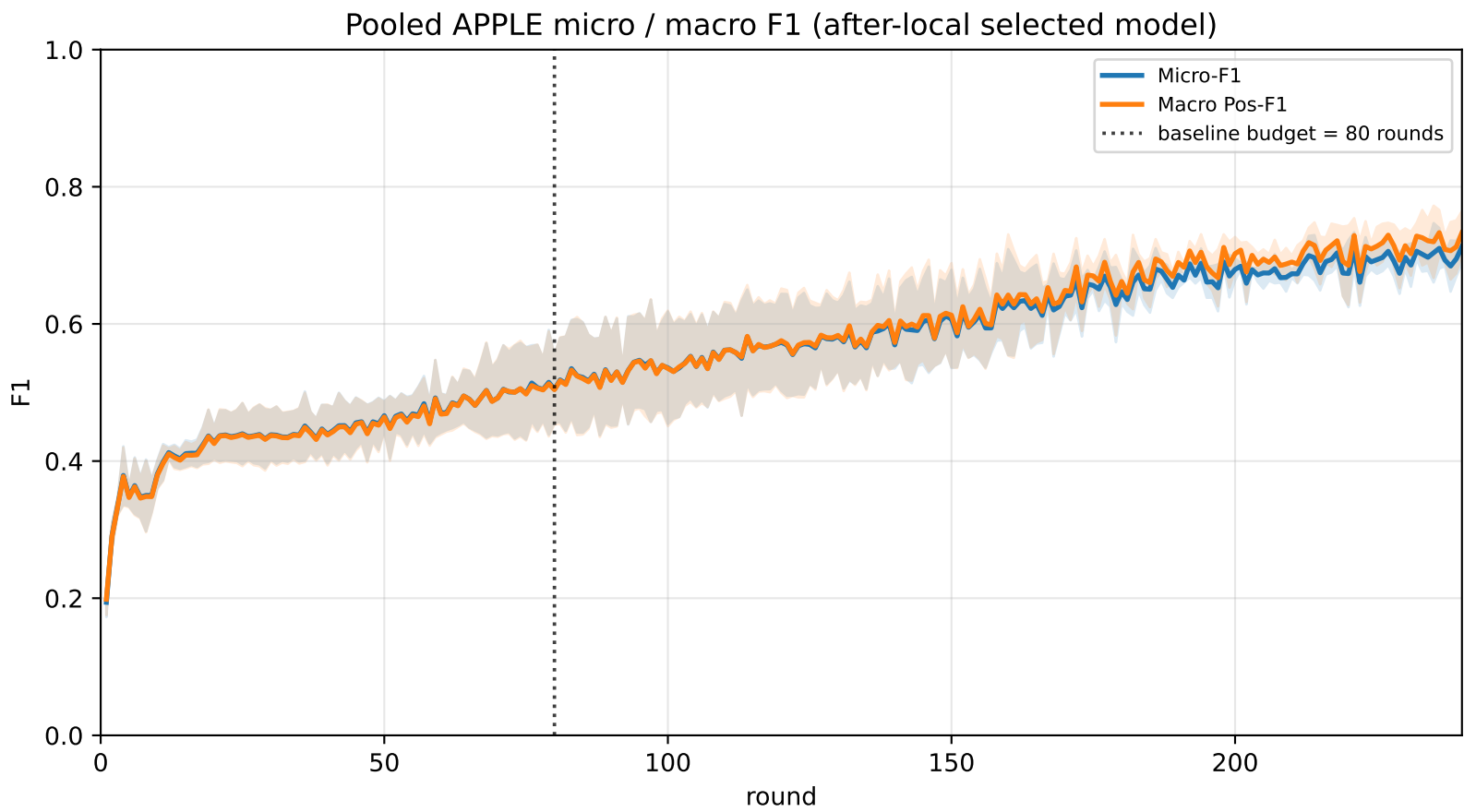
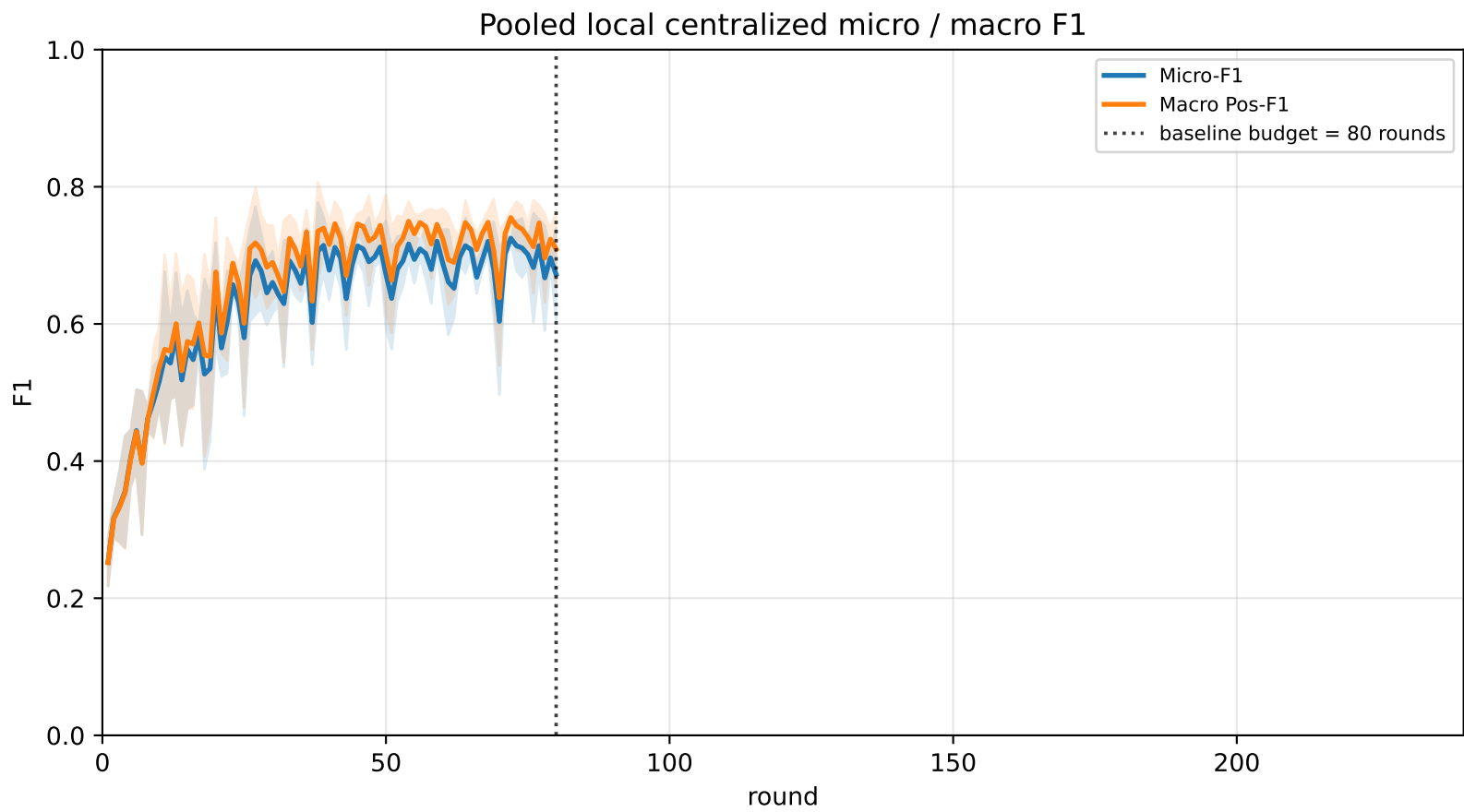
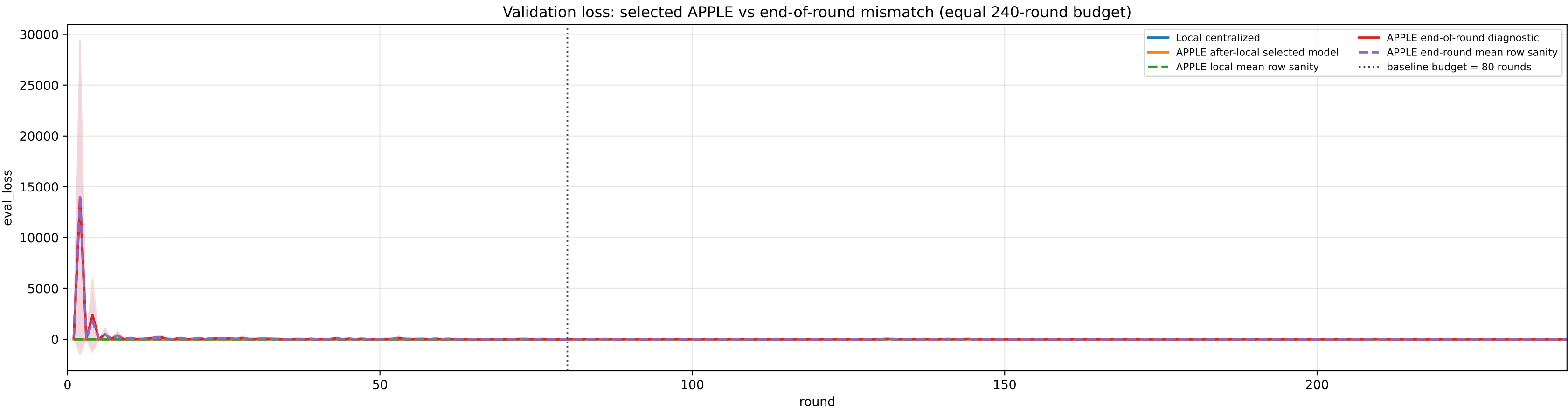
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	57.1% ± 7.9%	59.7% ± 8.6%	71.0% ± 8.9%	68.6% ± 10.4%	60.3% ± 10.1%	53.0% ± 10.3%	45.6% ± 4.2%
APPLE	70.1% ± 1.5%	72.7% ± 2.0%	89.7% ± 4.3%	78.3% ± 3.1%	68.1% ± 4.0%	68.8% ± 4.5%	58.7% ± 0.9%



family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3% | heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

Best test performance (APPLE selected by after-local validation; all methods 80 rounds)

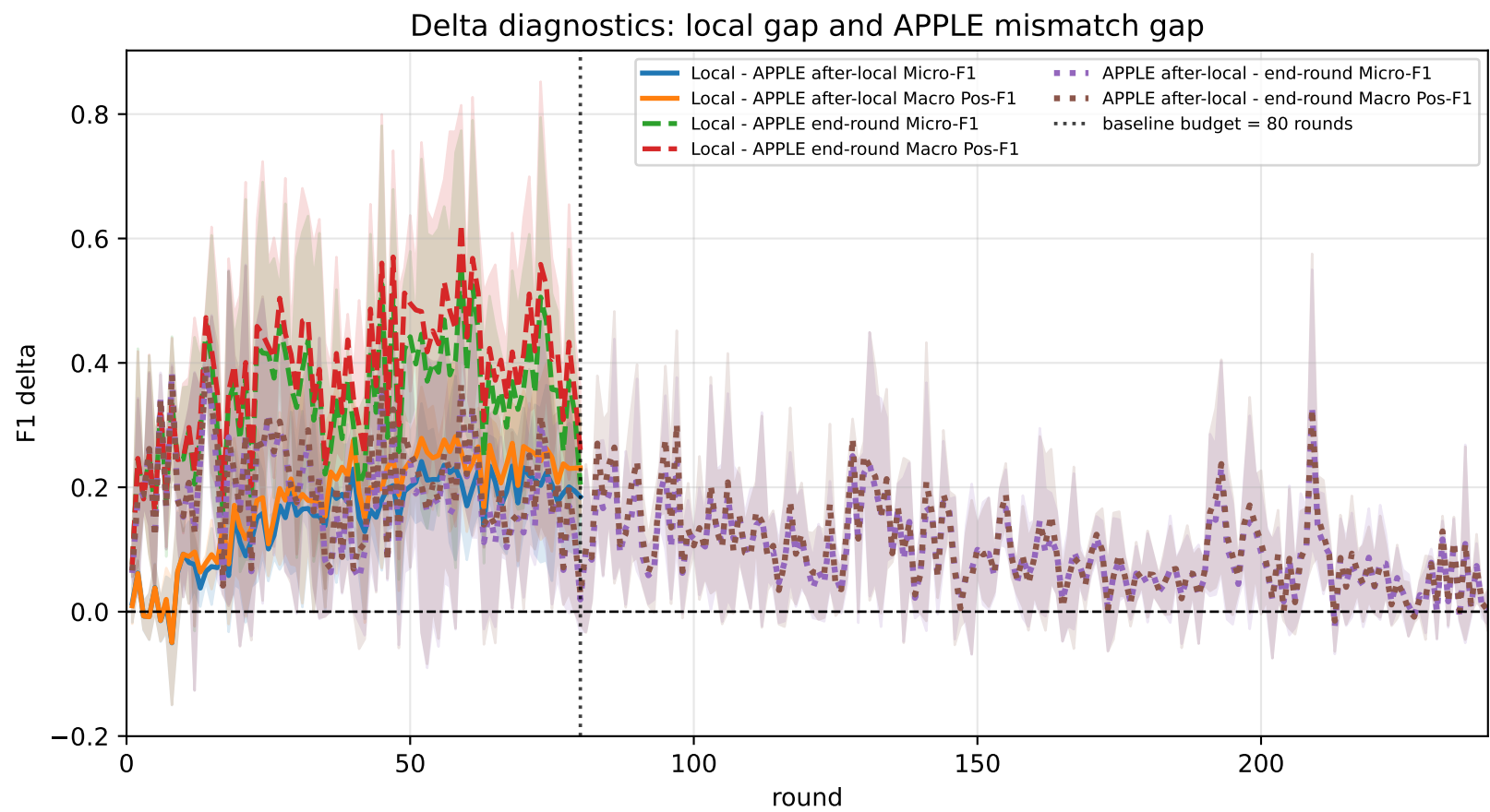
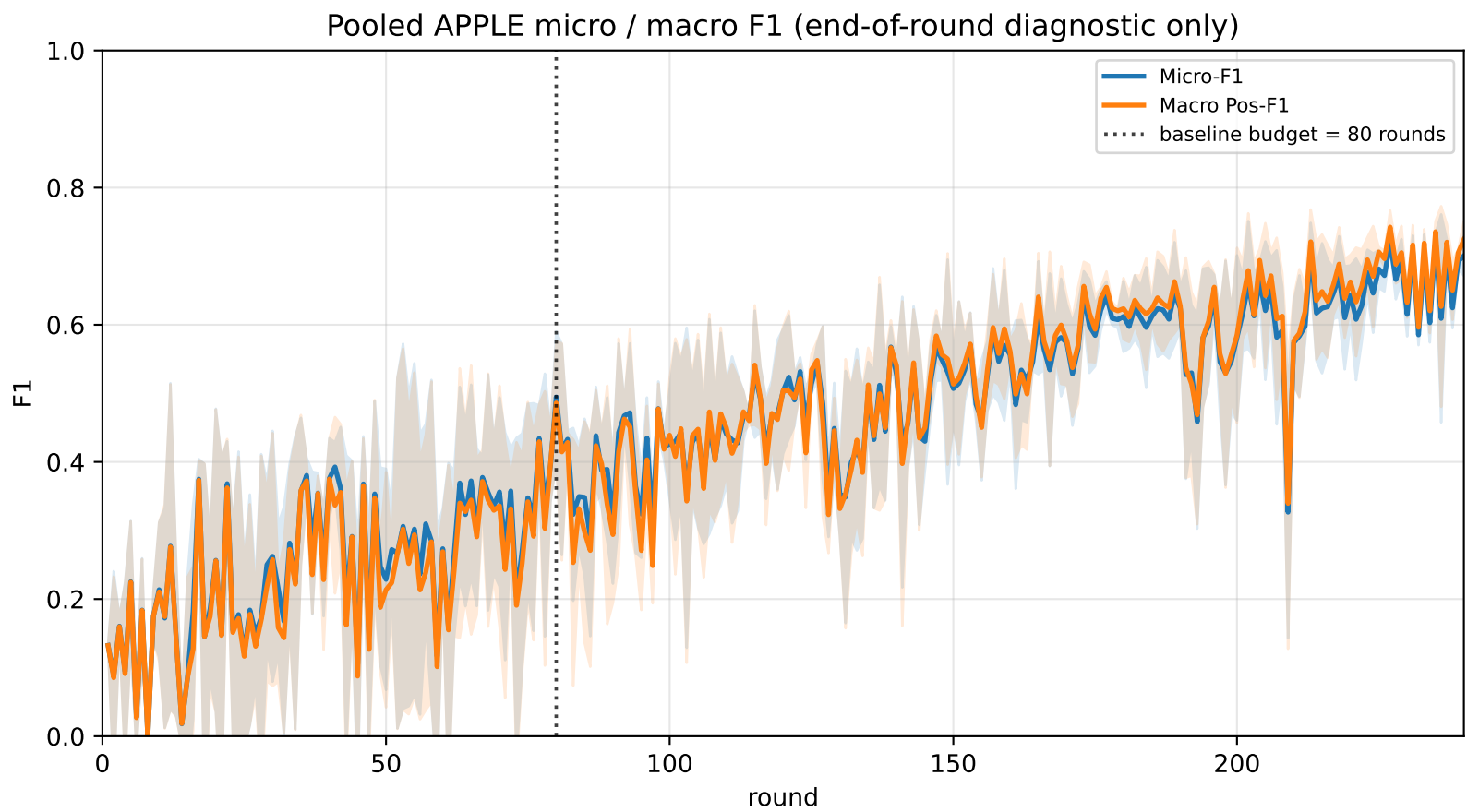
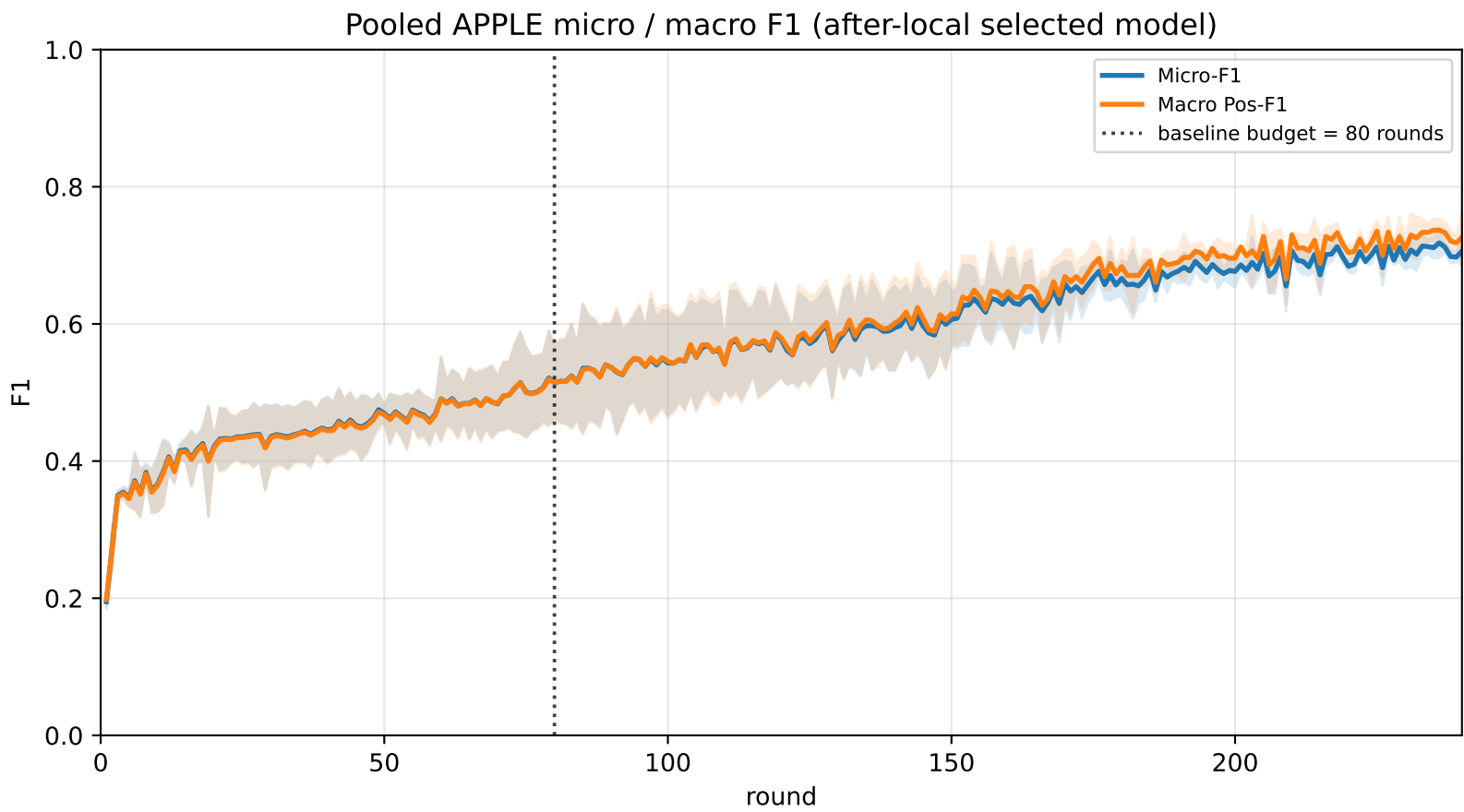
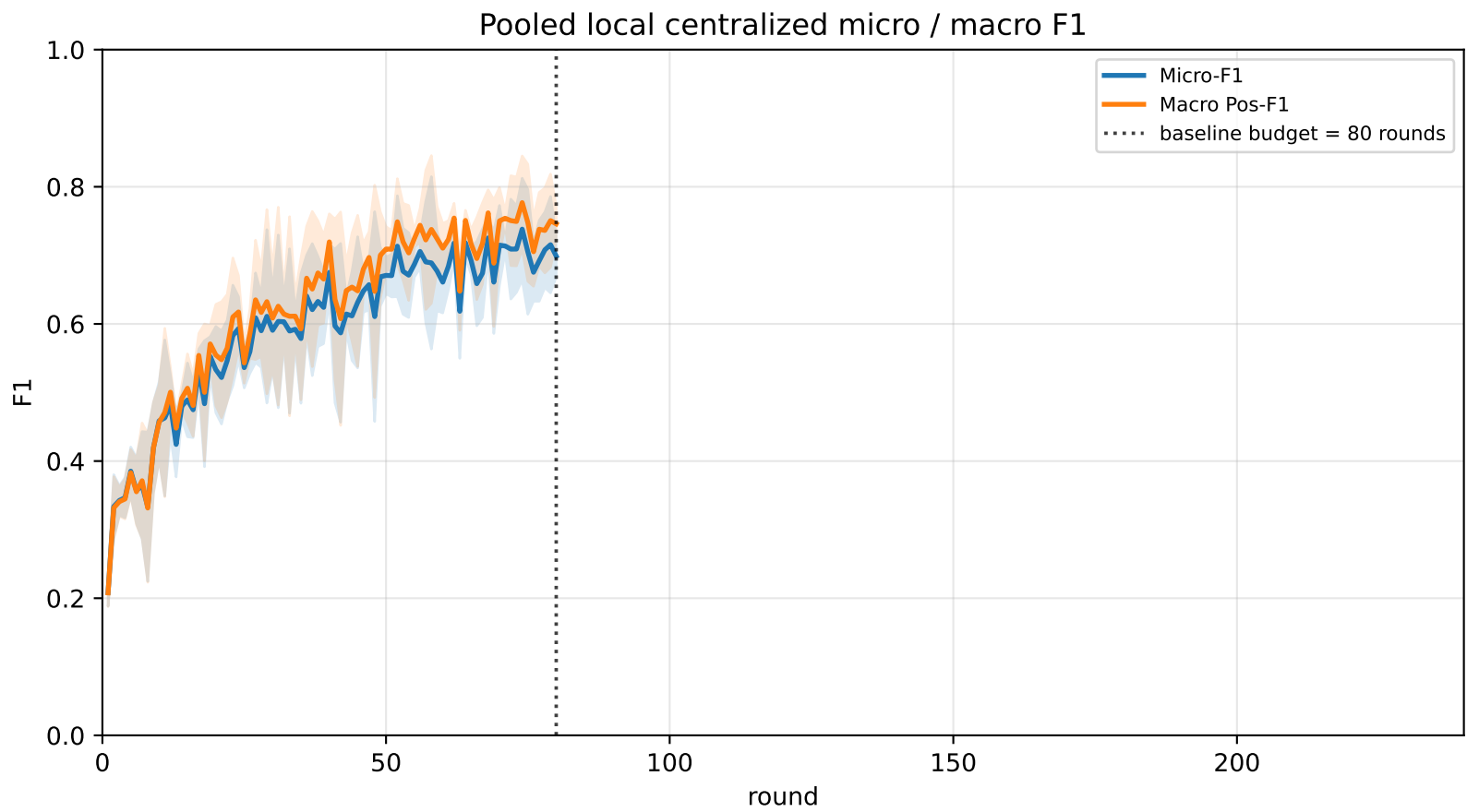
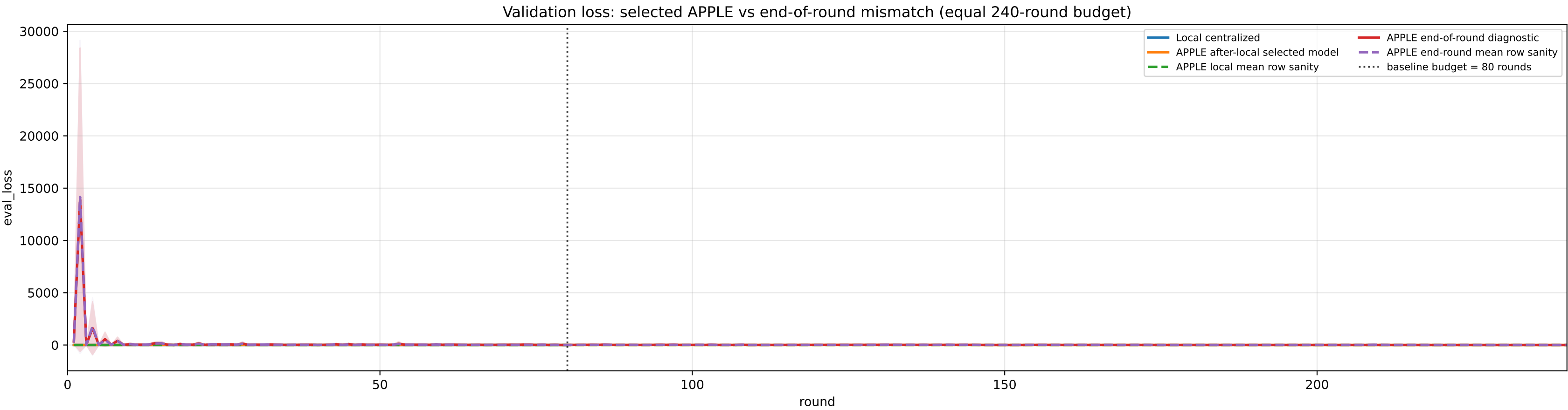
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	62.7% ± 2.4%	65.2% ± 2.6%	76.5% ± 5.8%	75.7% ± 4.0%	59.8% ± 2.9%	64.6% ± 3.0%	49.4% ± 2.2%
APPLE	66.9% ± 2.5%	68.4% ± 3.3%	74.9% ± 7.7%	75.5% ± 6.9%	70.5% ± 4.9%	63.6% ± 2.9%	57.7% ± 1.0%



family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

Best test performance (APPLE selected by after-local validation; all methods 80 rounds)

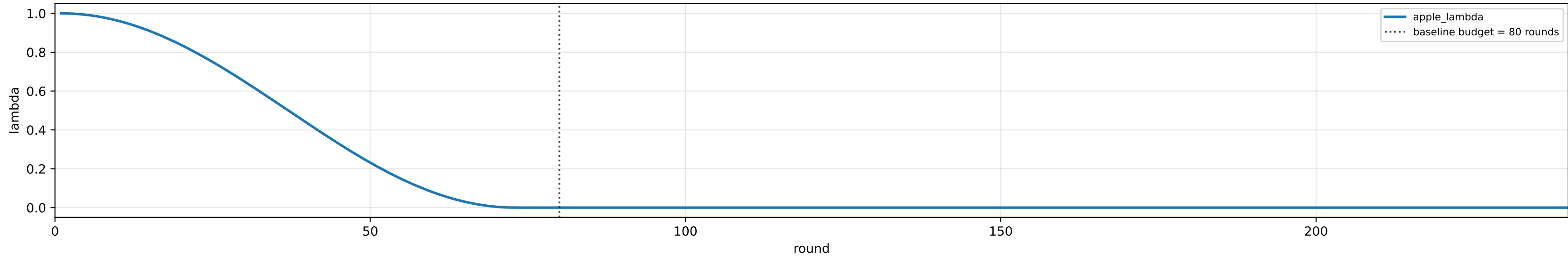
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	63.1% ± 3.5%	67.2% ± 3.7%	86.9% ± 6.6%	77.7% ± 9.8%	63.1% ± 2.3%	58.3% ± 3.3%	49.9% ± 3.9%
APPLE	69.8% ± 0.9%	71.8% ± 1.6%	81.7% ± 4.6%	80.9% ± 3.6%	69.6% ± 3.4%	68.7% ± 1.1%	58.3% ± 1.8%



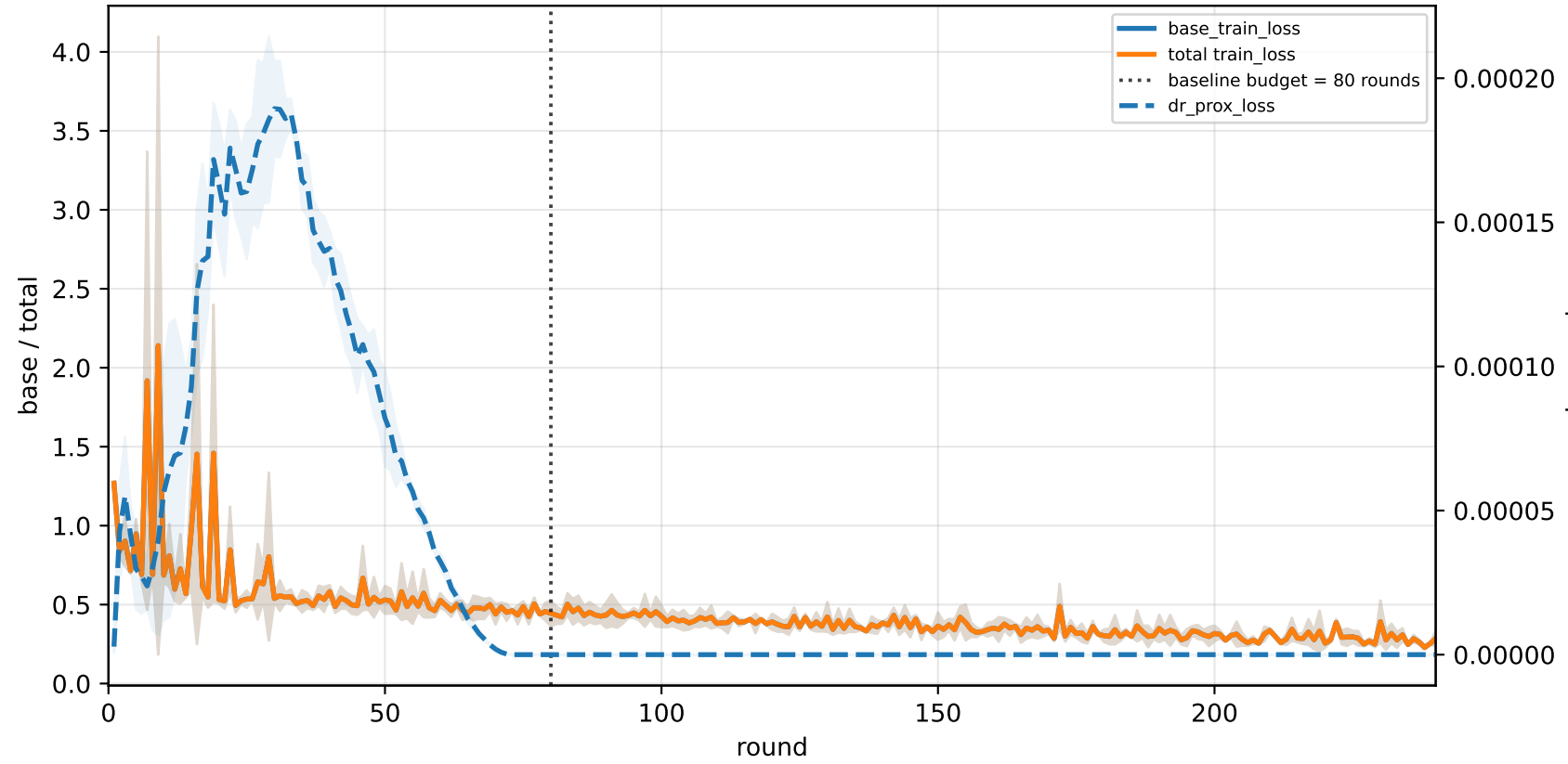
APPLE training-loss decomposition | Mid heterogeneity

family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

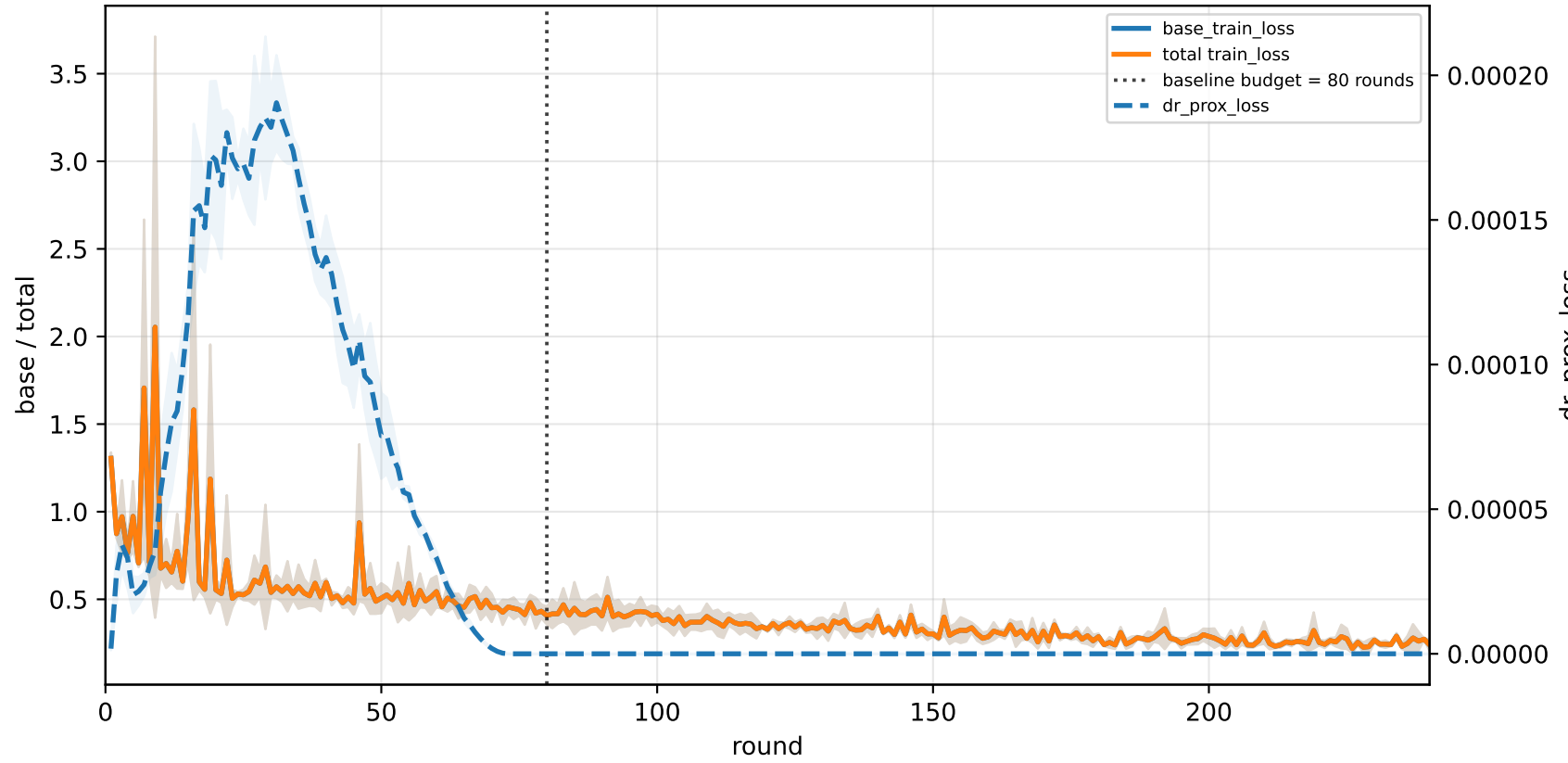
APPLE lambda scheduler | apple_mu=0.01



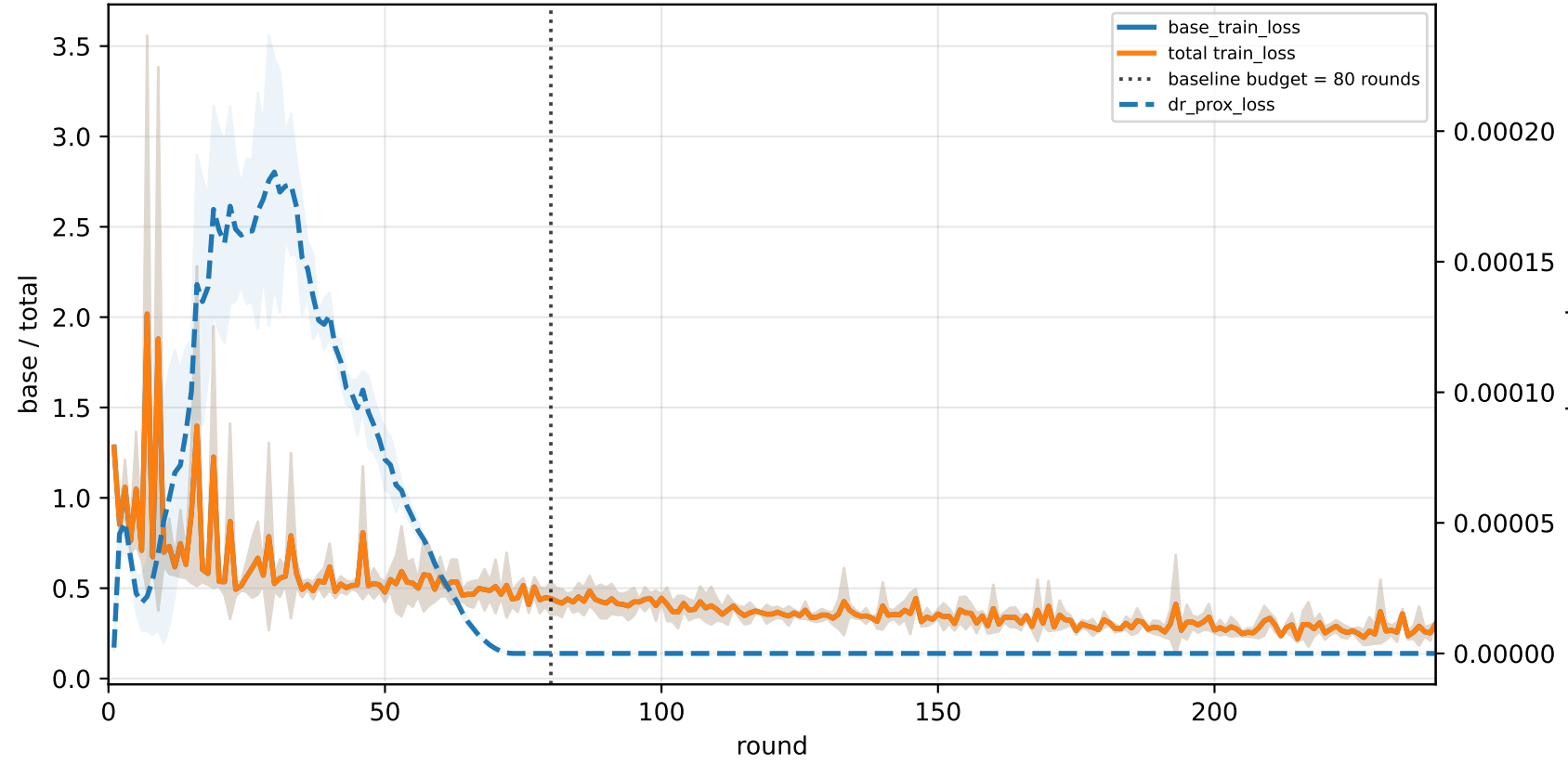
q_task_cycle2 | graph_id=3000005



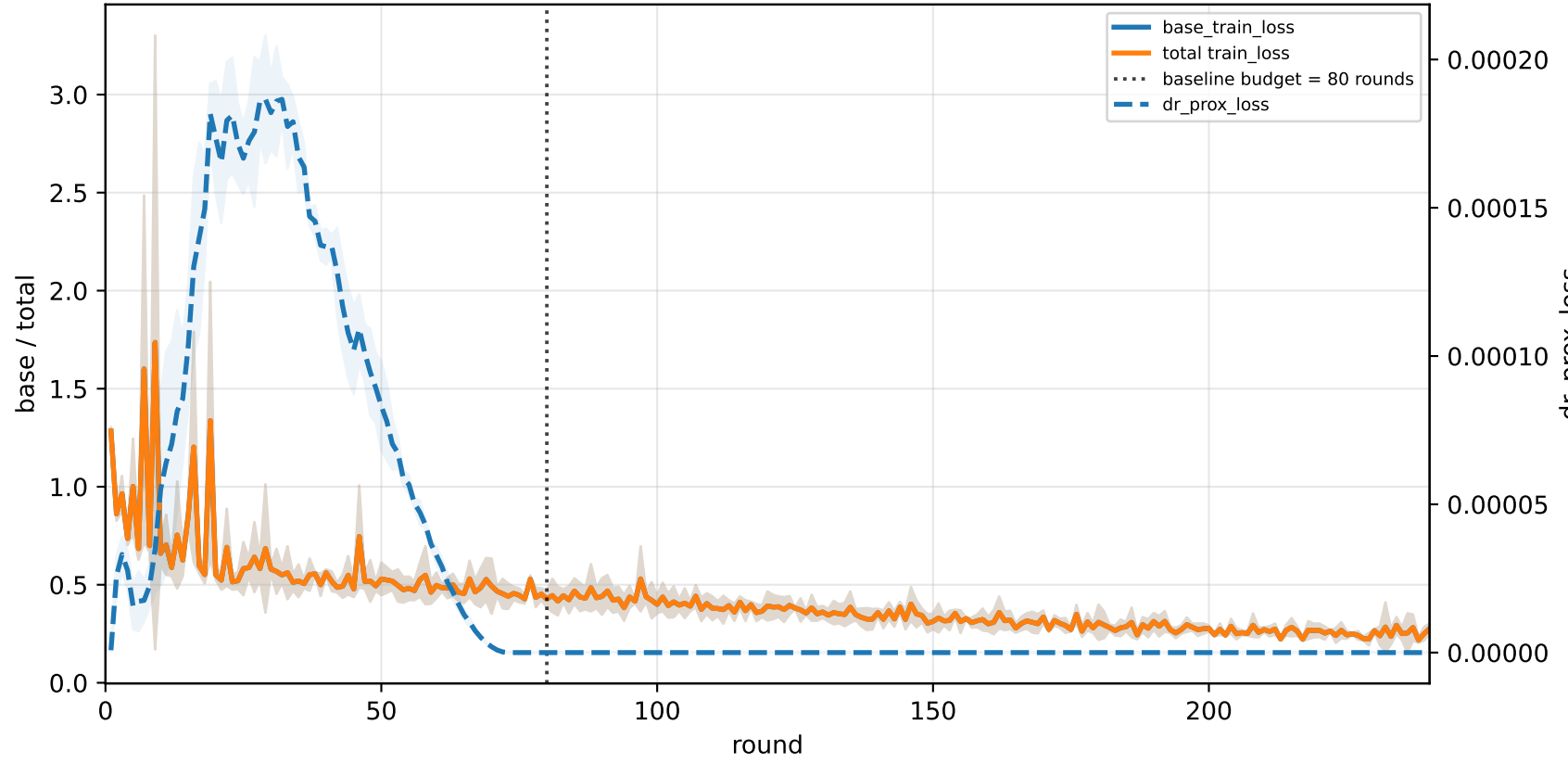
q_task_cycle3 | graph_id=3000006



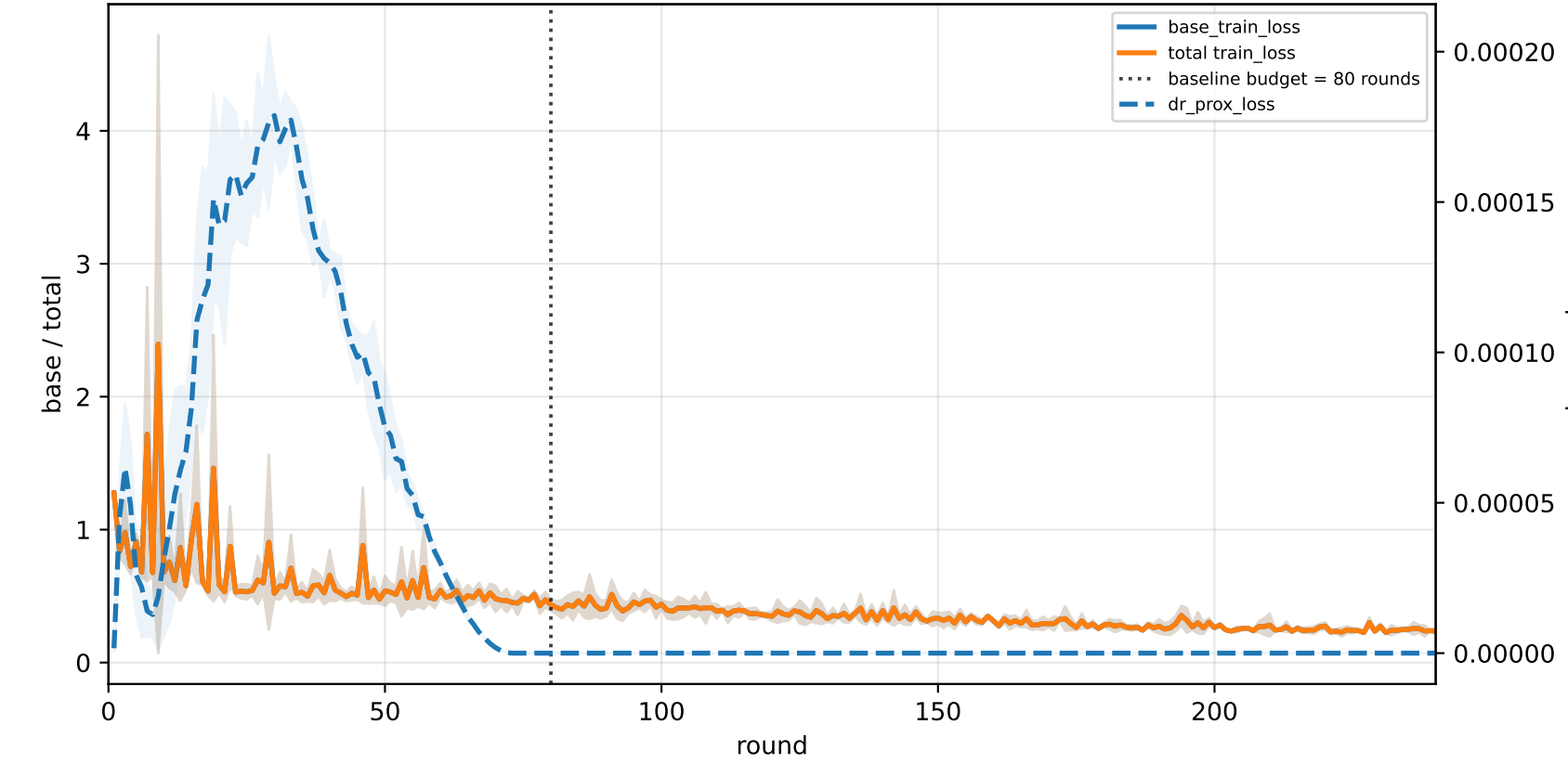
q_task_cycle4 | graph_id=3000007



q_task_cycle5 | graph_id=3000008

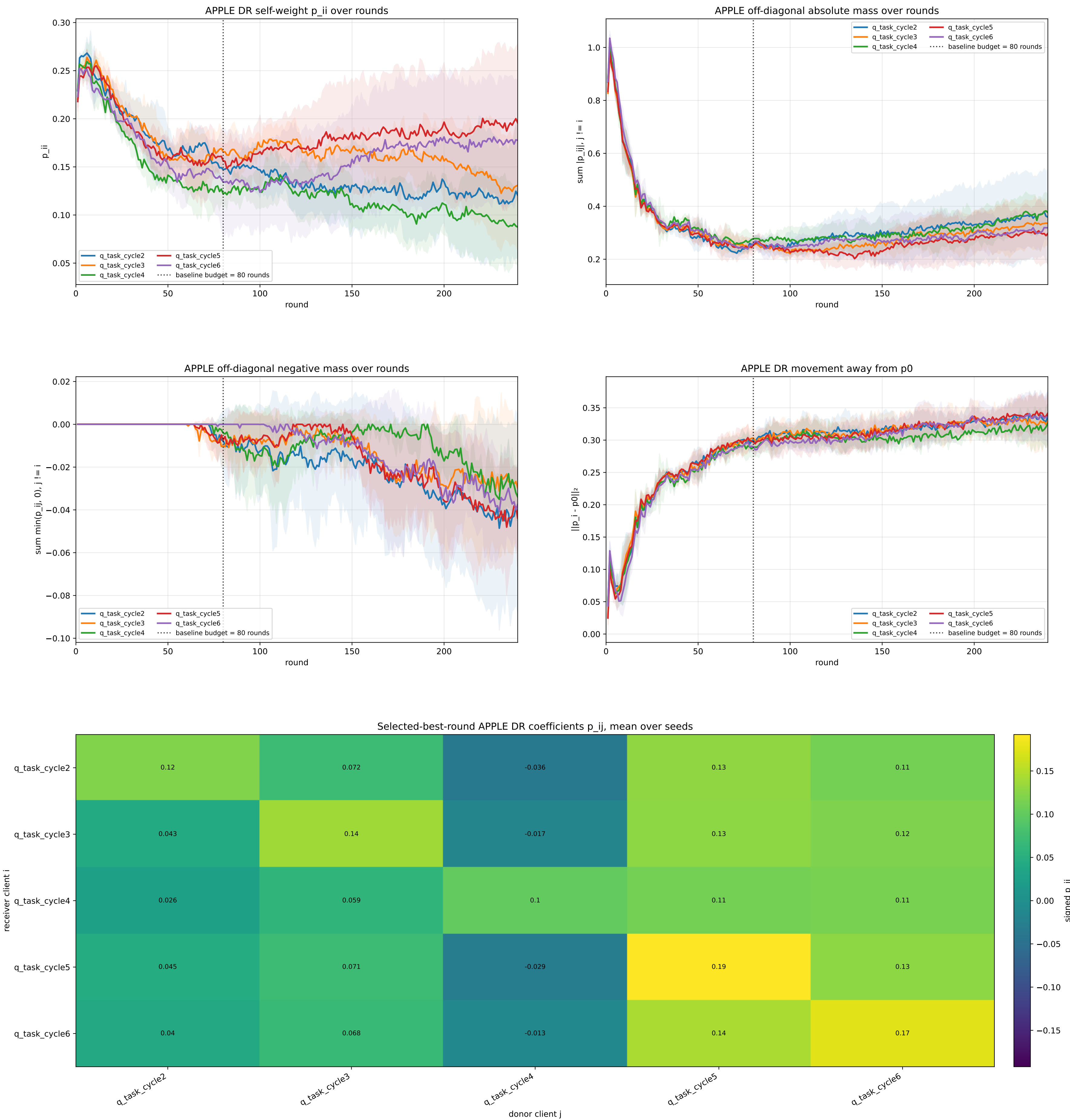


q_task_cycle6 | graph_id=3000009



APPLE DR behavior | Mid heterogeneity

family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

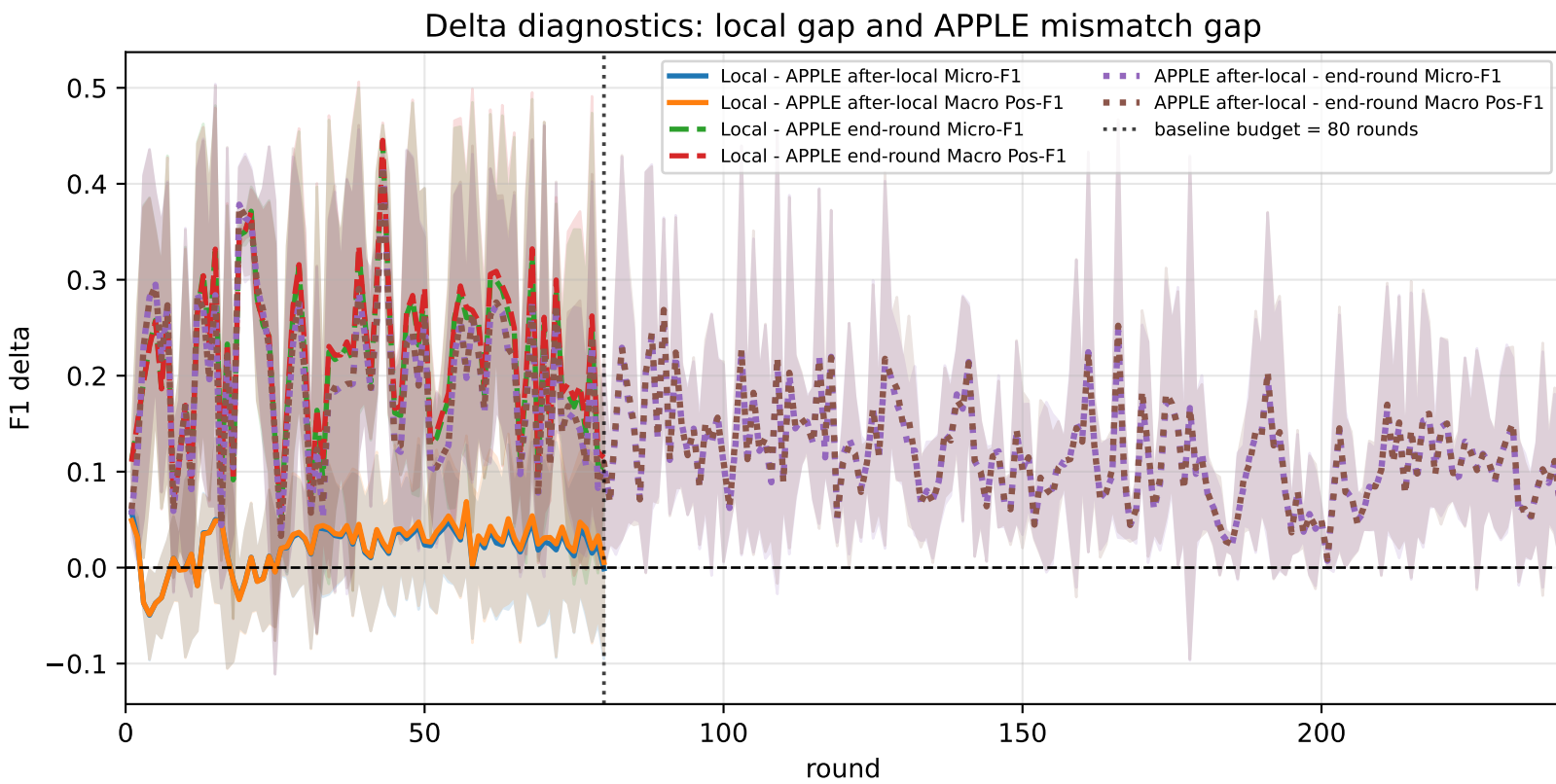
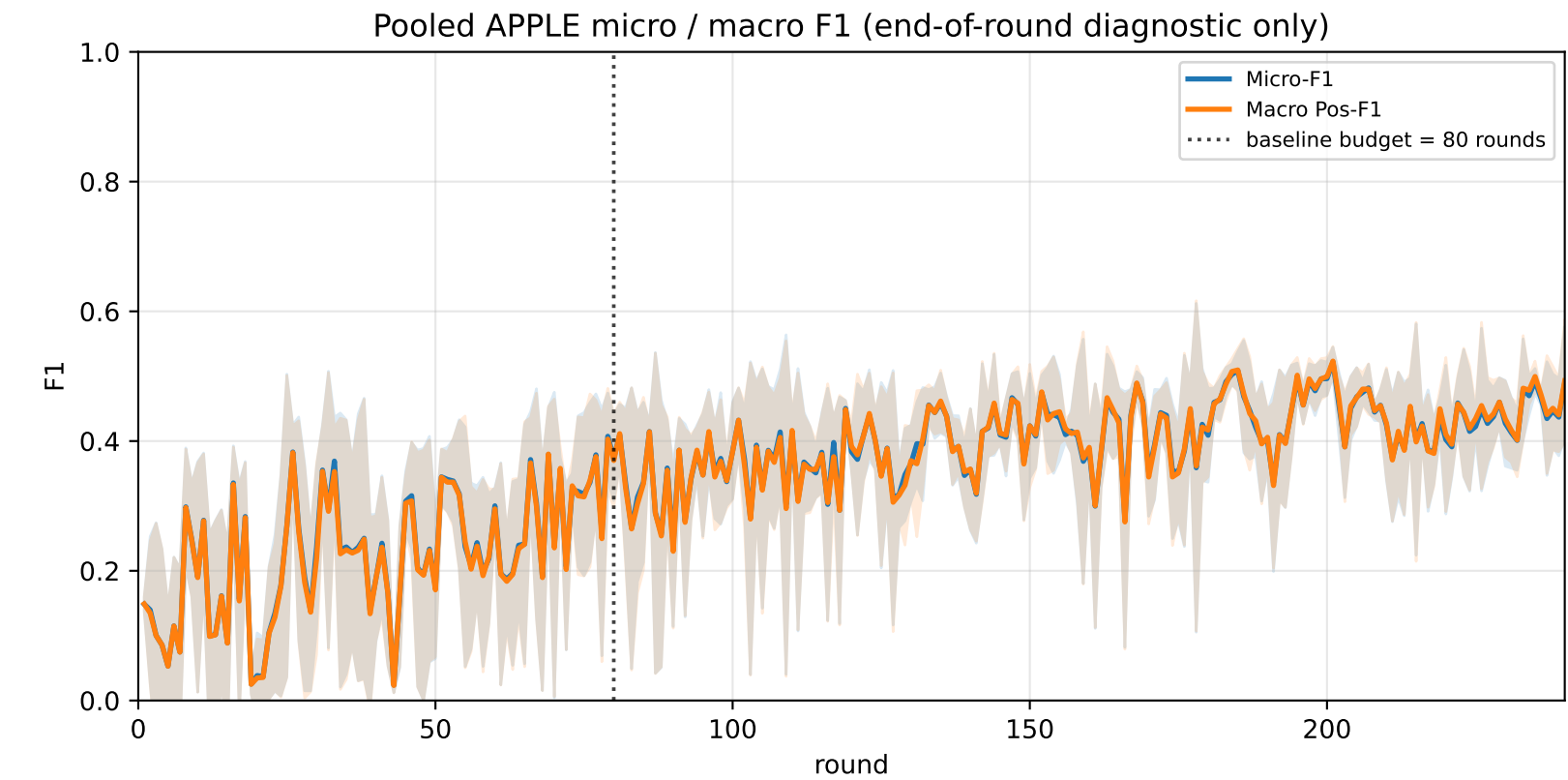
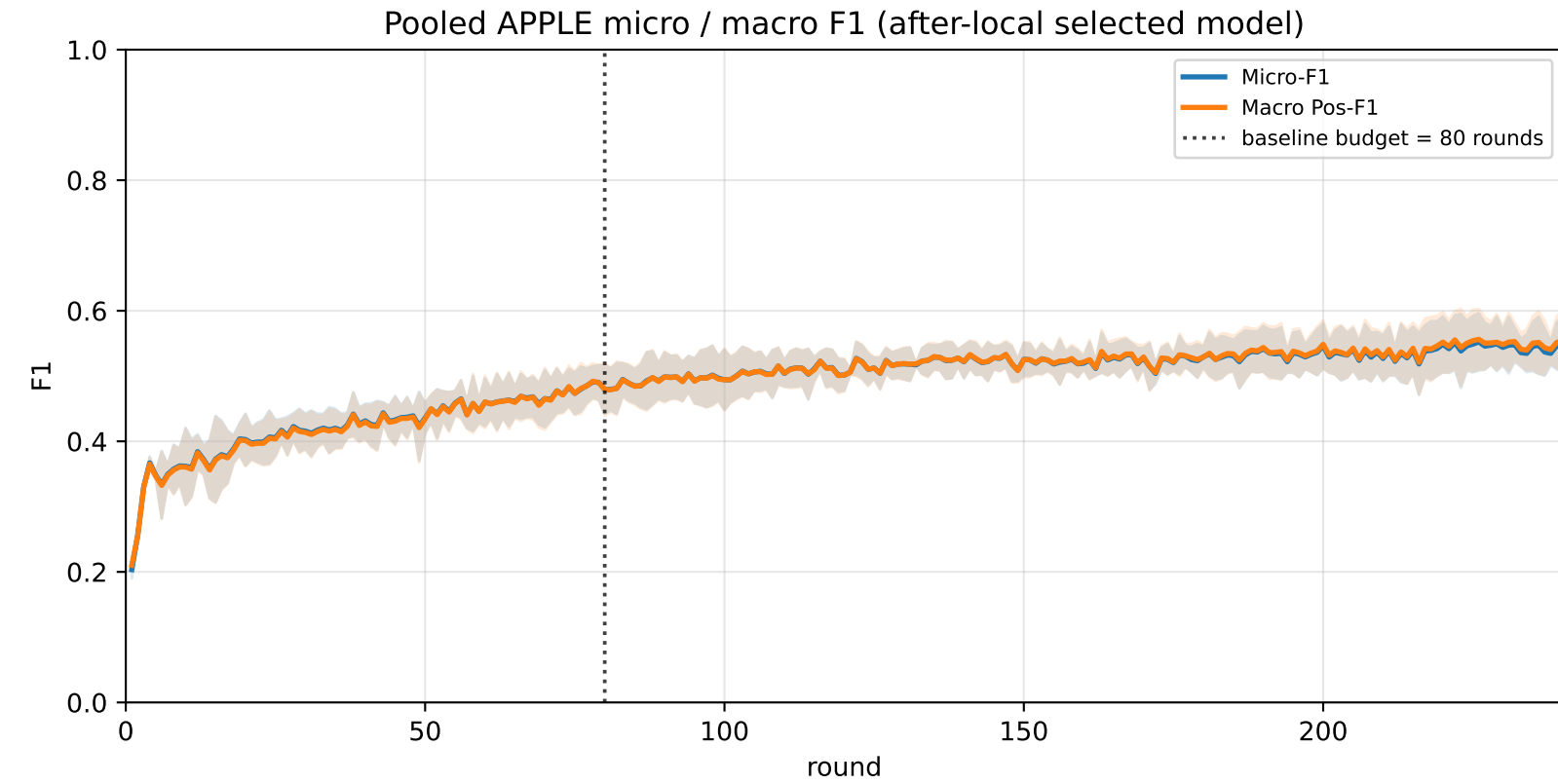
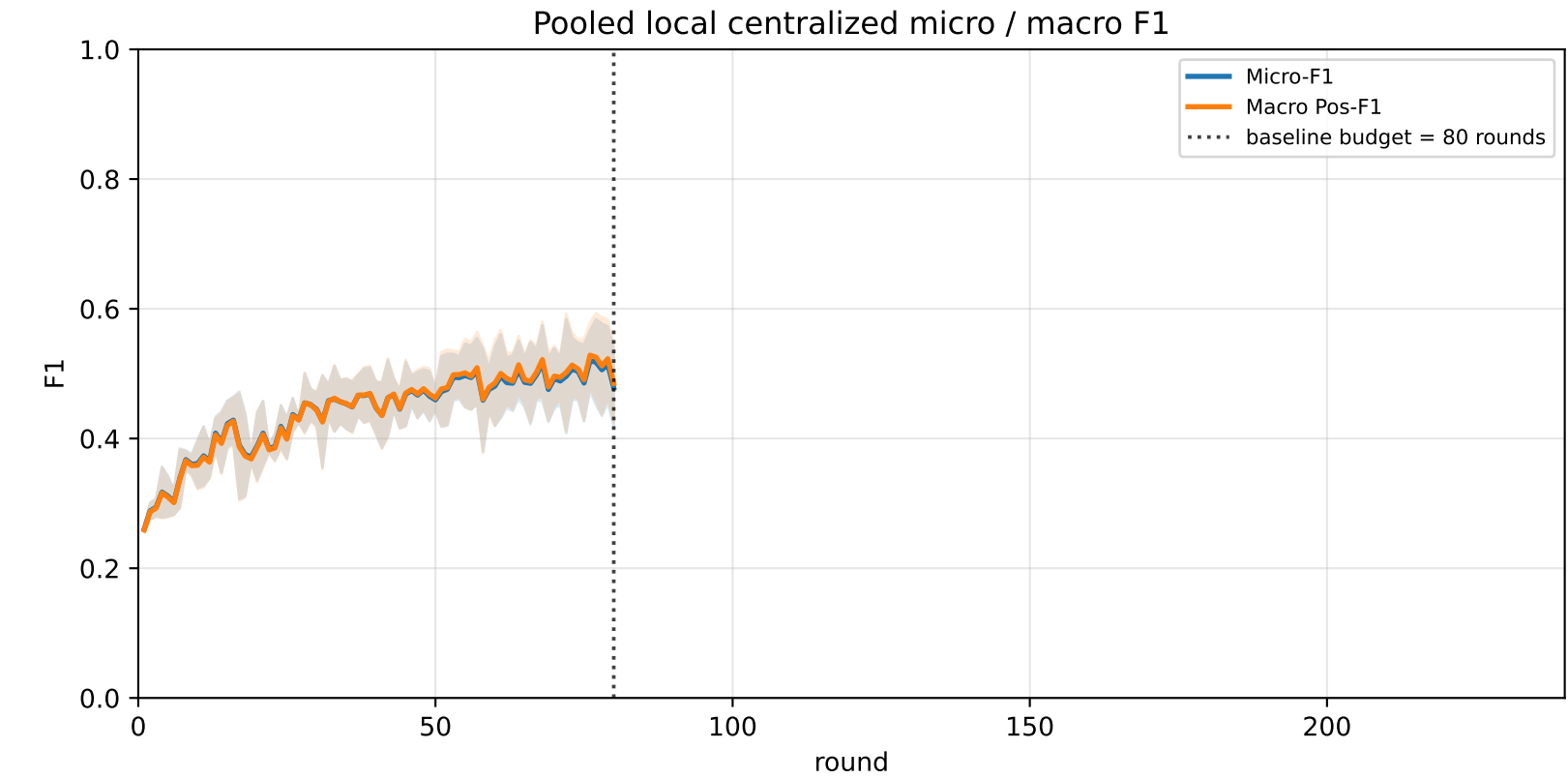
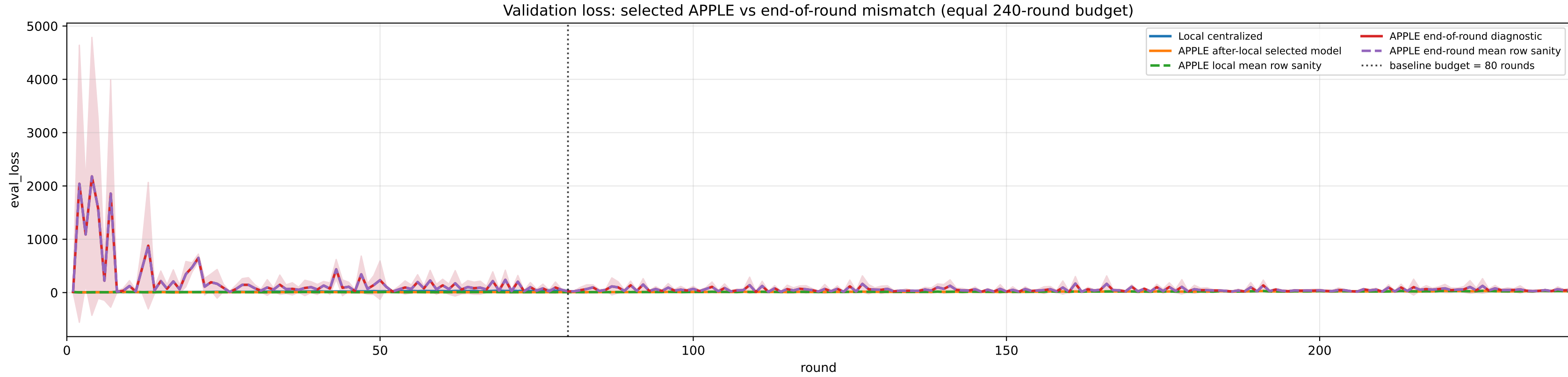


Controlled q-label heterogeneity: High | APPLE diagnostics | aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

Best test performance (APPLE selected by after-local validation; all methods 80 rounds)

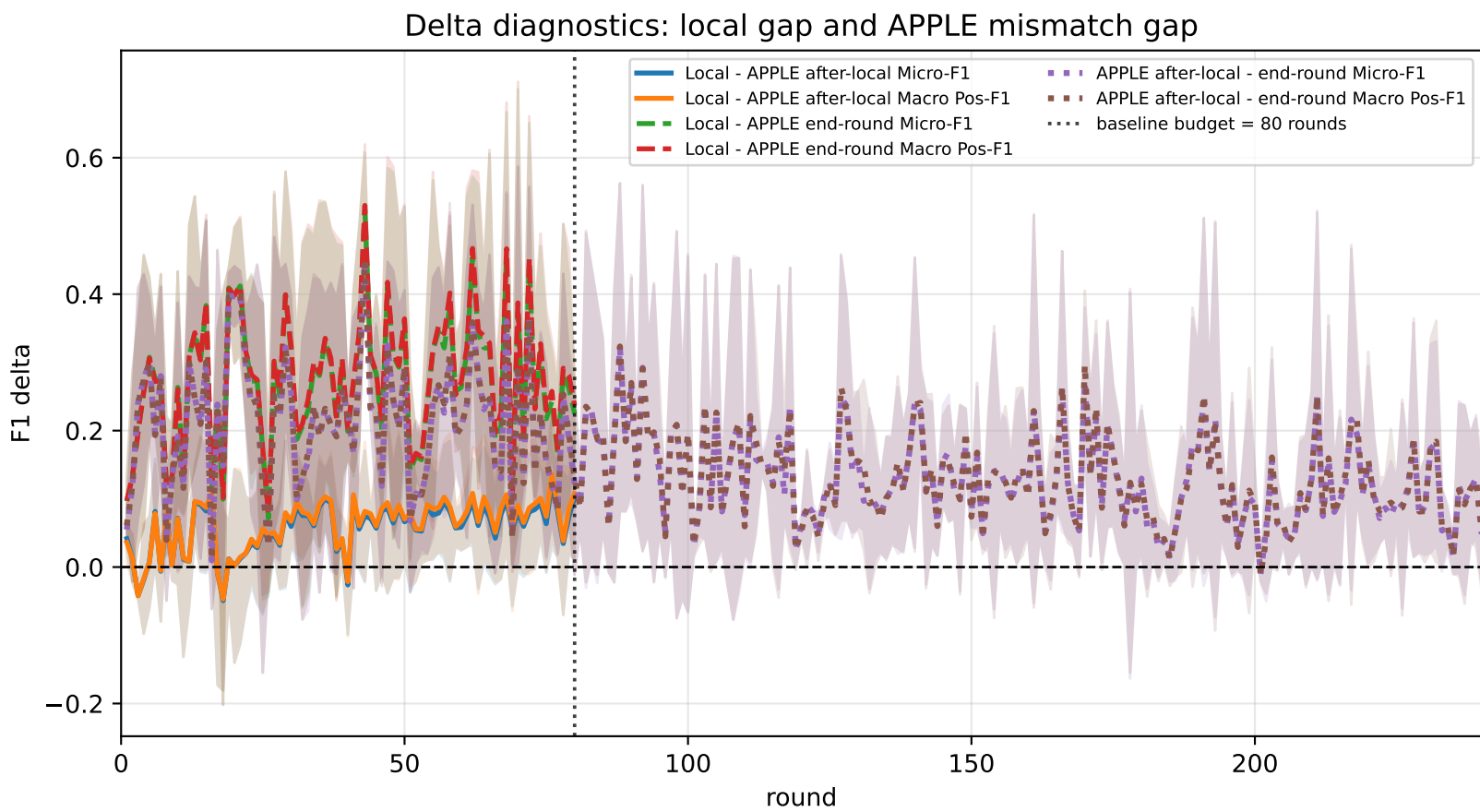
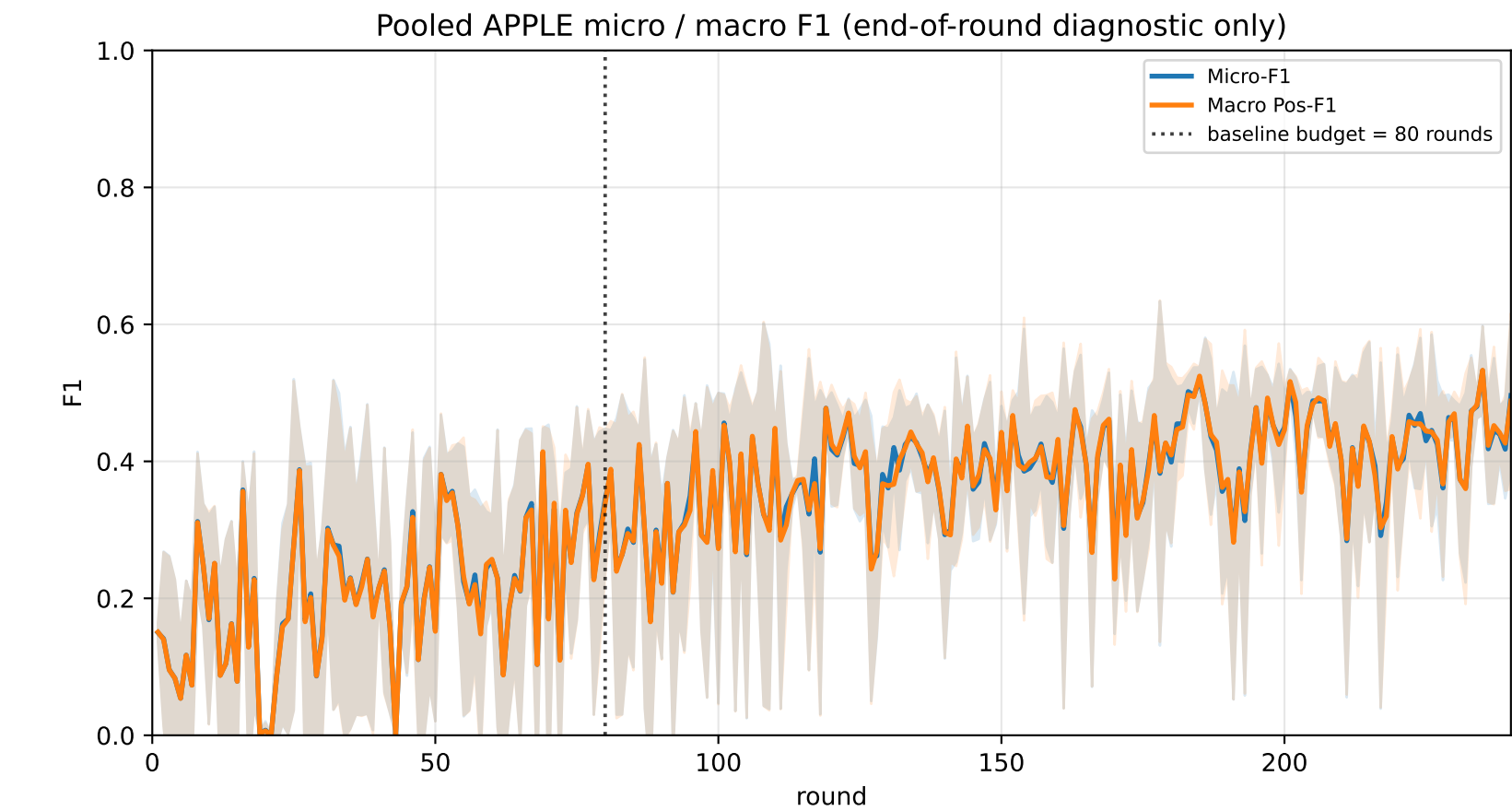
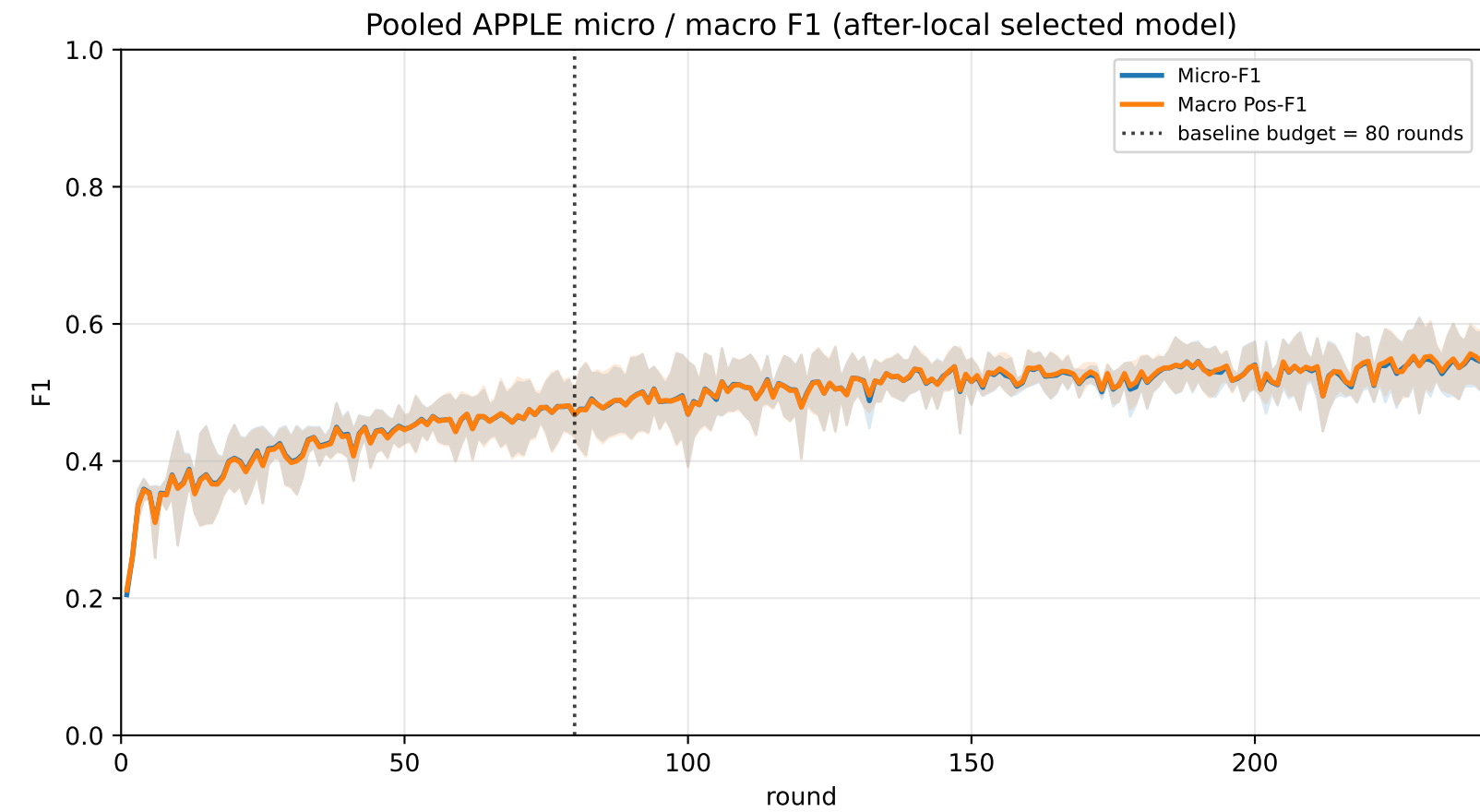
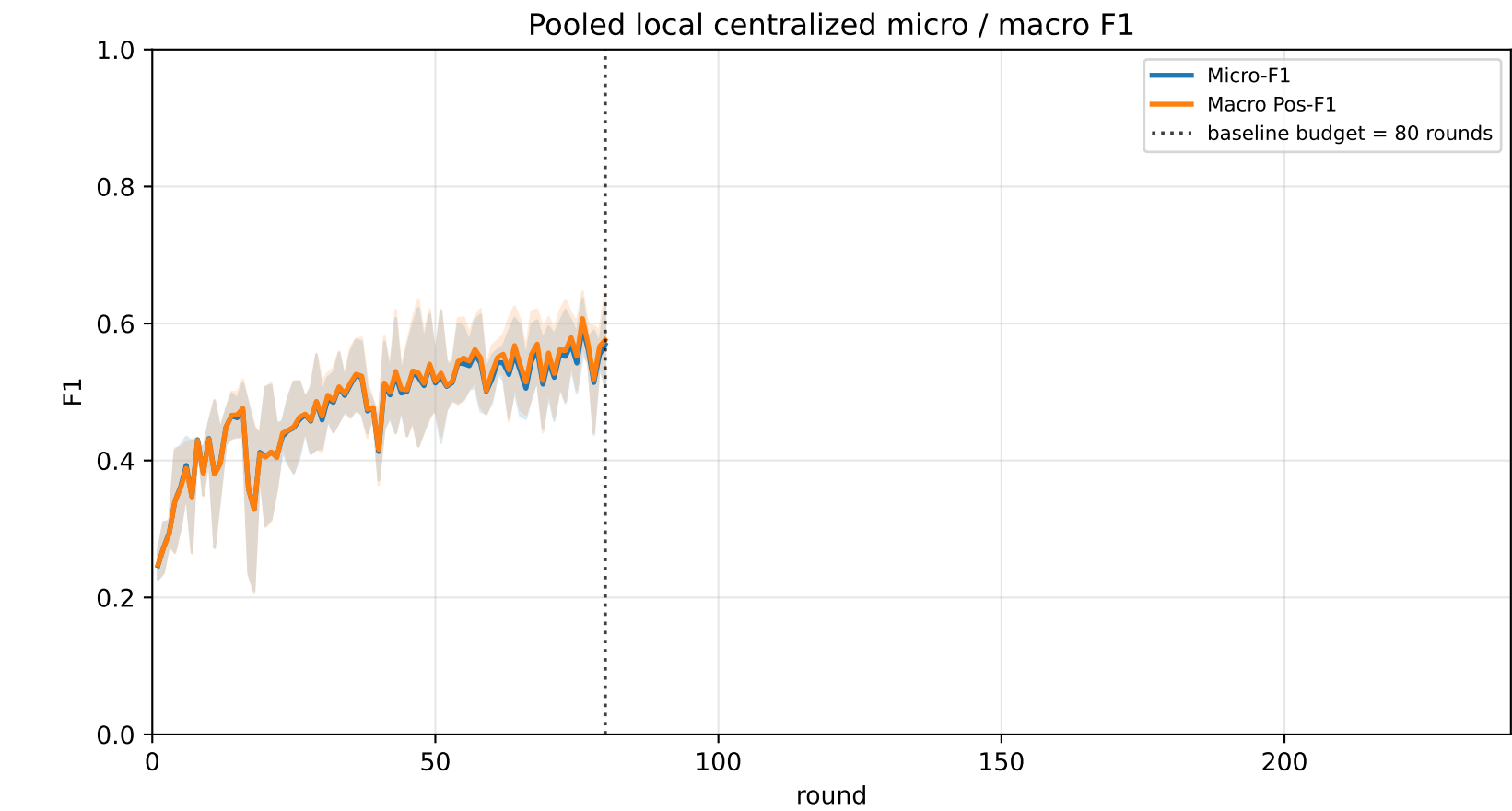
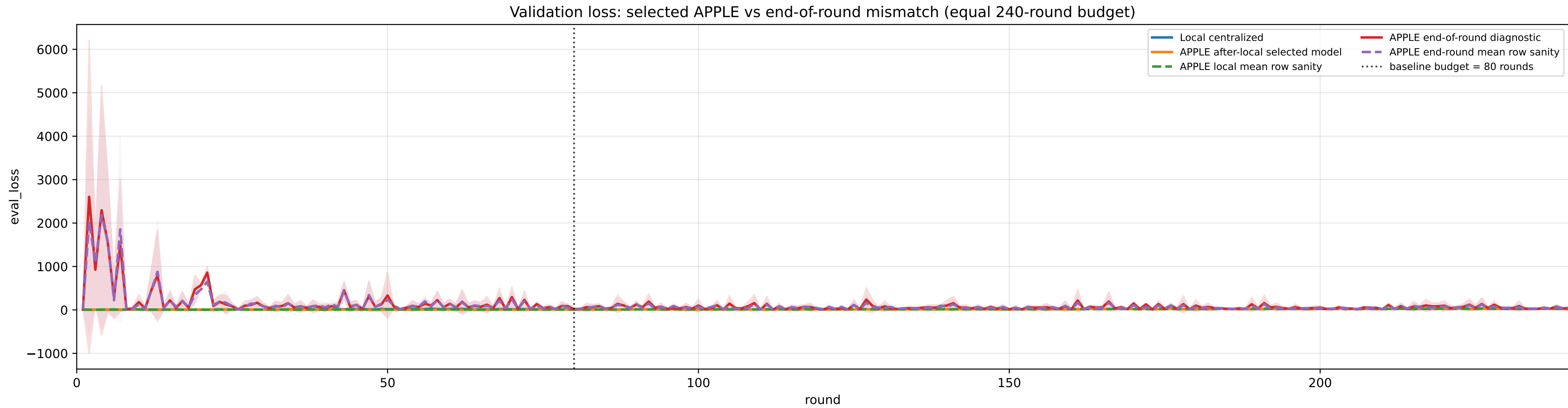
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	29.1% ± 5.1%	29.5% ± 5.3%	22.1% ± 2.8%	26.7% ± 3.4%	26.9% ± 3.1%	26.5% ± 3.3%	30.2% ± 3.0%
APPLE	31.3% ± 9.1%	31.3% ± 9.0%	26.4% ± 8.8%	31.7% ± 9.6%	32.3% ± 9.9%	31.3% ± 9.2%	34.4% ± 8.0%



family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

Best test performance (APPLE selected by after-local validation; all methods 80 rounds)

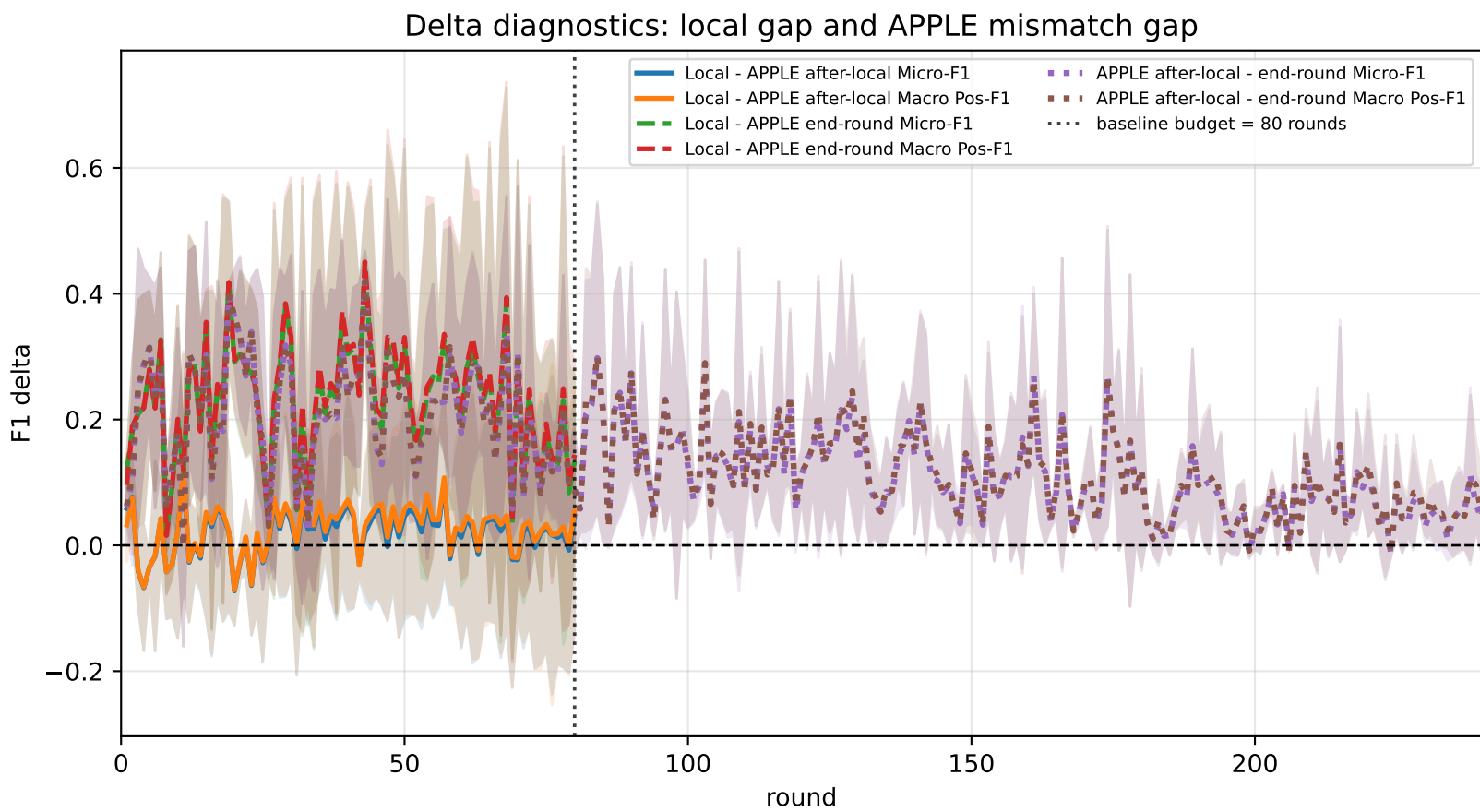
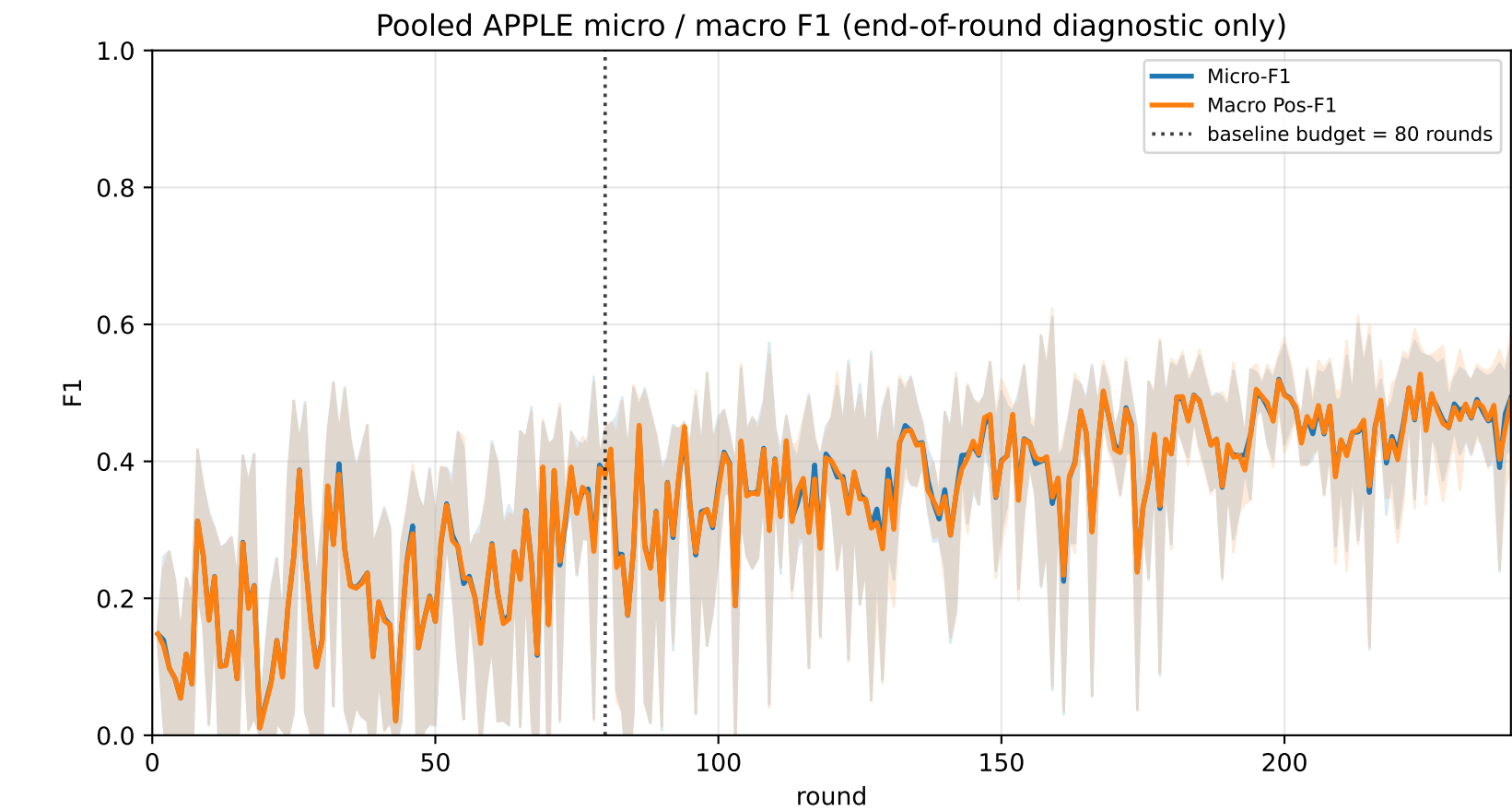
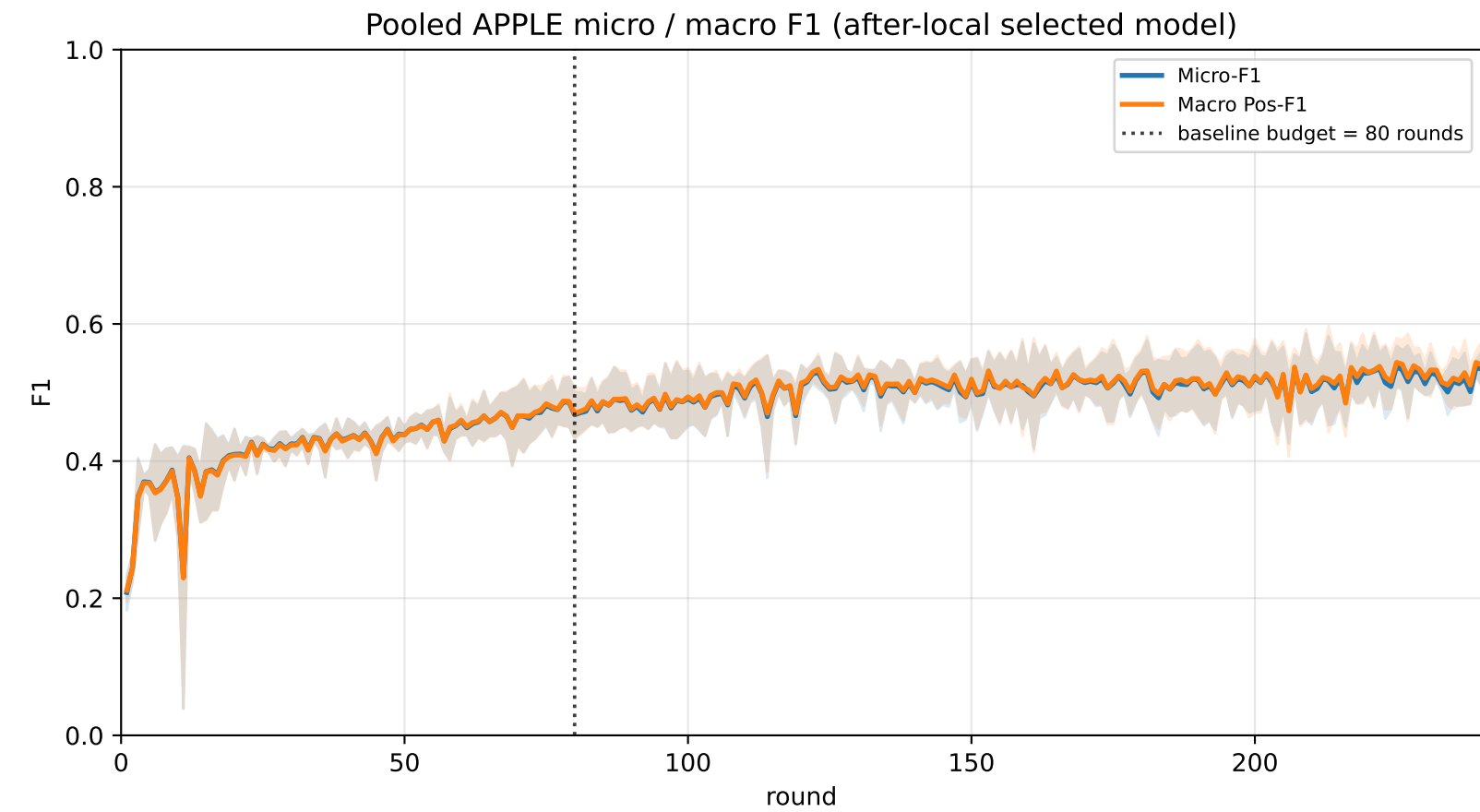
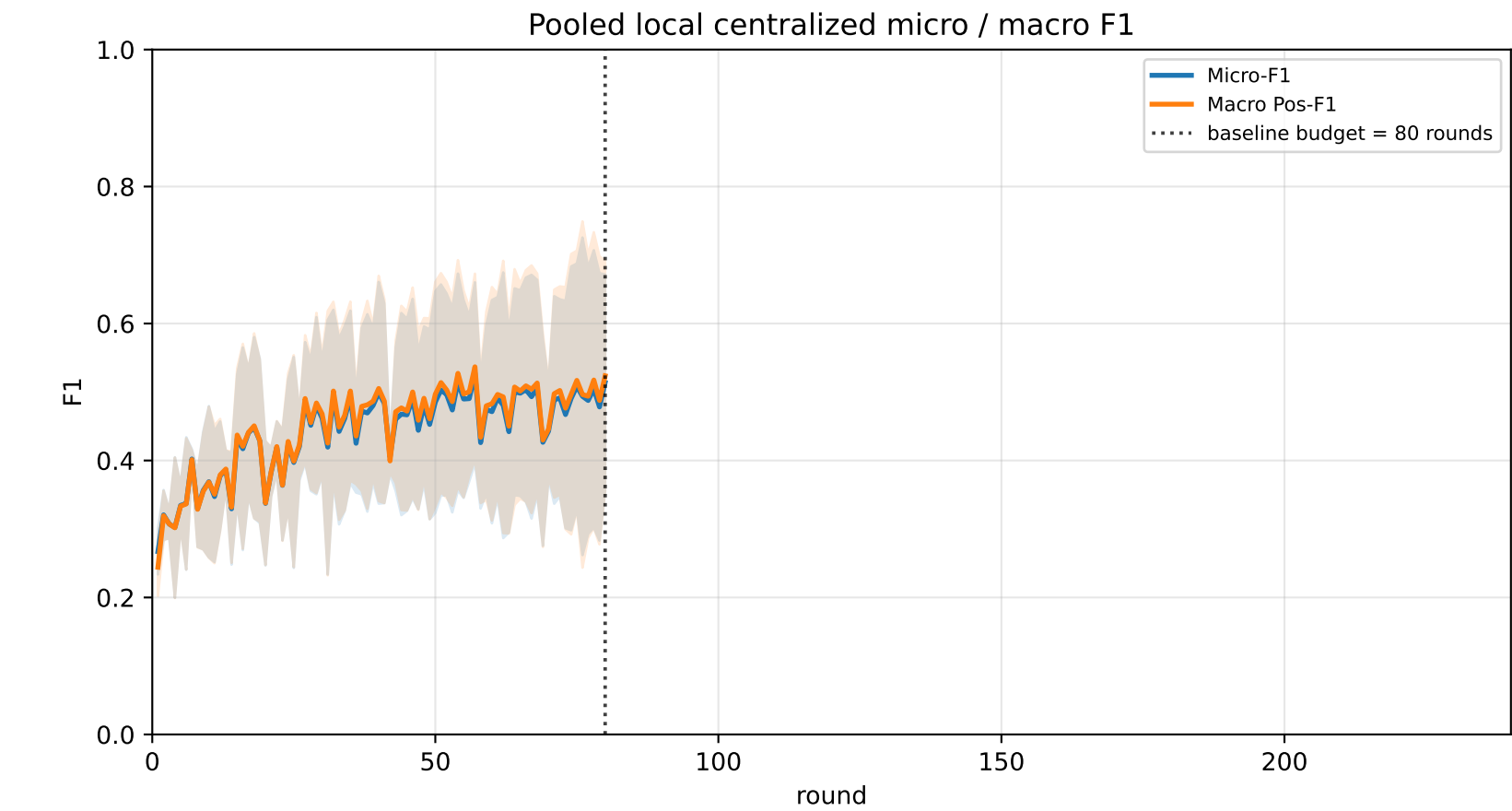
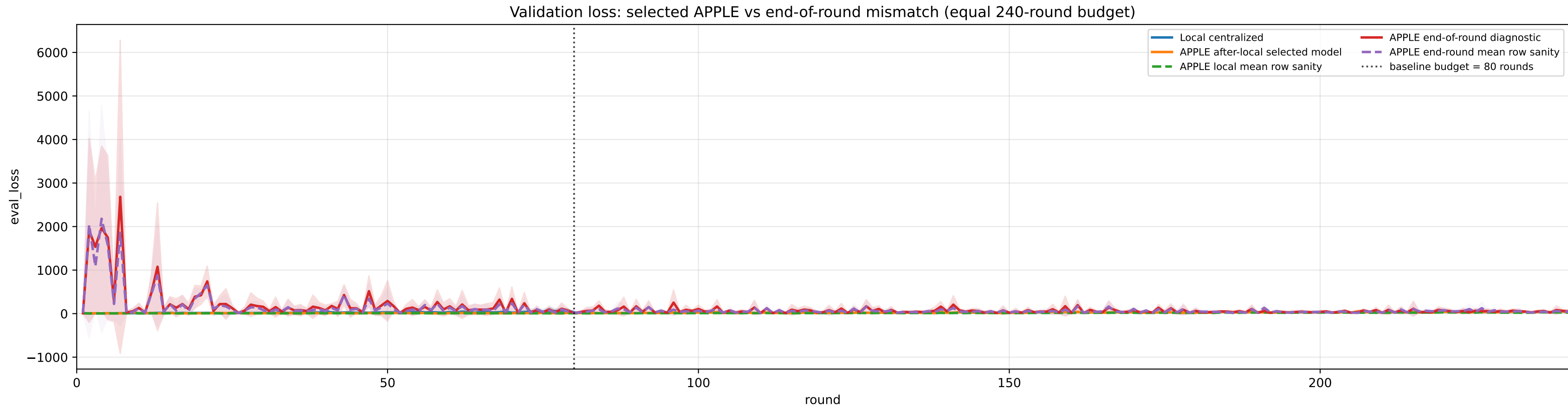
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	32.7% ± 12.1%	33.2% ± 12.8%	31.2% ± 16.3%	35.6% ± 15.5%	35.8% ± 15.4%	31.2% ± 10.3%	32.3% ± 6.7%
APPLE	32.0% ± 8.9%	32.0% ± 8.8%	26.7% ± 8.6%	32.6% ± 9.1%	32.8% ± 9.7%	31.8% ± 8.8%	35.9% ± 8.0%



family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

Best test performance (APPLE selected by after-local validation; all methods 80 rounds)

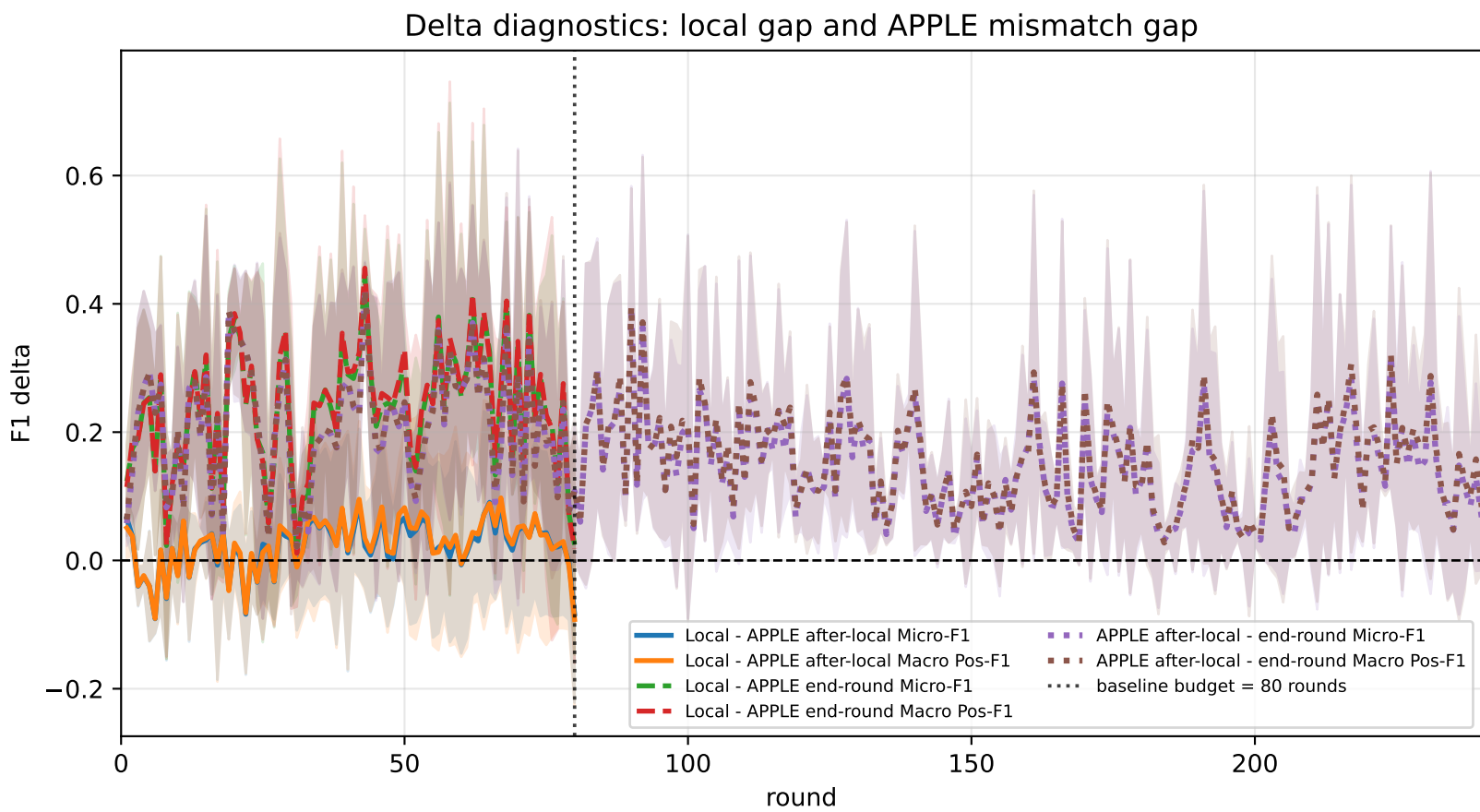
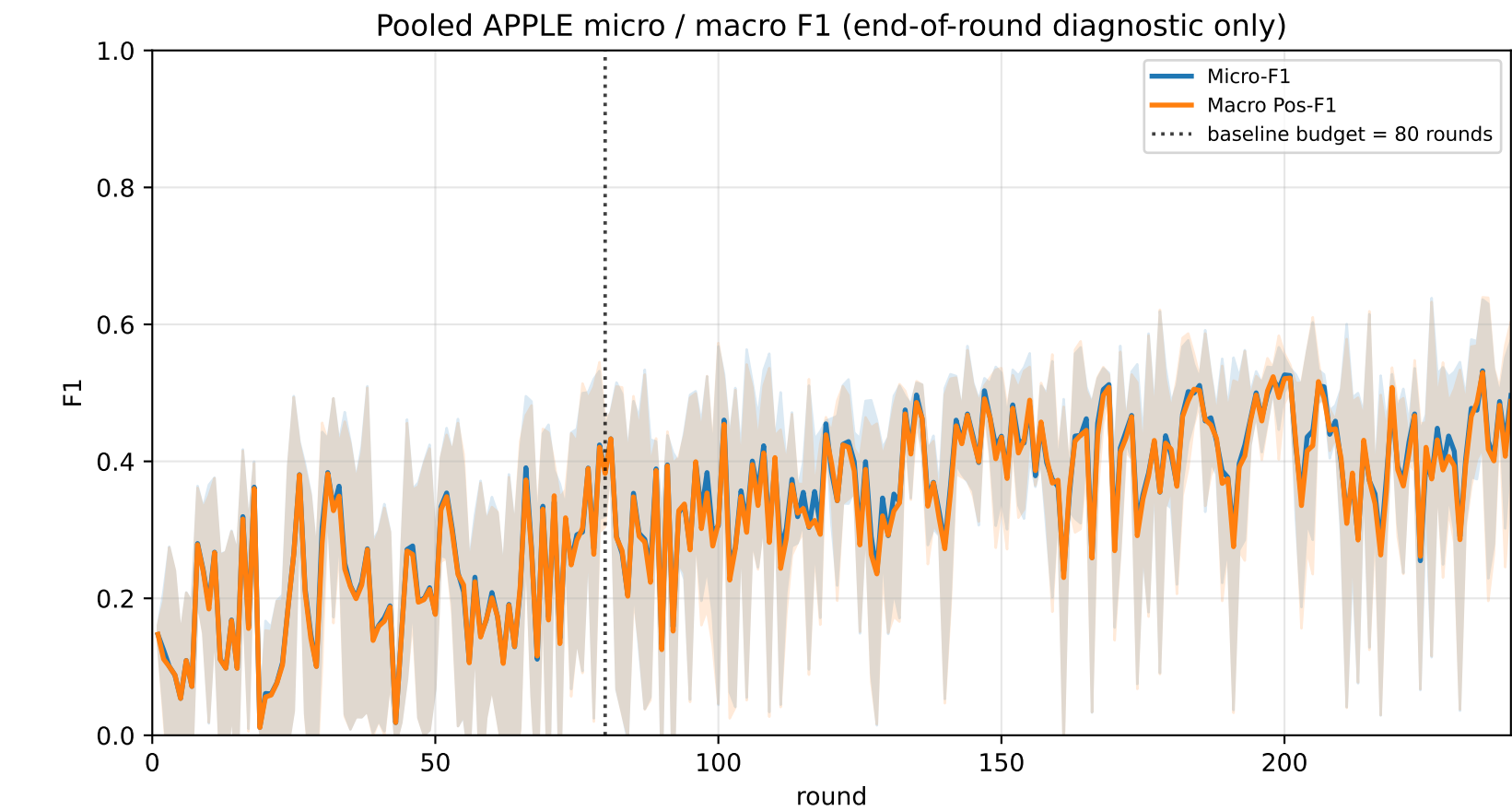
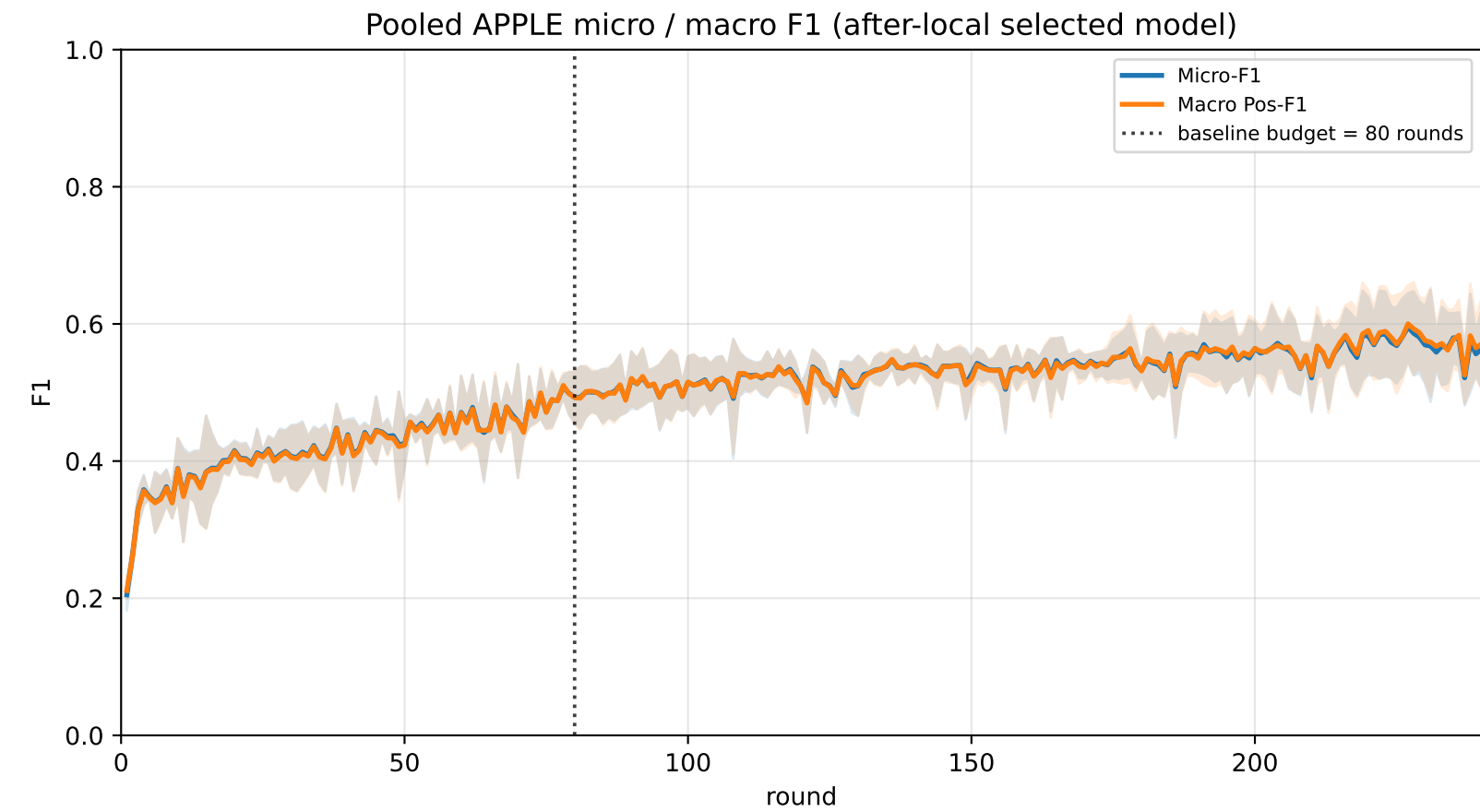
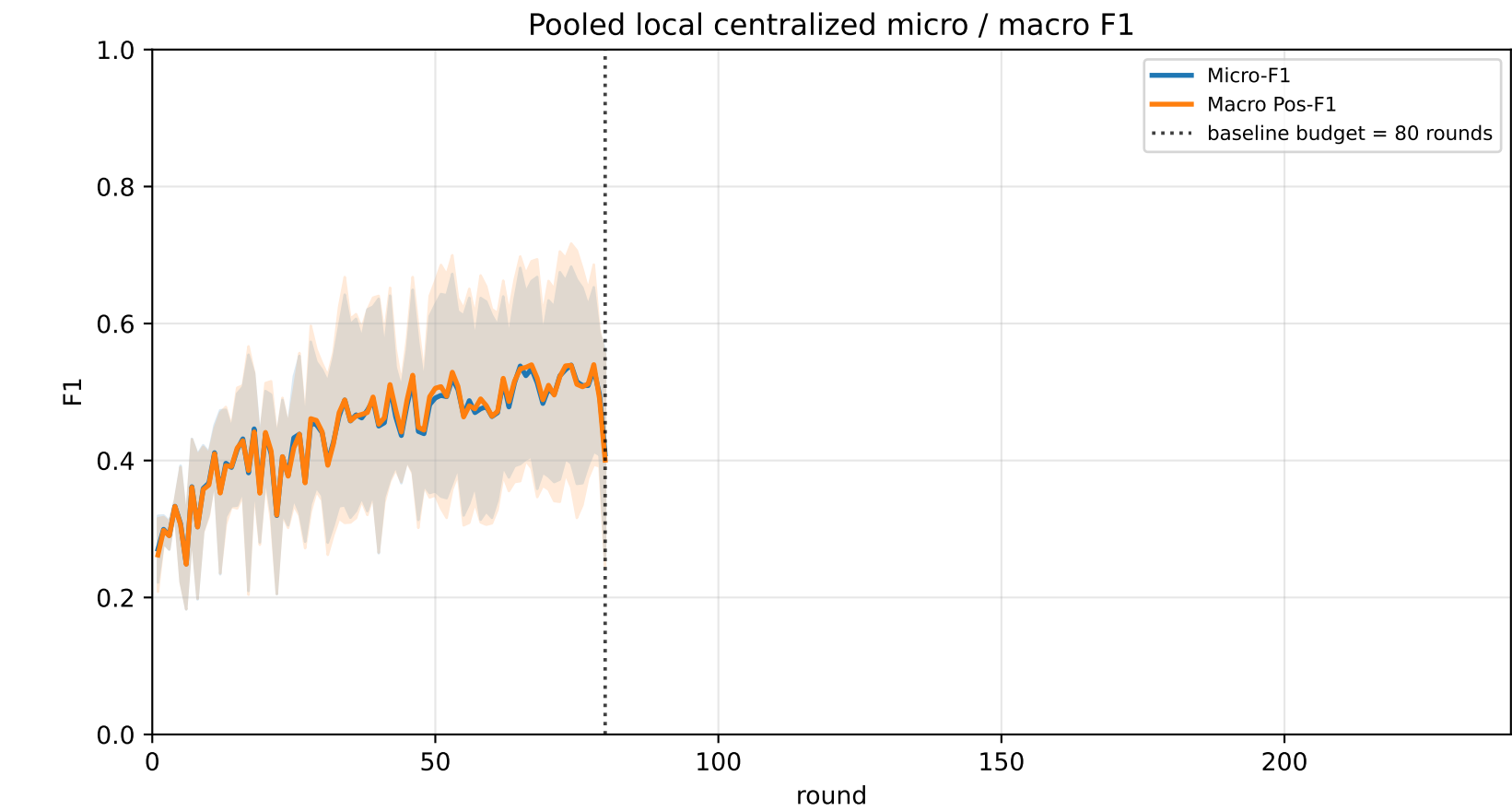
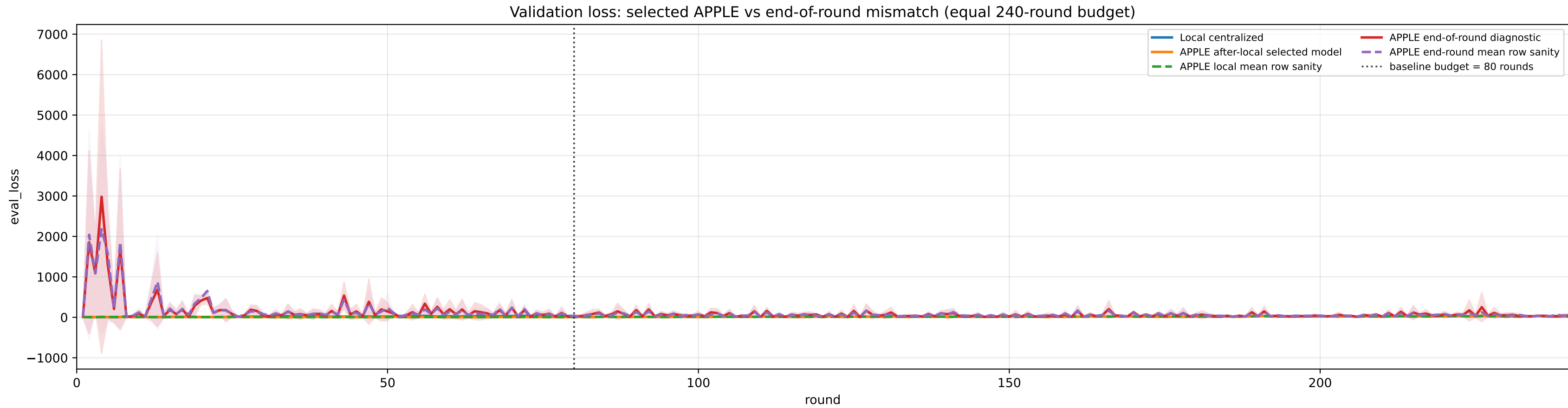
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	24.6% ± 3.5%	24.6% ± 3.5%	20.2% ± 3.0%	23.5% ± 4.2%	24.7% ± 3.6%	25.1% ± 3.4%	29.3% ± 3.3%
APPLE	31.4% ± 8.7%	31.3% ± 8.7%	26.8% ± 8.6%	31.5% ± 9.6%	32.5% ± 10.3%	32.2% ± 9.2%	33.7% ± 6.0%



family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

Best test performance (APPLE selected by after-local validation; all methods 80 rounds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	20.9% ± 2.3%	21.0% ± 2.5%	16.7% ± 1.9%	21.0% ± 1.6%	20.9% ± 2.8%	20.3% ± 1.5%	26.1% ± 4.9%
APPLE	31.1% ± 8.8%	31.0% ± 8.7%	26.5% ± 8.4%	31.4% ± 9.5%	31.9% ± 9.2%	31.1% ± 8.6%	34.4% ± 8.1%

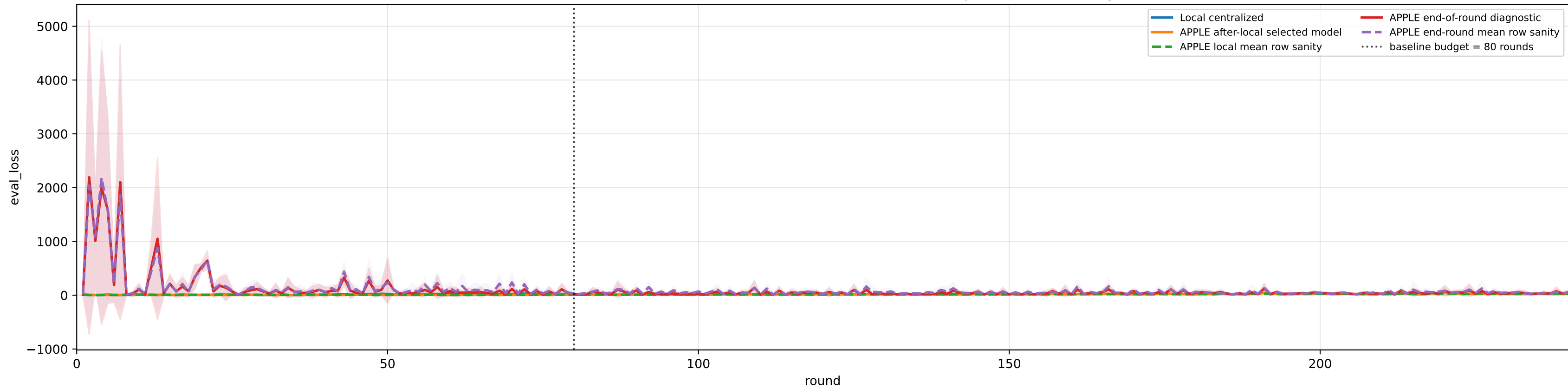


family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

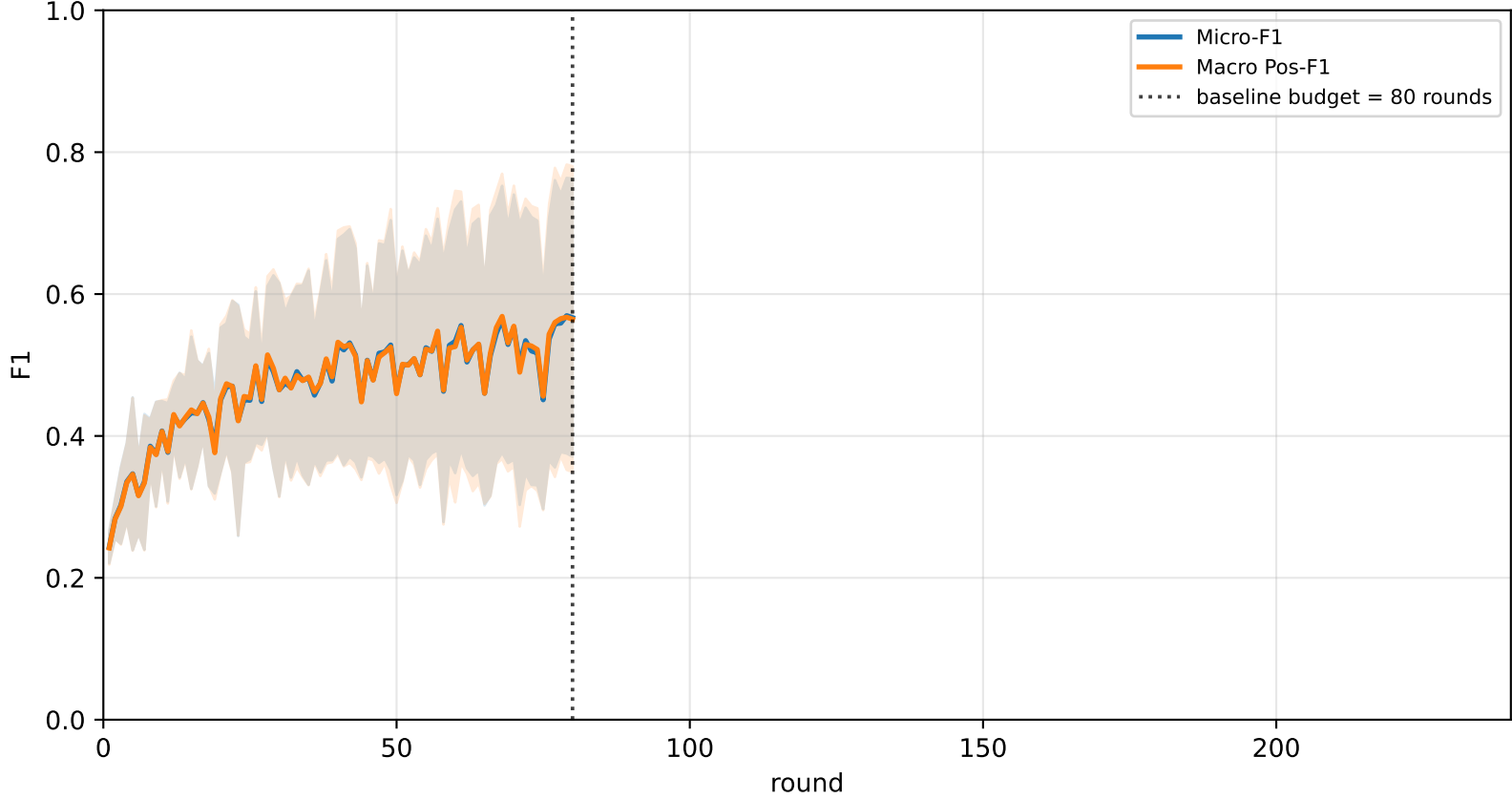
Best test performance (APPLE selected by after-local validation; all methods 80 rounds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	34.0% ± 13.0%	34.3% ± 13.3%	29.8% ± 14.7%	32.8% ± 11.9%	34.7% ± 10.6%	37.8% ± 19.0%	36.2% ± 10.7%
APPLE	30.5% ± 10.1%	30.4% ± 10.0%	25.8% ± 9.7%	31.3% ± 10.2%	31.5% ± 10.6%	30.0% ± 10.1%	33.5% ± 9.3%

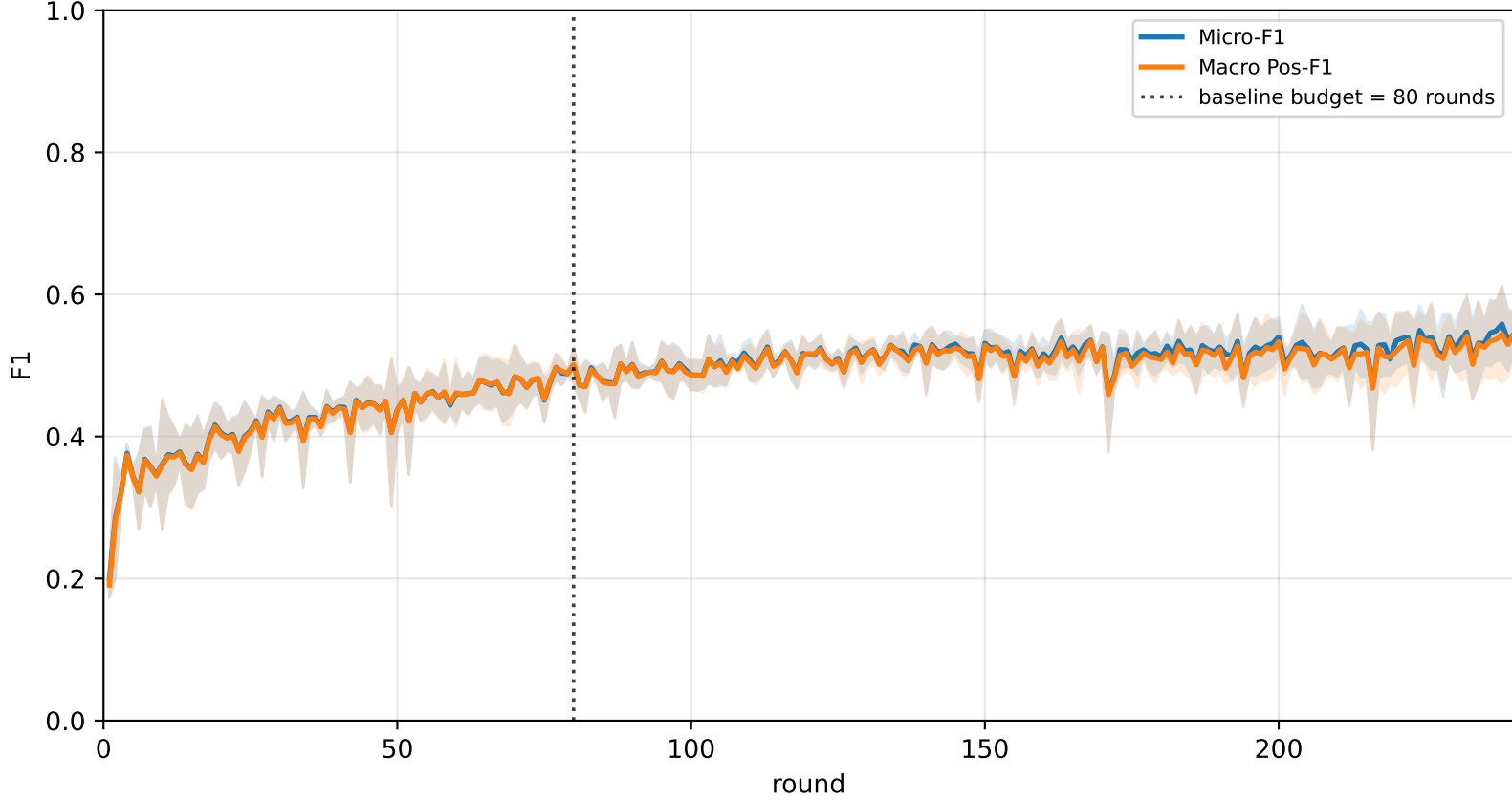
Validation loss: selected APPLE vs end-of-round mismatch (equal 240-round budget)



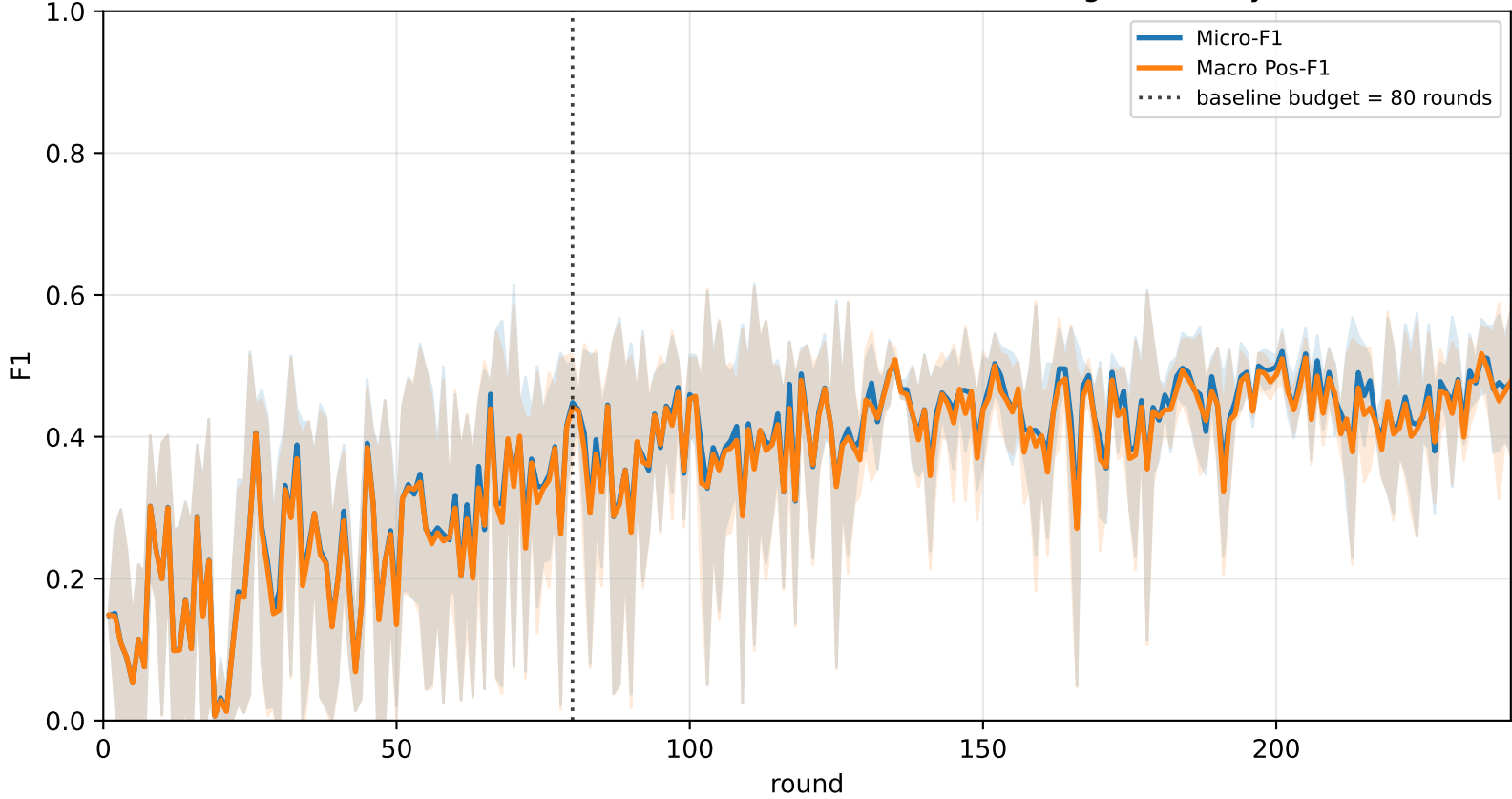
Pooled local centralized micro / macro F1



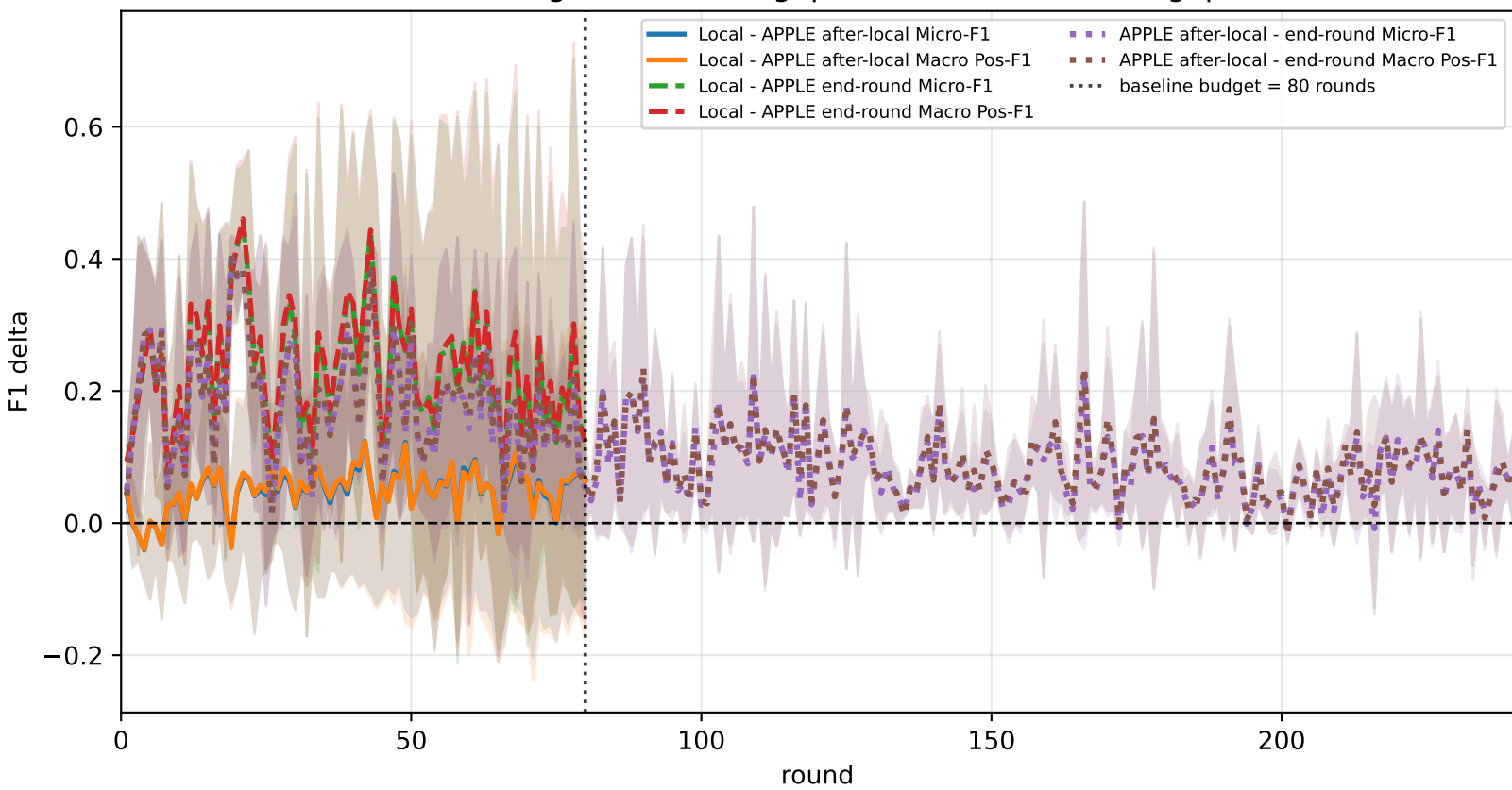
Pooled APPLE micro / macro F1 (after-local selected model)



Pooled APPLE micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: local gap and APPLE mismatch gap

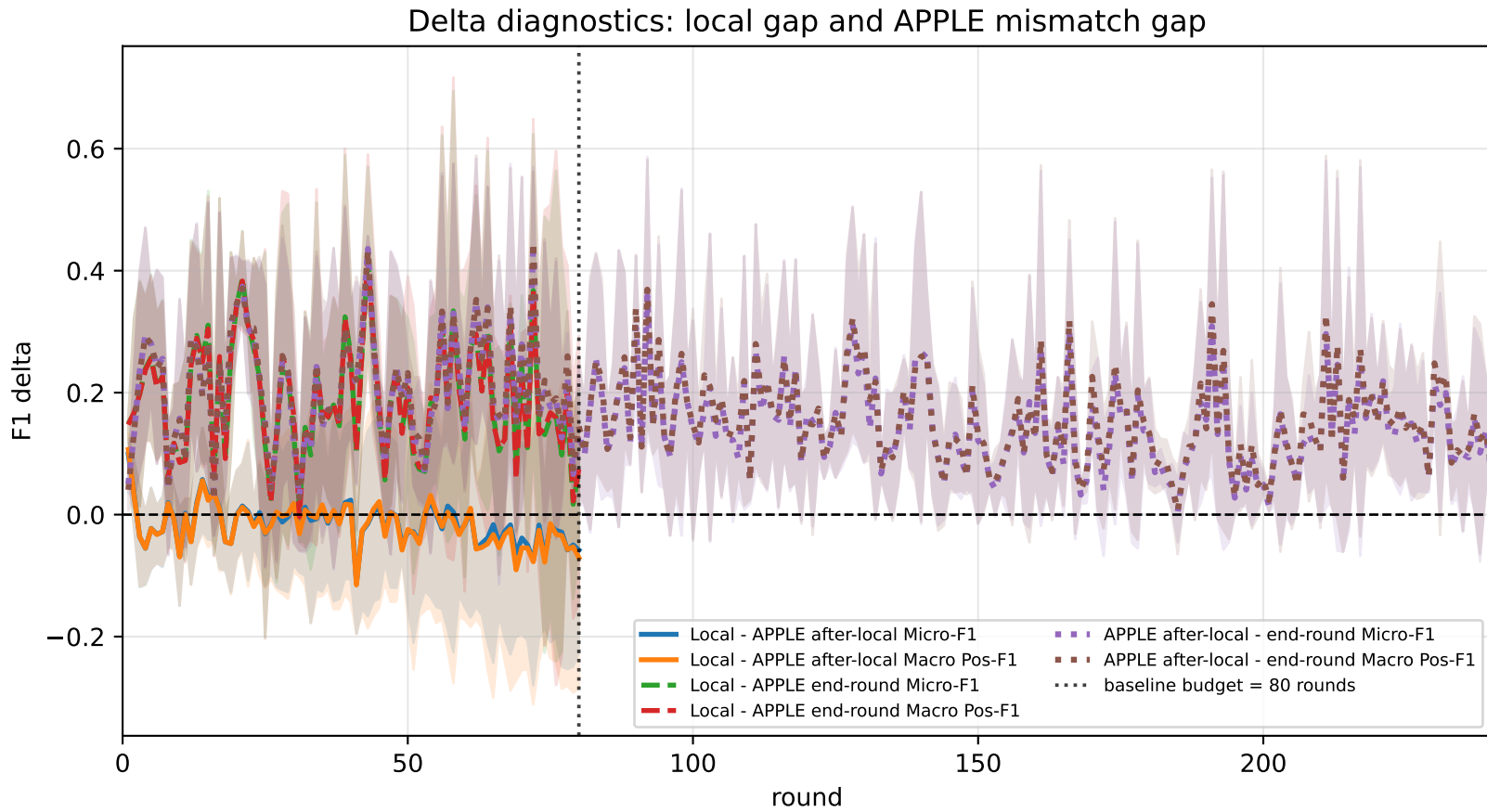
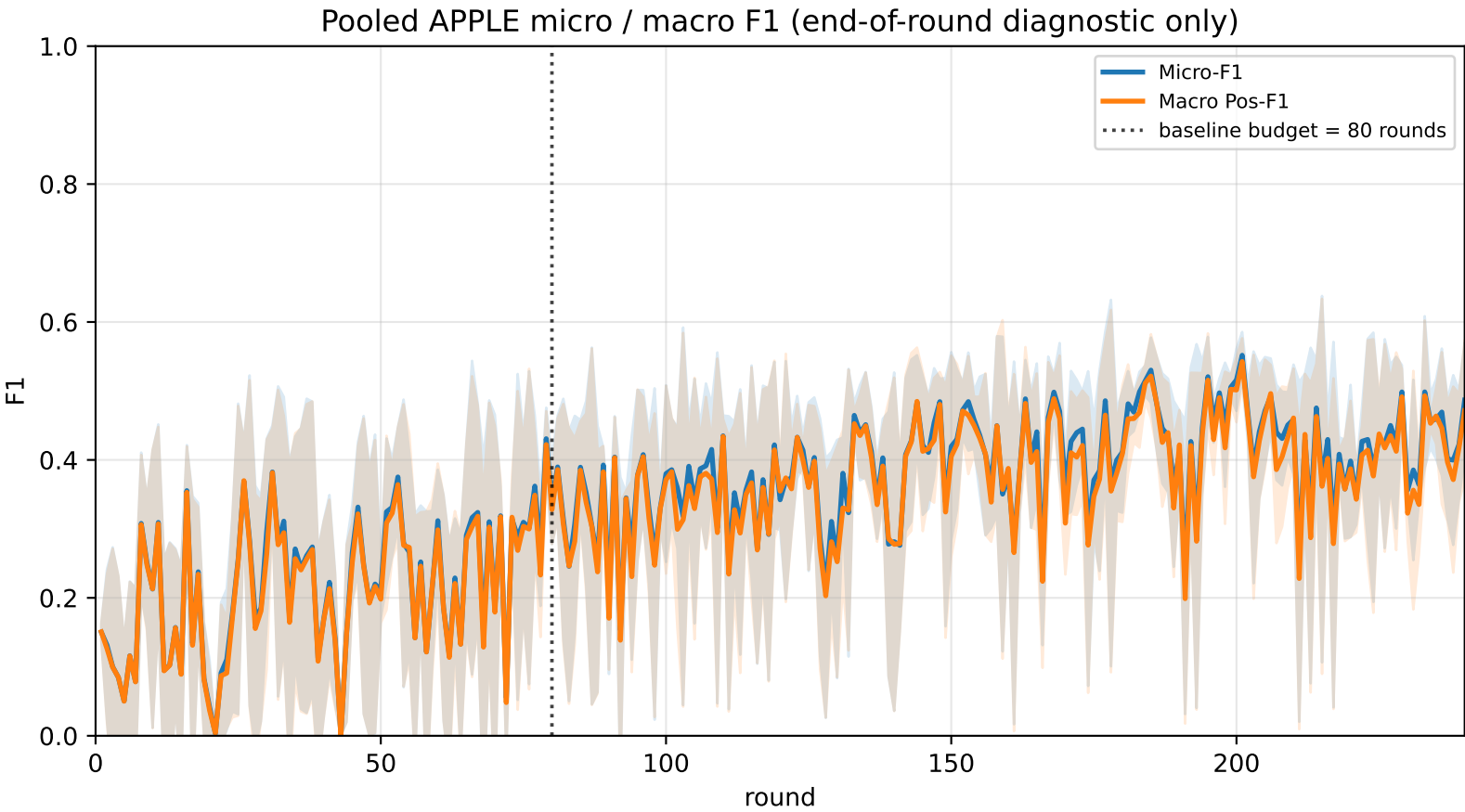
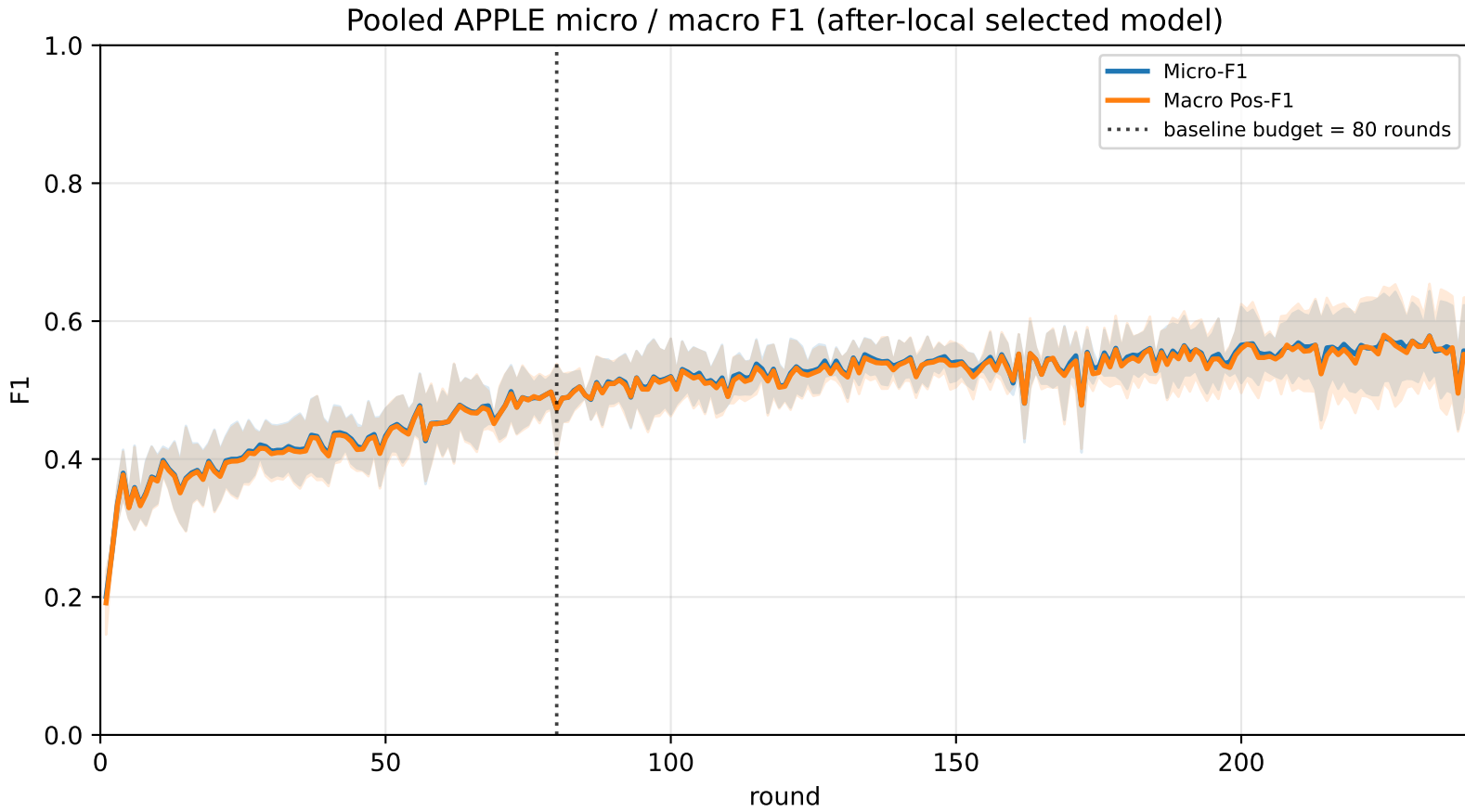
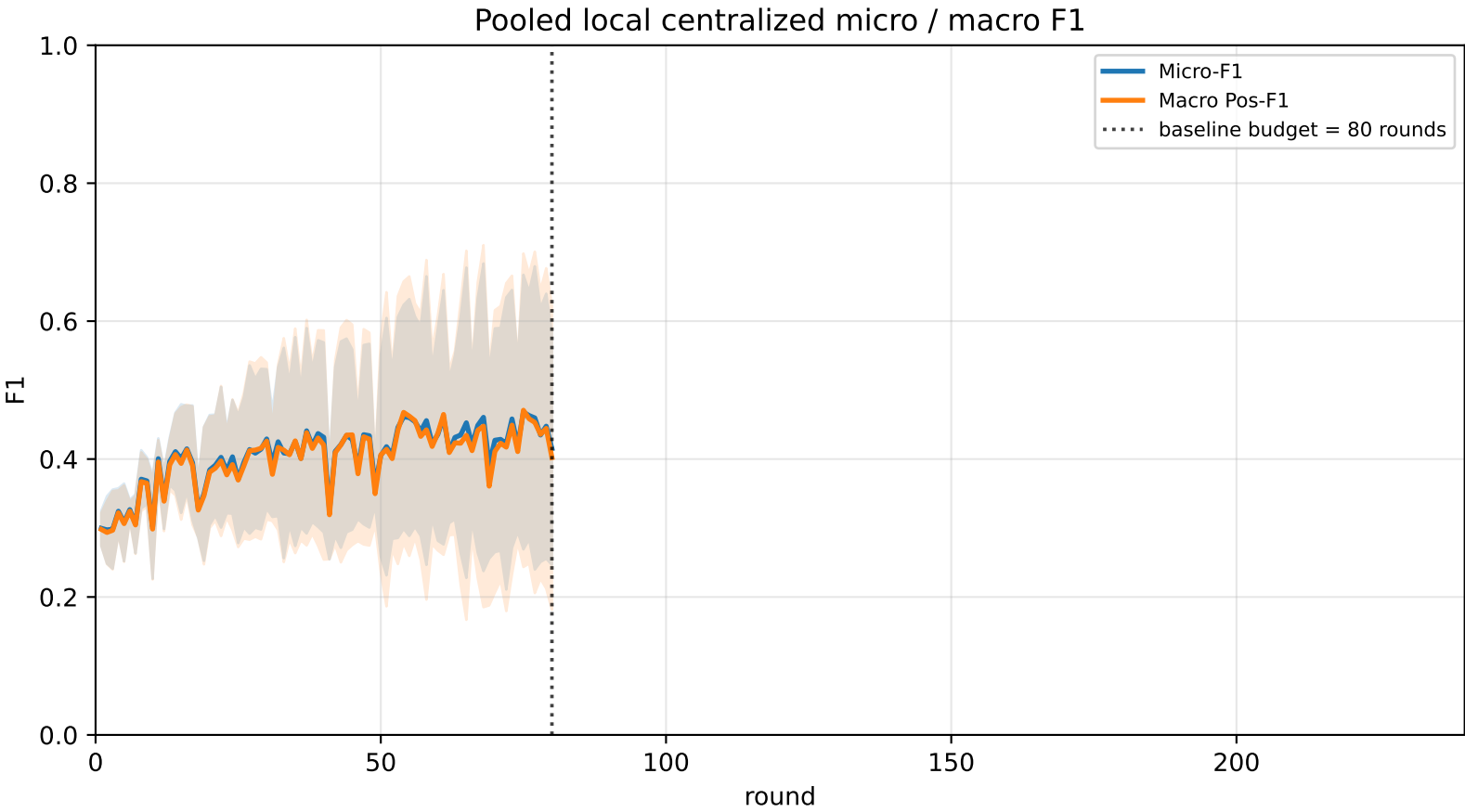
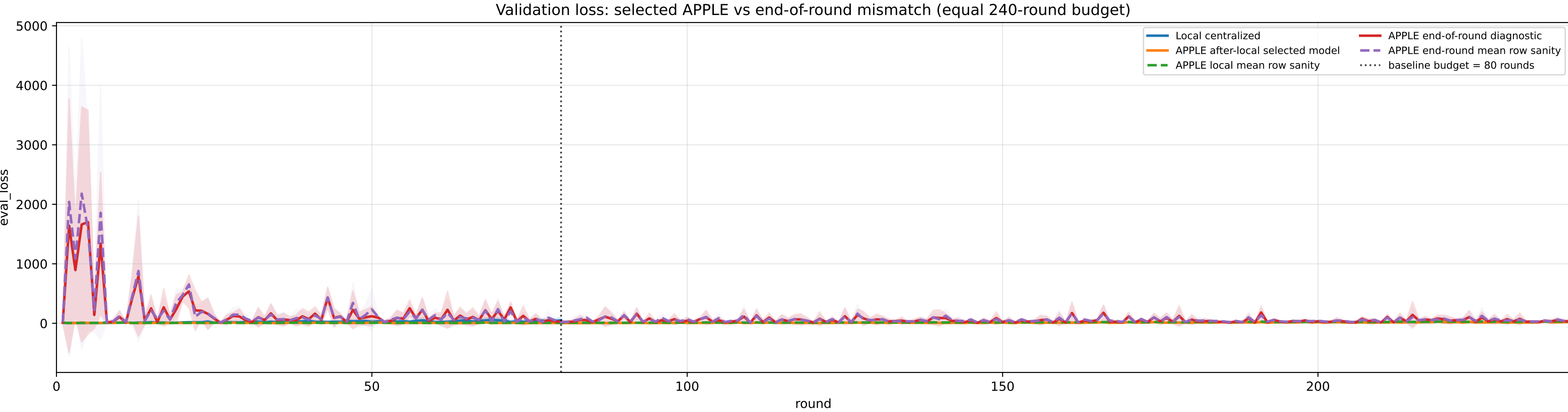


Controlled q-label heterogeneity: High | APPLE diagnostics | family=q_task_cycle6 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

Best test performance (APPLE selected by after-local validation; all methods 80 rounds)

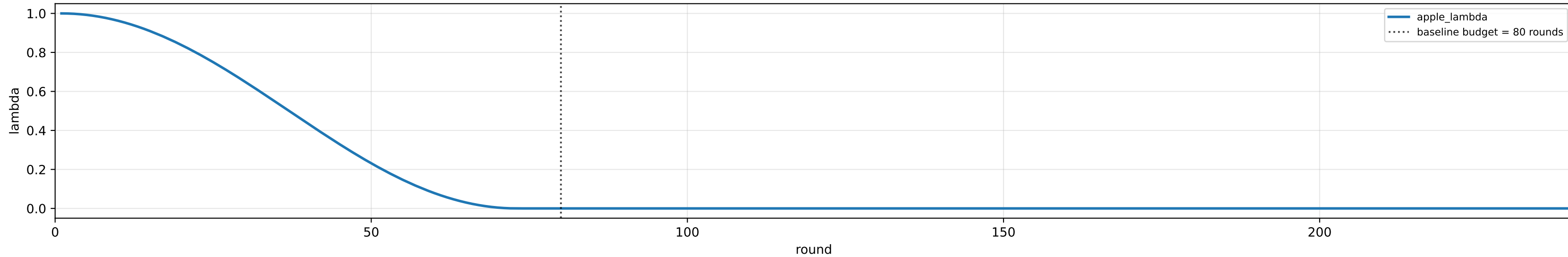
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	33.5% ± 12.2%	34.3% ± 13.3%	36.4% ± 23.0%	36.9% ± 15.6%	32.4% ± 10.8%	32.2% ± 9.6%	33.5% ± 7.5%
APPLE	31.6% ± 9.1%	31.5% ± 9.1%	26.5% ± 8.4%	31.9% ± 9.5%	33.0% ± 9.4%	31.6% ± 9.5%	34.7% ± 8.6%



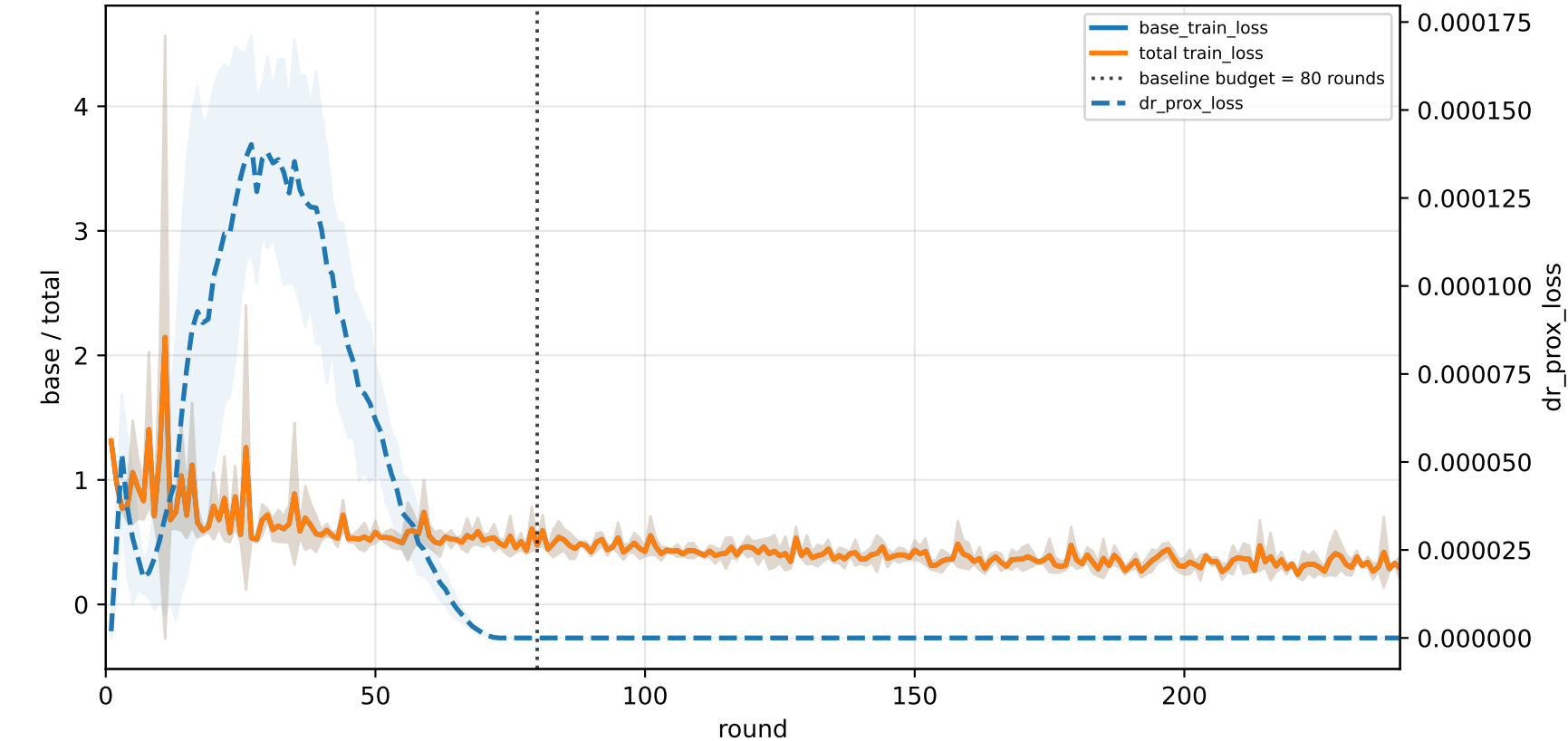
APPLE training-loss decomposition | High heterogeneity

family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
 family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
 family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
 family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
 family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
 heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

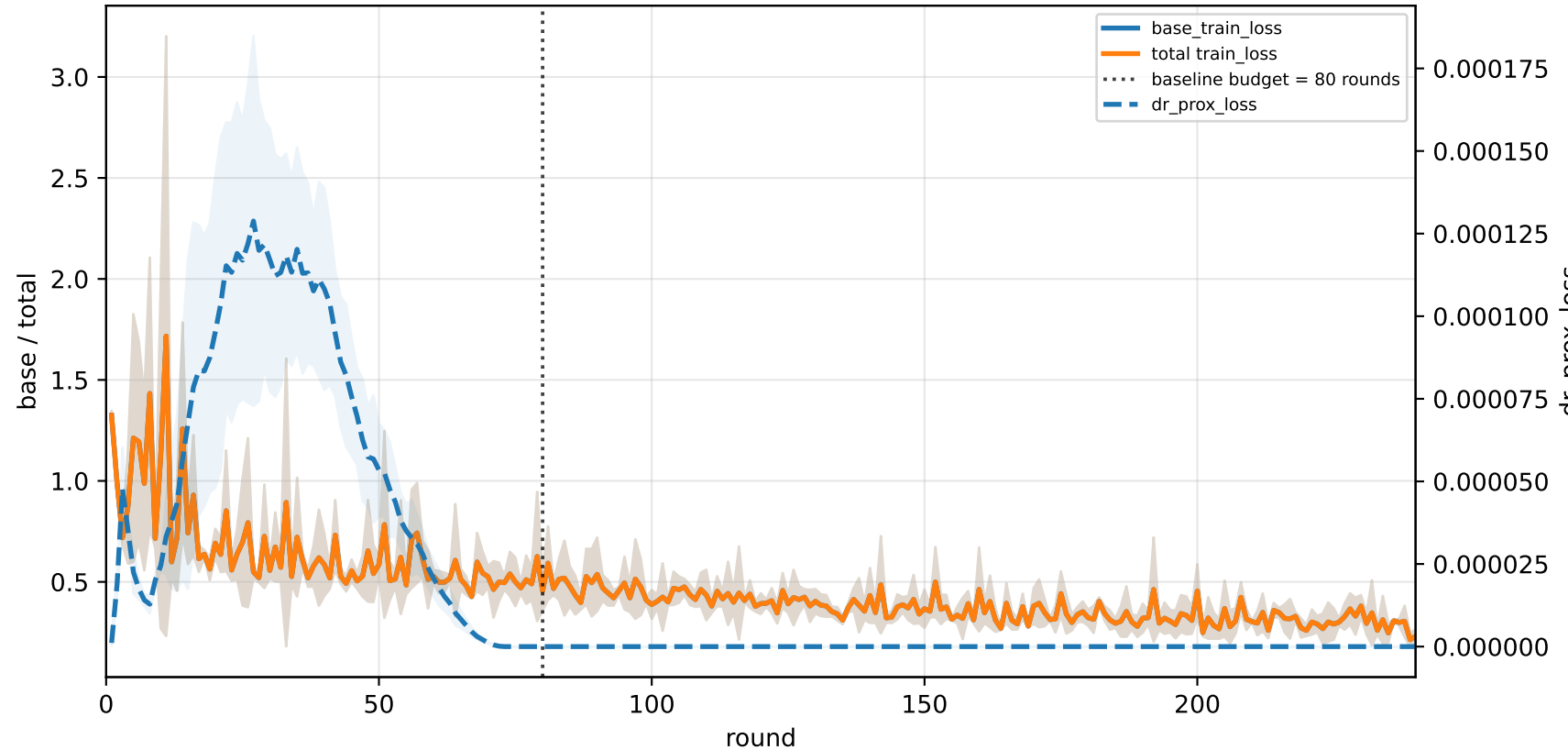
APPLE lambda scheduler | apple_mu=0.01



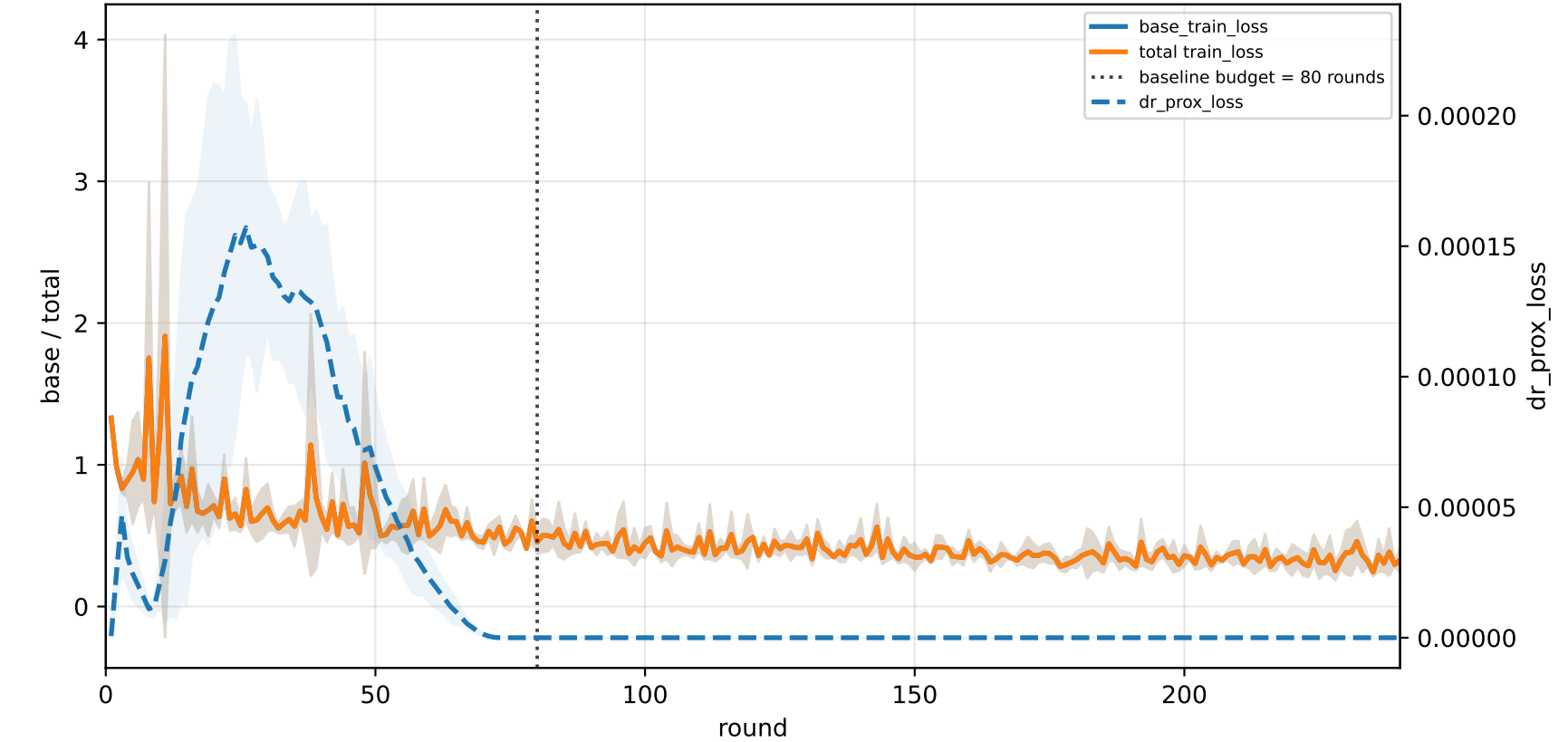
q_task_cycle2 | graph_id=3000010



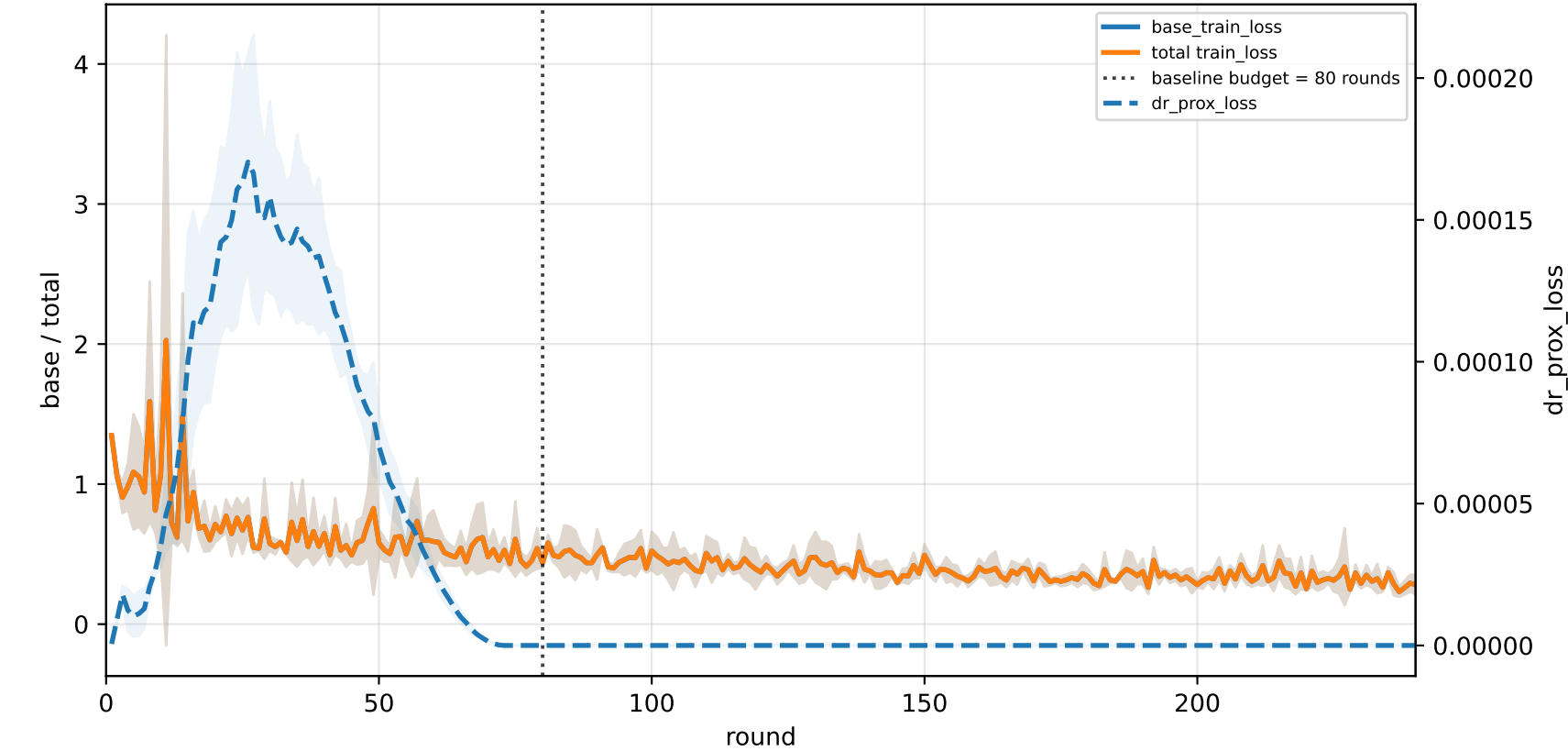
q_task_cycle3 | graph_id=3000011



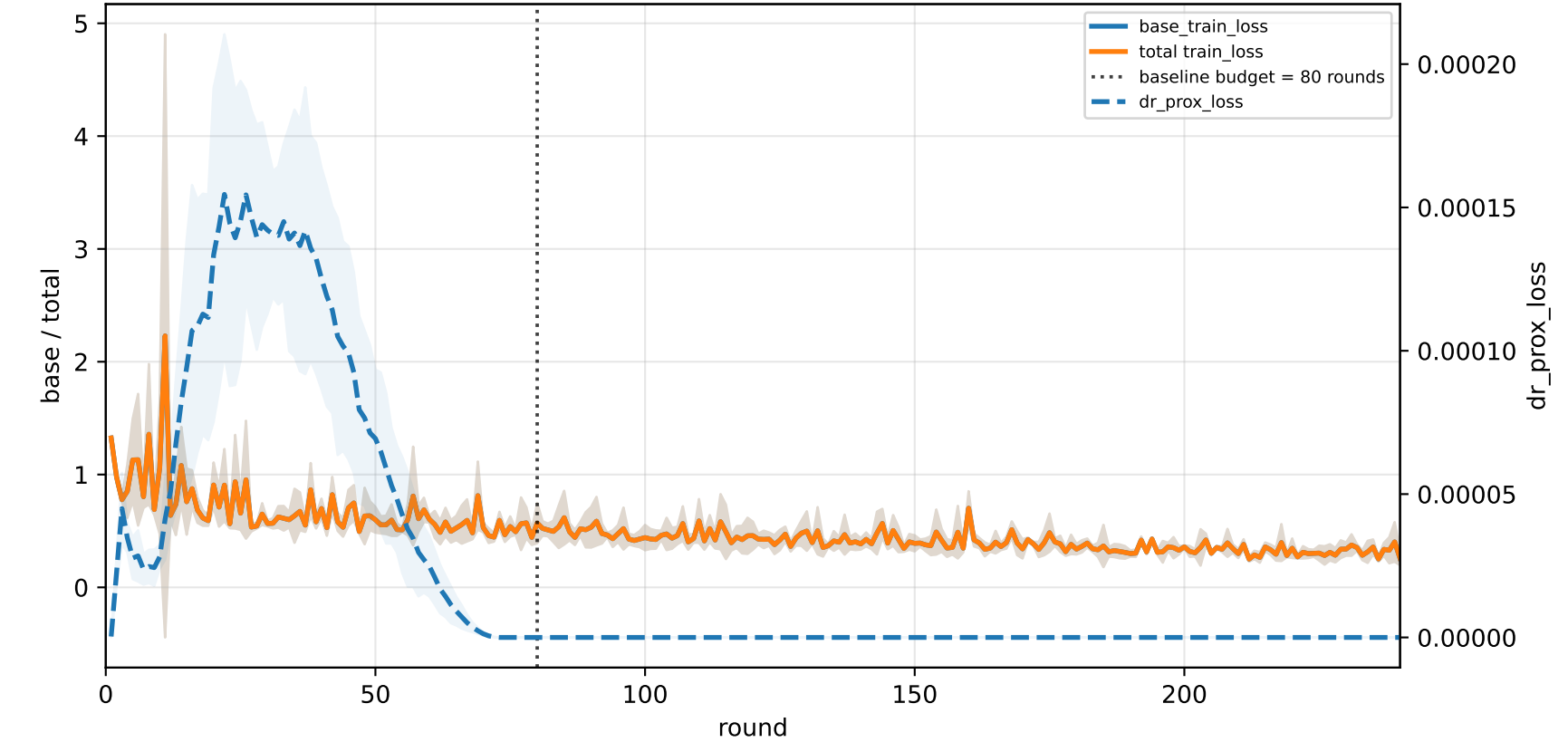
q_task_cycle4 | graph_id=3000012



q_task_cycle5 | graph_id=3000013



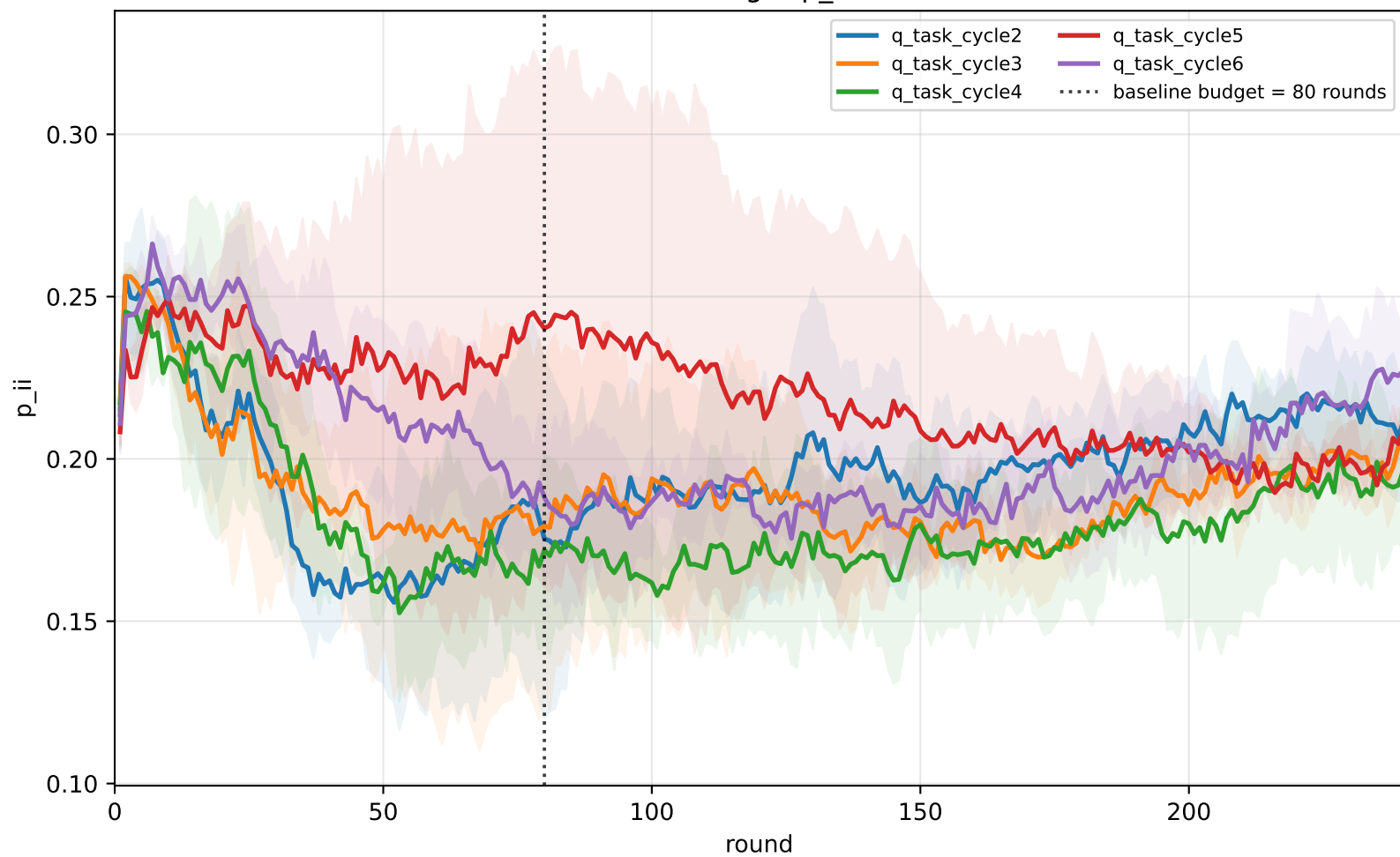
q_task_cycle6 | graph_id=3000014



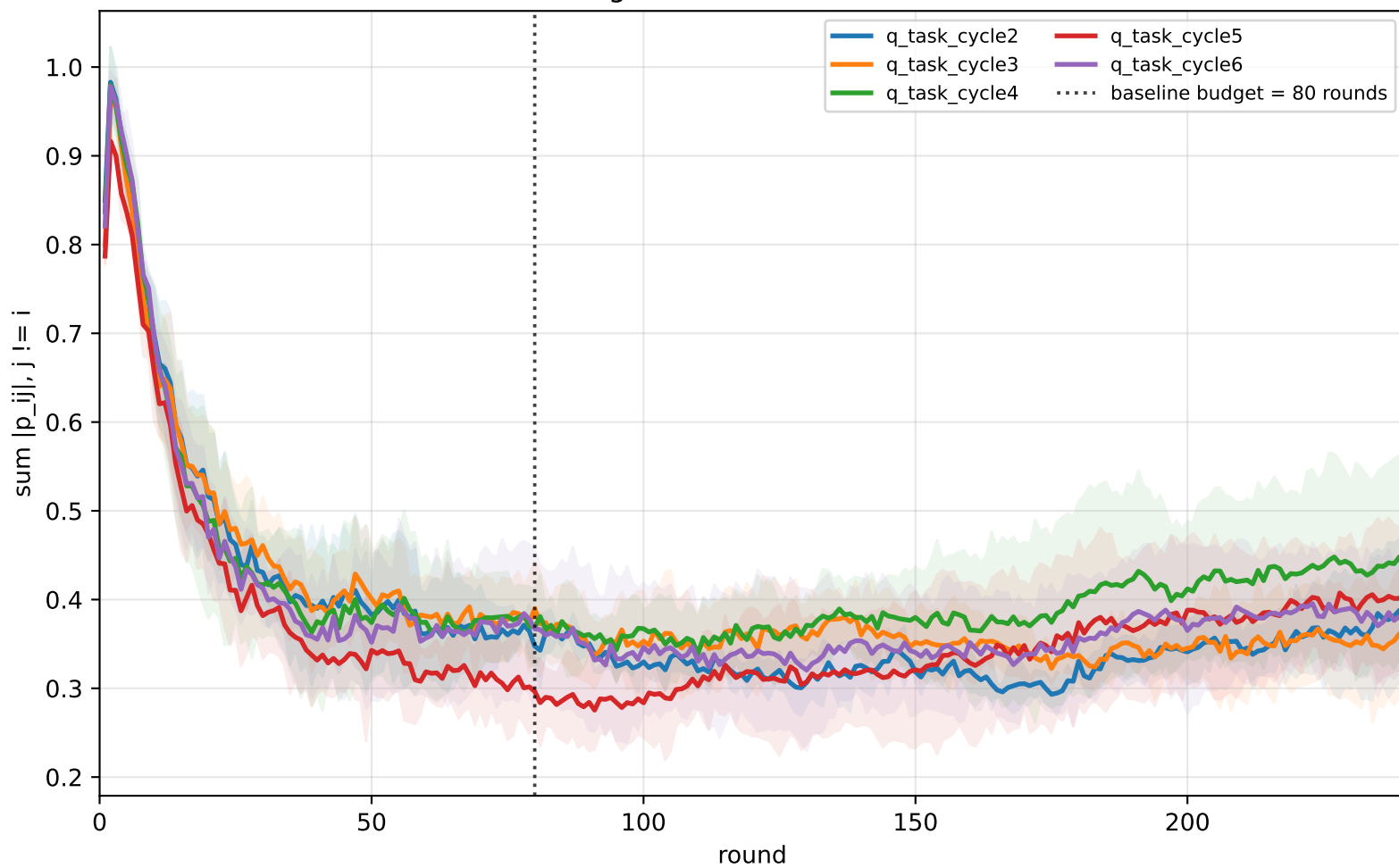
APPLE DR behavior | High heterogeneity

family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

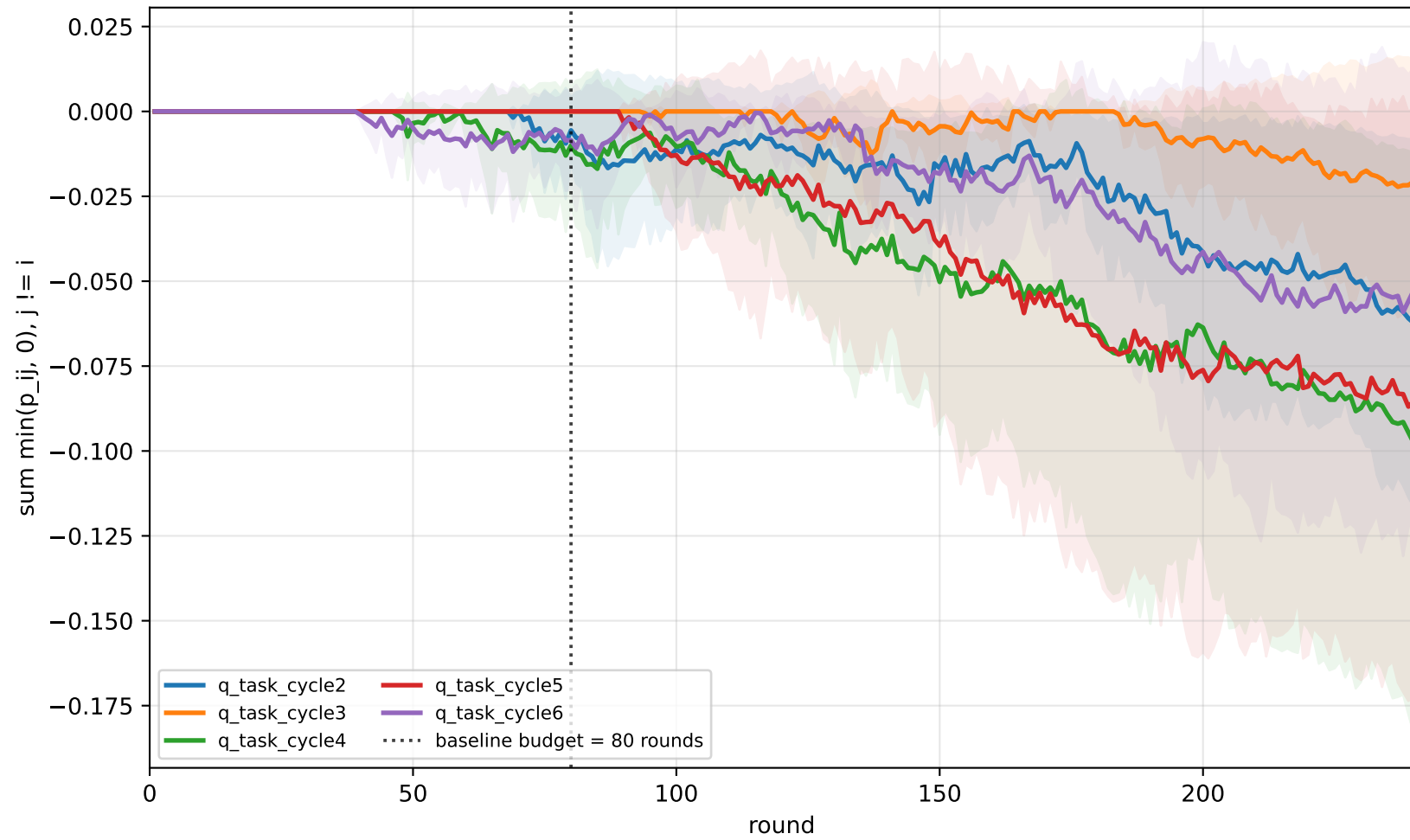
APPLE DR self-weight p_{ii} over rounds



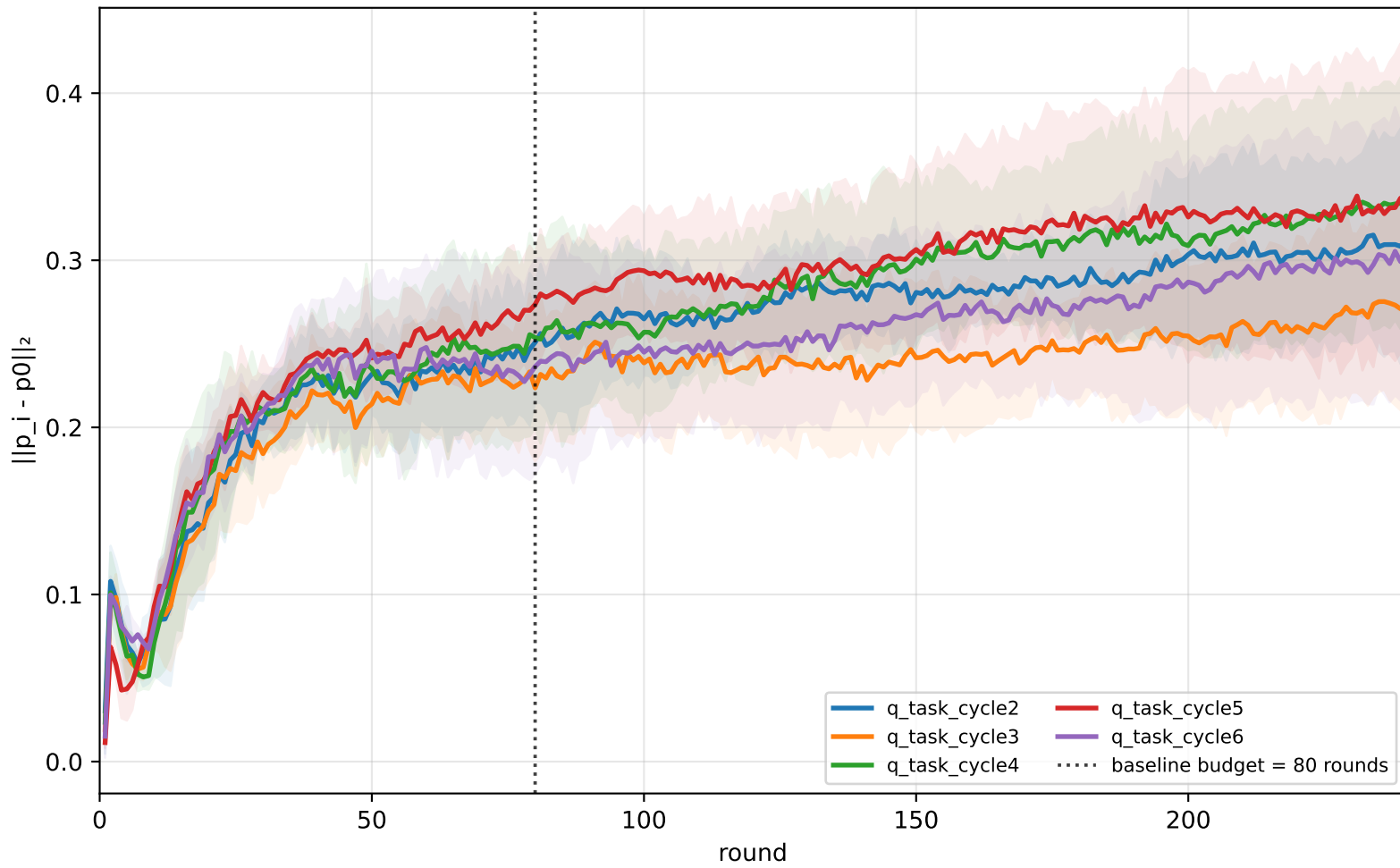
APPLE off-diagonal absolute mass over rounds



APPLE off-diagonal negative mass over rounds



APPLE DR movement away from p_0



Selected-best-round APPLE DR coefficients p_{ij} , mean over seeds

